

APPLIED PHYSICS REVIEWS

Local probe techniques for luminescence studies of low-dimensional semiconductor structures

Anders Gustafsson^{a)}

Division of Solid State Physics, Lund University, Box 118, S-221 00 Lund, Sweden^{b)} and Institut de Micro-et Optoélectronique, Département de Physique, Ecole Polytechnique Fédérale de Lausanne, CH-1015 Lausanne, Switzerland

Mats-Erik Pistol, Lars Montelius, and Lars Samuelson

Division of Solid State Physics, Lund University, Box 118, S-221 00 Lund, Sweden

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With the rapid development of technologies for the fabrication of, as well as applications of low-dimensional structures, the demands on characterization techniques increase. Spatial resolution is especially crucial, where techniques for probing the properties of very small volumes, in the extreme case quantum structures, are essential. In this article we review the state-of-the-art in local probe techniques for studying the properties of nanostructures, concentrating on methods involving monitoring the properties related to photon emission. These techniques are sensitive enough to reveal the electronic structure of low-dimensional semiconductor structures and are, therefore, able to give detailed information about the geometrical structure, including fabrication-related inhomogeneities within an ensemble of structures. The local luminescence probe techniques discussed in this review article can be divided into four categories according to the excitation source: (i) spatially localized microphotoluminescence spectroscopy using either strong focusing or masking; (ii) near-field optical microscopy to reach below the diffraction limitation of far-field optics, by either exciting, detecting, or both exciting and detecting in the near field; (iii) cathodoluminescence using focused energetic electrons in an electron microscope; and (iv) scanning tunneling luminescence, using low-energy electrons injected or extracted from the tip of a scanning tunneling microscope. © 1998 American Institute of Physics. [S0021-8979(98)08615-0]

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^{a)}Corresponding author; electronic mail: anders.gustafsson@ftf.lth.se

^{b)}Present and permanent address.

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I. INTRODUCTION

A. The need for local luminescence probes

Luminescence, i.e., radiative recombination of carriers, has traditionally been a very important method for studies of semiconductors. The inherent strengths of luminescence-based techniques are related to the characteristic band gap of semiconductors and that optical transitions across the band gap, either as intrinsic recombination or as recombination involving impurity states, are very efficient. The efficiency of the luminescence is, of course, one of the main reason for the rich variety of commercial optoelectronic devices, e.g., light-emitting diodes and injection laser diodes. Conventional optical spectroscopy uses excitation of large sample volumes and, thus, averages the properties over this volume. For a macroscopically homogeneous semiconductor this is insignificant, since the properties observed are the same on the length scale probed. However, for properties that vary over a typical length scale of $1\ \mu\text{m}$ or less, clearly a sub- μm spatial resolution is needed.

During the last decades there has been an unbroken trend towards smaller and smaller devices. Today, commercial devices and circuits employ lateral dimensions of the individual components down to $\sim 0.25\ \mu\text{m}$. Within the next decade, devices for photonic as well as for high-speed electronics will use characteristic dimensions of components around or less than $0.1\ \mu\text{m}$. In certain cases, quantum devices will use active device dimensions in the range $1\text{--}10\ \text{nm}$. Among such quantum devices are lasers, detectors, and modulators for photonic applications and resonant tunneling and ballistic transistor devices for high-speed electronics. It is obvious that these new quantum devices require alternatives to the conventional, macroscopic characterization methods. It is also reasonable to expect that optical methods will continue to be important to characterize electronic devices in a nondestructive and nonintrusive way.

In the remainder of this article we will first describe some aspects of the physics of low-dimensional structures, primarily to define the terminology for the readers with less experience in this field. We will then discuss the different local luminescence probe techniques in more detail, with special emphasis on the excitation mechanisms and characteristics of the respective methods. Finally, we will discuss a number of application areas for luminescence-based local probe techniques, ranging from observation of dislocations

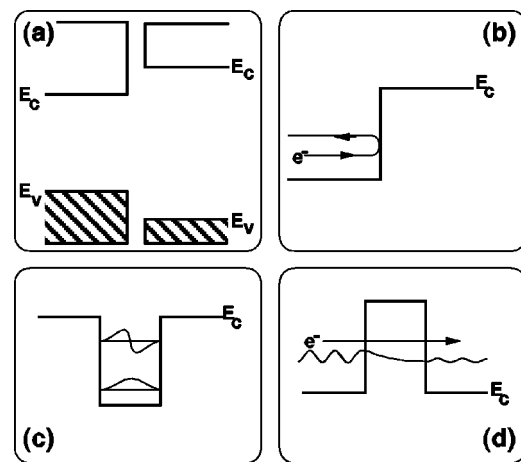


FIG. 1. (a) conduction- and valence-band edges of two semiconductors lined up such that electrons and holes experience lower band-edge energies in the left material. (b) heterostructure interface with a band alignment such that electrons are, at low temperatures, prevented from moving from the left-side semiconductor into the right-side semiconductor. (c) two heterostructure interfaces with a band alignment such that the conduction-band offset localizes the electron in the central semiconductor. Effectively, the structure of the conduction-band edge is that of a particle in a box, with finite potential barriers. (d) two heterostructure interfaces with a band alignment such that the conduction-band off-set localizes the electron in the two outer semiconductors. If the central semiconductor is thin enough, electrons may tunnel quantum mechanically from the left side, through the central barrier layer, and exit into the right side.

in strained layer structures to the study of monoatomic steps at the interface of a quantum well (QW), the imaging and characteristic luminescence spectra of individual quantum wire (QWR) and quantum dot (QD) structures, to the observation of emission from individual impurity atoms in quantum structures.

B. The physics of low-dimensional semiconductor structures

In this section we will introduce a few of the most fundamental phenomena, which are central for almost all the properties of low-dimensional semiconductor structures dealt with in this review article. A reader with experience in this field may choose to proceed directly to the next section. For more detailed reading the reader is recommended a textbook, e.g., *Physics of Semiconductors and Their Heterostructures*, by J. Singh.¹

Figure 1(a) illustrates how two semiconductors with conduction- and valence-band edges E_c and E_v can have different magnitudes of their forbidden energy gap, or band gap (E_g), and different lineup of the band edges. An electron at the conduction-band edge will have a lower total energy in the left material than in the right one. Figure 1(b) illustrates the phenomenon of a potential barrier which restricts the freedom of an electron incident from the left. Such a barrier is created, e.g., if the semiconductor on the left side in the figure has a lower conduction-band edge than that on the right. This situation is found for the conduction-band structure of GaAs (lower band gap) and AlGaAs (higher band gap). The electron will be forbidden to enter the right-hand

portion of the structure and will instead be reflected back by the potential barrier.

The next concept we want to introduce is that of confined states in a local potential well. A macroscopic semiconductor can be seen as a three-dimensional (3D) object in which the electrons move freely in all three dimensions. The kinetic energy E_e is thus given by

$$E_e = \frac{(\hbar k_x)^2}{2m^*} + \frac{(\hbar k_y)^2}{2m^*} + \frac{(\hbar k_z)^2}{2m^*}, \quad (1)$$

where $\hbar$ is Planck's constant, m^* is the effective mass, and k is the wave vector. Electrons with a kinetic energy of $\sim k_B T$ or less are considered as thermalized. Electrons with a kinetic energy well above $k_B T$ are generally denoted hot electrons. k_B is Boltzman's constant and T is the temperature.

The motion of electrons is controlled in low-dimensional structures by enforcing a restriction on them in terms of potential barriers. These barriers prevent an electron (hole) from moving in certain directions. A situation may be encountered similar to that Fig. 1(c), in which the electron can be trapped in between two barriers. This situation resembles that of the favorite textbook example of a particle, e.g., an electron trapped in a square-well potential. The well-known solution to this problem is that of localized states having the principle spatial distribution as sketched in Fig. 1(c). Hence, the potential structure has replaced the quasicontinuous energy-band structure of the free particles by a few discrete energy levels in the potential well.

Besides the confining quantum structures described above, different combinations of semiconductor materials can cause other types of quantum-mechanical phenomena. One important example is illustrated in Fig. 1(d), in which an electron (illustrated as a wave entering from the left) has a finite probability of quantum-mechanical tunneling through a sufficiently thin potential barrier. In the illustration it can be seen how the amplitude of the wave function decays in the forbidden area of the barrier and how the wave continues on the right-hand side of the barrier. Such potential structures are employed in (resonant) tunneling devices.

The main driving force for the interest in, and the development of, these new devices is related to the unique electronic structure of low-dimensional materials and to the corresponding density of states (DOS). An example of these low-dimensional structures is a QW, which is an extremely thin film formed by an epitaxial process, using the fact that the composition of the film can be changed within one or a few atomic layers. If the film is sufficiently thin, this leads to quantization in one dimension (see Fig. 2; 2D). As a consequence, the states of the electrons are pushed up in energy by the quantum confinement, resulting in only a small number of discrete energy levels, replacing the quasicontinuous distribution in a macroscopic crystal. In the case of the two-dimensional (2D) QW, the carriers are free to move in the plane of the film, which gives the total DOS a characteristic staircase shape, replacing the parabolic shape of the bulk DOS. QWs are practically used in semiconductor laser structures, where the increased overlap between electron and hole wave functions, as well as the more advantageous DOS are used to create laser diodes of higher efficiency and power.

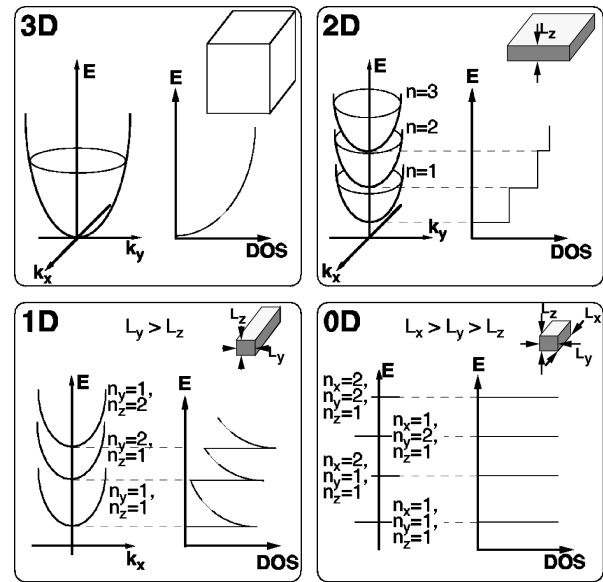


FIG. 2. In 3D (bulk) the carriers are free to move in all three dimensions, x , y , and z . The energy describes a four-dimensional parabola, generated by wave-vectors k_x , k_y , and k_z , illustrated by k_x and k_y in the figure. This leads to a DOS that is a square-root function of the energy. In 2D (QW) the carriers are free to move in two dimensions, x and y , and restricted in the third, z . This leads to a three-dimensional parabola generated by k_x and k_y . The restriction in z gives a quantization of the energy levels. This gives a series of identical parabolas, displaced from the zero level of the 3D case by the quantization energy for each quantum number, $n_z = 1, 2, 3, \dots$. The corresponding DOS is a stair-case function of the energy. In 1D (QWR), the carriers are free to move in one dimension, x , and are restricted in two dimensions, y and z . This leads to a two-dimensional parabola generated by k_x . The restriction in y and z gives a quantization of the energy levels. This gives a series of identical parabolas, displaced from the zero level of the 3D case by the quantization energy for each set of quantum numbers, $(n_y, n_z) = (1, 1), (1, 2), (2, 1), \dots$. The corresponding DOS has a $1/E$ dependence for each of the quantized energy levels. In 0D (QD), the carriers are restricted in all three dimensions. This leads to a series of discrete energy levels, and the corresponding DOS is a series of discrete states, at the energies of the quantized energy levels.

In the one-dimensional (1D) QWR case, the carriers are confined also in a second direction, and the carriers are free to move only along the QWR, 1D movement. As a consequence, the DOS has a $1/E$ dependence for each of the discrete pairs of states in the confined directions (see Fig. 2; 1D). Finally, if the movement of the carriers is limited in all three directions, one talks about a zero-dimensional (0D) QD for which the states are quantized in all directions and the DOS is a series of discrete, sharp states resembling that of an atom (see Fig. 2 0D). This comparison is the origin of the labeling of QDs as "artificial atoms." The tunable "atomic" energy levels of QDs give strong hopes for applications of QDs in novel devices. Besides the importance of intrinsic properties of these types of low-dimensional structures, the extrinsic properties of dopants also appear dramatically different. While macroscopic semiconductor structures, or devices, have a very large number of impurity atoms creating a background distribution of carriers and ionized dopant atoms, the minute volumes of low-dimensional devices will, on average, house only very few extrinsic atoms. Recently, it has become possible to study properties of individual dopant atoms and, clearly, luminescence methods are very suitable

to resolve properties of discrete impurity atom.

The quantization energy of the low-dimensional structures is a classical problem in quantum mechanics. This is the typical particle in a box problem. In its simplest version, the width of the box (the thickness of the layer) is less or equal to the deBroglie wavelength of the particle (electron or hole) and the potential barriers are infinite.^{1,2} This leads to quantized energy levels in the form

$$E_n = \frac{\hbar^2}{2m^*} \left(\frac{n\pi}{L_z} \right)^2, \quad n = 1, 2, 3, \dots, \quad (2)$$

where L_z is the width of the box (layer) and n is the quantization number. $n=1$ describes the ground, or first state, and $n=2$ is the second state, and so on. If the structure is a QW (QWR), the carriers are free to move in two (one) dimensions, adding the free-carrier energy for two (one) dimensions, Eq. (1), to the quantization energy. The states are, therefore, referred to as subbands, 2D in QWs and 1D in QWRs. In a QD, the carriers have no freedom of movement and only discrete states are formed, 0D states. Eq. (2) is simplified, compared to the real situation with finite barriers,¹ but it illustrates two very important features of low-dimensional semiconductor structures. The first point is the energy spacing between the ground state and the first excited state, $\Delta E = E_1 - E_2$. This number is of fundamental physical interest and is very important for device physics. If ΔE is larger than $k_B T$, only the first state is populated, and if it is smaller, higher laying states are populated as well. In the case of, e.g., lasers based on low-dimensional structures, it is essential to have as few states or subbands as possible populated, ideally only one.³

Another important observation can be made from Eq. (2). As L_z is reduced, a small fluctuation in the width, ΔL_z , will lead to a large fluctuation, δ , in the quantization energy, where δ is defined as $\delta = |E_1(L_z) - E_1(L_z + \Delta L_z)|$. This fluctuation can be detected in luminescence as a broadening of the emission peak. This has some implication on device physics as well. Even though the carriers are confined to the first subband, there will be a large spread in the energy with spatial position in the device. This adds a second criterion to the subband separation, where δ must be significantly smaller than the subband separation, ΔE .³ The thickness of a low-dimensional structure can be referred to as an integer number of monolayers (MLs), which is the smallest possible thickness difference. One ML is defined as the thinnest layer that contains a complete set of atoms, e.g., the correct mix of Ga and As atoms in one layer of GaAs. In the (001) direction of a zinc-blende structure, this is half the lattice constant, 2.83 Å. For thin layers, already a difference in thickness of 1 ML leads to a significant δ . This is illustrated in the Fig. 3 for the QW case.

C. An introduction to general luminescence

Luminescence is the process where a system (a semiconductor in this review article) in an excited state emits photons in order to relax to a lower-energy state. Many different methods of exciting the semiconductor are possible. Some examples of different methods are: sound (sonolumines-

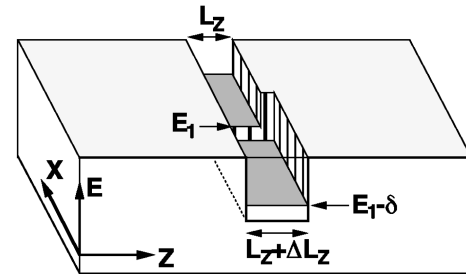


FIG. 3. Illustration of the effect of a thickness variation of a QW. The figure shows the conduction-band edge perpendicular to and parallel to the QW. As the thickness L_z of the QW is increased to $L_z + \Delta L_z$, along the QW, the quantization energy E_1 is decreased by δ to $E_1 - \delta$ [cf. Eq. (2)]. In a thick QW this leads to a broadening of the emission peak. In a thin QW, even a thickness variation on a scale of 1 ML introduces a significant difference in the quantization energy. Under ideal conditions this can give rise to a series of discrete emission peaks from areas separated by a thickness of 1 ML.

cence), light (photoluminescence), heat (thermoluminescence), electron beams (cathodoluminescence), etc. Figure 4 illustrates the process of photoexcitation, as well as different processes that may cause light emission. If the semiconductor is pure, the recombination will be between electrons in the conduction band and holes in the valence band. If the temperature is sufficiently low (e.g., less than 25 K for GaAs) the electron-hole pairs (e-h pairs) will form a bound state, in a hydrogen-atom-like fashion, forming a particle called an exciton. The recombination of an exciton will give rise to sharp-line luminescence, with an energy of the band gap minus the binding energy of the exciton. If the semiconductor contains impurities, several new recombination paths, via the impurity states, open up. Electrons from the conduction band may recombine with neutral acceptors, which after recombination become negatively charged. Neutral donors may recombine with holes in the valence band, becoming positively charged. At higher impurity concentrations electrons bound to donors may recombine directly with holes bound to acceptors, giving rise to donor-acceptor pair luminescence. A third category of impurities is isoelectronic impurities, where the impurity has the same valency as the

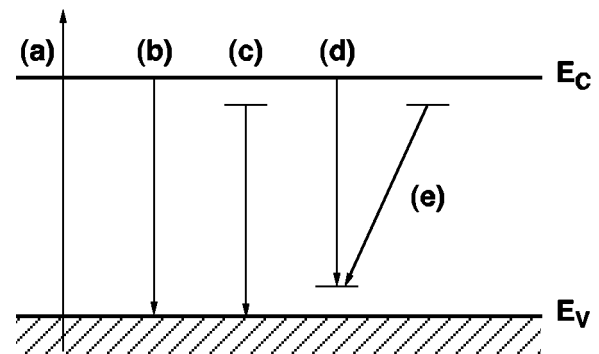


FIG. 4. Illustration of different processes that can give rise to light emission in semiconductors. Initially, an electron and a hole are generated (a), which subsequently relax to their respective band edges. Some of the carriers are also captured by donors and acceptors. Recombination subsequently proceeds through band-to-band recombination (b), from donors to valence band (c), from conduction band to acceptors (d), or from donors directly to acceptors (e).

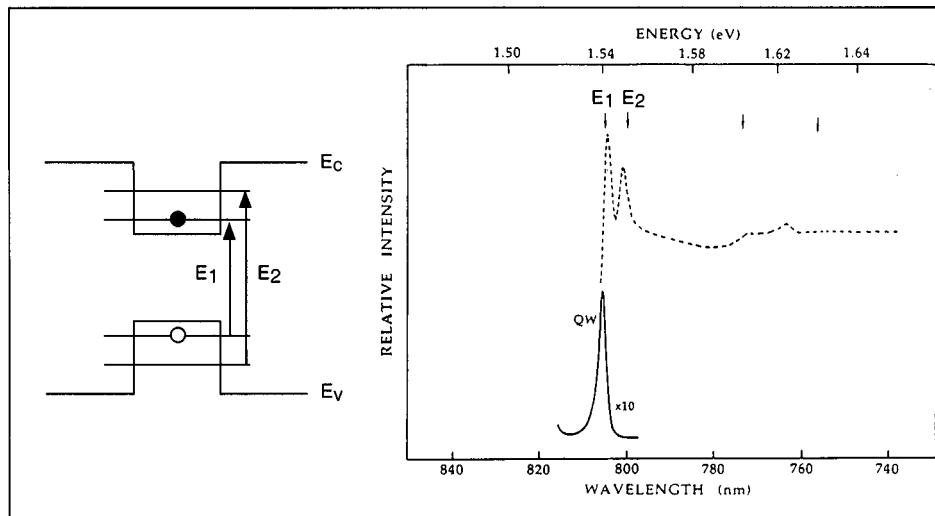


FIG. 5. An electron-hole pair generated in the barrier of a QW sample will with high probability of being captured into the QW and subsequently recombining. The carriers will relax to the ground state of the QW, e.g., electrons will relax to the level denoted E_1 . The inset shows a PL spectrum (solid line) from a GaAs QW in between barriers of GaPAs grown on a GaAs substrate. The emission from the QW is dominated by a single peak, corresponding to the E_1 transition. The PLE spectrum (dashed line), where the emission from the QW is monitored while the excitation energy is varied, however, shows absorption from higher lying states in the QW, e.g., the E_2 transition. The transitions are indicated by arrows in the spectrum.

atom it replaces in the host lattice. The isoelectronic impurities may bind excitons, which can give luminescence, substantially below the energy of free excitons.⁴ Luminescence studies are, in general, a very powerful method for obtaining information about impurities, also at low concentrations. It should also be noted that not all recombination between electrons and holes results in light emission, since there may also be efficient nonradiative recombination paths.

In this review article we will not treat impurities in bulk or bulklike materials, but we note that low-dimensional structures can be viewed as isoelectronic defects having dimensions ranging from 2 to 0. We will also concentrate on luminescence techniques which have a high spatial resolution, i.e., photoluminescence (PL), near-field scanning optical microscopy (NSOM), cathodoluminescence (CL), and scanning tunneling luminescence (STL). The spatially resolved PL is generally referred to as micro-PL (μ -PL), in contrast to conventional PL. The generation mechanisms are similar in CL and PL, where an energetic electron or a photon generates one or more e-h pairs. The carriers will relax to their respective band edges (the conduction band for electrons and the valence band for holes) in the barrier material. At low temperatures the electrons and holes will be captured by the structure (e.g., a QW) and then relax to the lowest-energy state of the structure. Figure 5 illustrates the sequence of events for a QW, along with an experimental PL spectrum of a typical QW. The emission from the QW is much stronger than the emission from the barrier material, although the QW fills less than 10% of the volume in which the carriers are generated. This demonstrates the strong capture of the carriers by the quantum structure. Even for QDs, where the QD-to-barrier volume ratio can be as low as 10^{-4} , the PL spectra are dominated by the emission from the QDs. Examples of this will be shown in the application section below. In STL, the excitation is somewhat different at low bias. In contrast to CL and PL, which in general generate e-h pairs, the tunneling tip can be operated under conditions, where only minority carriers are injected into the sample, which subsequently recombine with majority carriers.

At low temperature and at low excitation power density, the emission from a low-dimensional structure is usually

dominated by the recombination of excitons in the ground state. At higher excitation density, there might also be emission from excited states. At high concentrations, the carriers will screen the Coulomb interaction and subsequently break up the exciton. High temperatures (>30 K) will also break up the exciton through thermal dissociation. The binding energy of excitons varies with, e.g., the width of the QW,¹ and is of the order of 10 meV in direct-band-gap semiconductors, which corresponds to a $k_B T$ of a temperature of ~ 100 K. In QDs, which we argue are fundamentally different from the higher-dimensional analogues, the situation is different. In order to have a binding energy of an exciton, there must be a possibility to ionize the exciton, i.e., to get a free electron and a free hole. In QDs, this is impossible, unless one of the carriers is excited to the continuum, i.e., the barrier where the wave function is not localized in the QD. In other words, there is no possibility to turn off the Coulomb interaction between carriers localized in a QD.⁵

It is often necessary to determine the energy position, as well as character of the higher lying states, in order to obtain a more complete understanding of the electronic structure of a quantum system, as illustrated in Fig. 5. These excited states are not usually observed in emission. One method to observe these higher lying states is to use absorption spectroscopy. An absorption-based method that is closely related to luminescence is photoluminescence excitation spectroscopy (PLE). In PLE the wavelength (energy) of the excitation source is varied, while the emission from the relevant structure is monitored, as illustrated in Fig. 5. An advantage of PLE over absorption is that the emission from the relevant structure is monitored and, unless there is a transfer of carriers from other features, the excited states linked to the monitored structure are measured. Since there is no emission if there is no absorption, PLE is also a very sensitive technique. In conventional absorption measurements the interesting absorption is often a small reduction in a large transmission, which means that the signal of interest is a small signal on a large background signal. This reduces the sensitivity, unless a detector with a very high dynamic range is used. On the other hand, the advantage of absorption over PLE is that it is quantitative, and can measure transition

probabilities or oscillator strengths. Although we concentrate on direct luminescence studies in this review article, we expect to see a lot more activity in the area of spatially resolved PLE (μ -PLE). Especially since this technique has a potential for probing the excited states in low-dimensional structures.

The polarization dependence of the emission/absorption in an ideal QW is derived in Appendix B. In practice, the situation is more complicated. The confinement, as well as the strain, will cause the states to be linear combinations of the states given in Eqs. (B2), (B3), and (B4), which will change the polarization rules. This mixing is even more significant for very narrow QWs and for strained systems. In addition, if we are dealing with QDs or QWRs with inhomogeneous strain, it is necessary to also include the strain field in a more general way, where the gradient of the strain field is accounted for.⁶ For wide and weakly strained QWs, it is possible to determine whether the light-hole state or the heavy-hole state is lowest in energy by using polarization-dependent PLE. For narrower and more strained QWs, the polarization dependence has to be calculated, e.g., using the model developed in Refs. 7 and 8. If the strain is sufficiently high, it might also be necessary to add the strain-dependent spin-orbit coupling⁹ as well as strain terms, which couple the valence and conduction bands.¹⁰ More sophisticated models have recently become available,^{11,12} however, still based on the $\mathbf{k} \cdot \mathbf{p}$ model (See Appendix A).

In a QWR, the mixing of the valence-band states is more significant and the general conclusion is that the light emitted is polarized along the wire.^{13,14} It must, however, be remembered that the coupling of the light into the sample depends on the wire orientation. This is especially important if the QWR is fabricated by etching and has the same dimensions as the wavelength of the exciting light. It is easiest to avoid artificially induced polarization by measuring in a true back-scattering mode, i.e., where both the exciting light is introduced, and the emitted light is collected normal to the surface. Otherwise, the emitted light will be polarized due to the different reflectivities of the surface-to-air interface for different polarizations.¹⁵

The luminescence from a (001)-oriented QW is not polarized when the emission is detected normal to the (001) surface. It is expected that for asymmetric QDs, the luminescence will be polarized. An example of asymmetry is an elongated pyramid-shaped QD, with the edges of the hexagonal base plane oriented along the [110], [100], and [010] directions.¹⁶ Polarization-dependent PL, CL, or STL from individual QDs, should thus be an interesting method to (indirectly) measure the main features of the shape of QDs. It is sometimes possible to get circularly polarized emission from QWs, provided that the exciting light is circularly polarized.¹ This is due to electron-spin effects¹⁷ and allows spin effects to be probed on a local scale.

D. A brief introduction to fabrication techniques

In this section we will briefly introduce the various techniques used to fabricate the low-dimensional structures discussed in this review. Any detailed description of the techniques is far beyond the scope of this review, instead, we

will limit ourselves to giving the reader an overview of the number of acronyms used in the rest of the review.

Most of the low-dimensional structures described in this review have at some stage in the fabrication process been formed by epitaxial growth. For historical, rather than practical, reasons these growth techniques are divided into three categories. In molecular beam epitaxy (MBE) the sources are solids or liquids and the substrate is subjected to beams of small molecules or atoms; in liquid phase epitaxy (LPE) the layers grow from a melt; and in vapor phase epitaxy (VPE) the sources are gaseous and the growth takes place from the gas phase. The techniques all have their pros and cons, in terms of growth rate, interface quality, material purity, economy, safety, etc. One of the techniques, LPE, has a significantly higher growth rate, making the necessary thickness control on a ML scale difficult. This means that very few low-dimensional structures are grown by this technique.

There are numerous variations of VPE and MBE, where the borders between the two are more a matter of design of the epitaxy equipment rather than significant differences in the actual growth process. The most commonly used VPE technique uses metalorganic gases, metalorganic vapor phase epitaxy (MOVPE), also known as metalorganic (organometallic) chemical vapor deposition (MOCVD or OMCVD). This technique is further divided into subcategories according to pressure in the reactor during growth. Growth at or around atmospheric pressure (AP) is denoted AP-MOVPE and a pressure below 150 mbar is denoted low-pressure (LP-) MOVPE. In the ultimate low-pressure limit, the situation is close to MBE, and the technique is known as chemical beam epitaxy (CBE). There are also some variations on the MBE theme. If gas sources are used, the technique is called gas source (GS) MBE and specifically if metalorganic sources are used, it is known as MOMBE. In a special growth mode, the group III (II) and group V (VI) sources are switched on and off in sequence. This way, half of a ML is grown during the opening of each source, resulting in the growth of atomic layers and the growth mode is known as atomic layer epitaxy (ALE) in MBE and flow modulation in MOVPE.

The fabrication of many types of low-dimensional structures involves etching in some phase of the processing. This can be to prepare a substrate for subsequent (additional) growth or to reduce the size and/or dimensionality of existing structures. The first step in this process is to transfer a pattern to the surface of the sample. This is mainly done by lithography in resist, using photolithography, or x-ray lithography in combination with a mask; using light holography; or writing the pattern with an electron beam. The etching is done by chemical (wet) etching or by plasma (dry) etching. The wet etching, generally has a higher etch rate in certain crystallographic directions. In this case, the resulting structures are generally determined by well-defined planes, e.g., {111}A planes. In the zinc-blende structure, e.g., most III-V semiconductors, the "A" indicates a group-III-terminated plane, as opposed to the group-V-terminated "B" plane. The plasma etching, on the other hand, produces a one-to-one transfer of the pattern. Again, there are many variations on the plasma etching based on the chemistry involved. The

gases involved are, typically, inert gases like argon and hydrogen and reactive gases like Cl_2 , freon (CFC) or I_2 . In the case of reactive ions, the technique is known as reactive ion etching (RIE).

Another type of fabrication of QWRs and QDs from a QW is ion implantation with subsequent annealing. This can create interdiffusion between the QW and its barrier, which generally leads to an increase of the band-gap material in the implanted QW region. This can be used to create additional lateral barriers within the QW. The lateral variations in the implantation can be controlled in two ways. The simplest is a mask to block out the global implantation and the more complicated version is to write the pattern directly with a focused ion beam (FIB).

II. LOCAL LUMINESCENCE PROBE TECHNIQUES

A. Generation and injection mechanism and excitation volume

In this review article we describe different experimental tools available to the experimentalist to investigate, mainly optical, properties of single quantum objects. All these tools are based on detection of light and have the possibility of a spatial resolution better than $1\ \mu\text{m}$. The methods that will be described are $\mu\text{-PL}$; NSOM; CL; and STL.

In the PL, PLE, and NSOM methods, the excitation takes place with the absorption of photons, creating e-h pairs. This absorption takes place from the surface and into the sample in the area illuminated by the incident light. When these e-h pairs recombine radiatively, the generated photons can be detected, see Fig. 6(c). The depth of the primary excitation can be varied by changing the wavelength of the incident light. The generated carriers will have the possibility to diffuse before they recombine. The excitation power can also be varied for observation of band-filling (or state-filling, in the case of QDs) effects, and the excitation can be pulsed to obtain information about the temporal behavior. In the PLE mode, a certain wavelength can be detected while scanning the wavelength of the excitation. The simplest method to obtain the local excitation is to use a focused laser spot, where the size of the spot is limited by diffraction to about the wavelength of the excitation light, illustrated by excitation through a microscope objective in Fig. 6(b). A simple way to increase the spatial resolution is to use a mask on the sample with submicron-sized holes, where the resolution is given by the size of the hole. The mask approach has a disadvantage in the lack of control of the position of the holes, with respect to features of interest in the sample. However, there will be no effect of drift and no ambiguity as to the position of the excitation. It is also possible to return to the same spot for a second measurement. The ultimate limit of spatial resolution of photoexcitation is to work in the near field using NSOM. There are several different modes of NSOM based on the near-field concepts, see, e.g., Refs. 18 and 19. In this review article, we will concentrate on the aperture-based NSOM, referred to as a-NSOM. In the a-NSOM the resolution is no longer limited by diffraction, but set by the size of the aperture used. The principle is to excite and/or detect through an aperture,

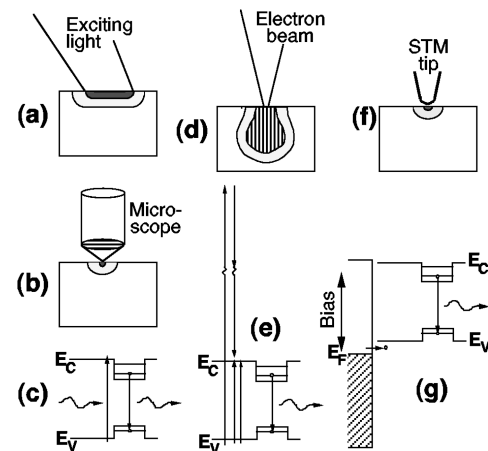


FIG. 6. (a) Shaded area indicates the effective excitation volume in PL experiments, including finite depth excitation (dark grey) and diffusion of generated carriers before they recombine (lighter grey). (b) like in (a) but with a strongly focused excitation spot (or light collection spot) using an optical microscope. This technique, referred to as micro- $\mu\text{-PL}$ may allow spot size control better than $1\ \mu\text{m}$ in diameter. (c) Schematic band diagram of PL and $\mu\text{-PL}$ experiments with conduction- and valence-band edges, describing excitation processes in which photons with an energy larger than the band gap are absorbed. Also indicated are the confined electron and hole states from which the carriers recombine after they have been captured into the quantized states. (d) Hatched area indicates the volume in which e-h pairs are generated from the slowing down of incident energetic electrons. The grey area (volume) shows the added diffusion of the carriers. (e) Band diagram to illustrate the process of generating e-h pairs from step-wise reduction in the energy of incident electrons. As in (c), recombining e-h states in a QW are shown. (f) Indication of the excitation volume in STL, with a very small primary excitation volume resulting from the injected tunneling carriers. Also in this case, the grey area indicates the added volume resulting from the diffusion of the carriers. (g) Band diagram to explain mechanisms of luminescence generation by the local injection of minority carriers (in this case, holes) from the tip of the STM, with properly adjusted tip-sample potential to allow tunneling. The case shown here is for electrons being extracted from the valence band of the sample, which is equivalent to holes being injected into the semiconductor, leading to detectable radiative e-h recombination in the QW.

which can be scanned over the samples, where the aperture is smaller than the wavelength of the light passing through it. For practical reasons, the typical aperture size used in experiments is $\sim\lambda/3$, but apertures as small as $\lambda/40$ have been demonstrated.²⁰

Spatially resolved CL is usually performed using the focused high-energy electron beam from a scanning electron microscope (SEM) or a scanning transmission electron microscope (STEM). As a consequence of the relaxation of the primary electrons, Fig. 6(e), secondary electrons are formed, resulting in spreading of the excitation in the sample, Fig. 6(d). This leads to the formation of an excitation volume, or generation volume, which is substantially larger (depending mainly on the incident electron energy) than the primary beam diameter. The possibility to change the incident electron energy can be used to vary the penetration depth from a few tenths of a μm to several μm . The diffusion process of the created e-h pairs is also present, preventing extremely high resolution, but in some cases with specifically designed samples the resolution may be better than $0.1\ \mu\text{m}$. Also, in this case the excitation can be pulsed and the excitation density varied.

TABLE I. Absorption coefficients ($\times 10^4 \text{ cm}^{-1}$) for various semiconductors at 300 K, after Refs. 21 and 22.

Material	Energy (eV)										
	1.0	1.2	1.4	1.6	1.8	2.0	2.2	2.4	2.6	2.8	3.0
GaAs	0	0	0	1.48	2.75	4.3	6.1	9.0	14.2	28.1	59.2
InP	0	0	0	3.5	4.9	6.4	8.5	11.1	15.2	22.3	37.9
Si	0	0.006	0.04	0.12	0.24	0.45	0.72	1.47	2.4	4.6	8.2
Ge	0.8	1.7	2.4	5.6	9.1	18.9	45.7	59.7	60.8	61.9	65.2
GaP	0	0	0	0	0	0	0.004	0.09	0.3	3.3	6.8

In order to overcome the limitations with a rather large generation volume in the CL method, the low-energy electron (or hole) source of the scanning tunneling microscope (STM) can be used. The STM can be operated under conditions where the incident carriers have such low energy that they do not create secondary carriers and, thereby, no e-h pairs. This gives a very high lateral precision in the actual point of excitation, Fig. 6(f). The only limiting process is the diffusion of the injected minority (or majority) carriers before they can recombine. One advantage with STL is the possibility to change the energy of the injected carriers by changing the applied voltage between the tip and the sample, thus allowing carriers to be injected resonantly, Fig. 6(g). It is also possible to vary the current in order to increase the excitation density for studying heavy pumping effects. Furthermore, by increasing the applied tip-sample voltage substantially, it is possible to enter into the field-emission mode where the excitation mechanism is similar to CL above, although the energies are much lower (in the range of 5–100 eV). In this mode, e-h pairs are created, which will diffuse before recombination, which can lead to a reduced spatial resolution.

1. Conventional and microphotoluminescence and photoluminescence excitation spectroscopies

In PL, the incident light is absorbed in the sample, generating e-h pairs. It is of considerable importance to know where the carriers are generated. If the structure of interest is buried deep in the sample, it is necessary to use an exciting wavelength which is able to penetrate to a sufficient depth. The intensity of the light decreases in a semiconductor as $I(x) = I_0 e^{-\alpha x}$, where x is the distance from the surface, α is the absorption coefficient, and I_0 is the intensity transmitted through the surface (the incident light intensity minus the reflection in the surface). Typical values of α are given in Table I. An absorption coefficient of 10^4 cm^{-1} corresponds to a penetration depth of 1 000 nm (i.e., 1 μm), where the intensity is $I_0/e = 0.37$, i.e., 37% of the original intensity. At a depth of 3.9 μm , the remaining intensity is 2%, corresponding to an energy deposition of 98%, cf. Eq. (5) below.

For a direct band-gap semiconductor, a typical absorption coefficient is 10^4 – 10^5 cm^{-1} for a photon energy above the band gap. For an indirect band-gap semiconductor, this absorption coefficient is 1–100 cm^{-1} . In a direct band-gap III–V QW, the absorption is $\sim 1\%$ per state, independent of the thickness of the QW (see Appendix A). It has to be understood that the width of a QW is usually much less than the wavelength of light, which makes it quite meaningless to define an absorption coefficient having units of cm^{-1} .

The penetration depth of the light is often considerably shorter than the typical diffusion length of carriers and excitons. The lateral resolution is not only limited by diffusion of the carriers but may also be limited by diffraction of the exciting light. This is the well-known Rayleigh criterion for the minimum spot size d of a focused laser beam:

$$d = 1.22\lambda / (2n \sin \theta), \quad (3)$$

where λ is the wavelength of light, n is the refractive index of the medium directly in front of the sample, θ is the half angle of the solid angle passing through to the focusing lens, and $n \sin \theta$ is referred to as the numerical aperture. In practice, the lower limit of d is usually considered to be $\lambda/2$. Consequently, the depth resolution, as well as the lateral resolution, is in most cases limited by the diffusion length, generally, in the order of 1 μm . One practical factor affecting the resolution limit is the spherical aberration in the cryostat windows, which has to be corrected by optical means. Lenses for this purpose are commercially available. If the exciting light has an energy which is lower than the band gap of the barrier material, which is usually the case in PLE, the penetration depth can be very large (several millimeters or more) and the depth resolution in this case is obtained through the selective absorption by the region (e.g. a QW) of interest. If a very surface selective excitation (10–50 nm or $\alpha \sim 2$ – $5 \times 10^5 \text{ cm}^{-1}$) is needed, it may be necessary to use a short-wavelength laser, e.g., the UV lines from an Ar^{++} laser for a GaP sample (see Table I).

2. Near-field scanning optical microscopy

The diffraction limitation of the spatial resolution²³ [cf. Eq. (3)] of PL and μPL can be overcome by working very close to the sample surface, where the diffraction limit is not yet established. This region is generally known as the near field, as opposed to the far field where the diffraction limit has been established. One way to exploit the near-field region is to use a-NSOM. The basis for this technique is to utilize a small subwavelength-sized aperture placed very close to the sample, as will be described in the Sec. II C on instrumentation. If both the aperture diameter and the aperture-to-sample distance are made smaller than the wavelength of light, the normal optical diffraction limits are no longer valid.^{24,25} In this case, the resolution is primarily given by the size of the aperture. Using a subwavelength-sized aperture, it is possible to obtain subwavelength spatial resolution.^{20,26,27} To date, the best resolution obtained, using optical excitation, is $\lambda/40$.²⁰ Although the principle has been known for a long time,²⁸ technological requirements have

hindered most of the practical solutions. In the following, a brief discussion of the near-field concept will be given, specifically addressing the possibilities to study low-dimensional structures.

Many different modes of near-field microscopy have been implemented experimentally, although detailed theoretical treatments have been limited to a few publications.^{19,29–33} Near-field microscopy concerns the interaction between electromagnetic radiation that varies with the gap between the probe aperture and the sample, and the optical properties of the sample. In order to understand the near-field phenomena one can separate the problem into three parts: the source, the zone of interaction, and the detector. The theory is quite complex, since both near-field and far-field phenomena have to be considered simultaneously. The observed contrast depends on both the real and imaginary parts of the refractive index of the sample, which further complicates the interpretation of the obtained images. In order to treat the problem of resolution and contrast quantitatively, one has to solve Maxwell's equations in the presence of complicated boundary conditions. This type of analysis, however, goes beyond the scope of this article and we refer the interested reader to other more specialized articles for a detailed theoretical description, such as Refs. 18, 19, and 29–32. Recently, a strong coupling between topography and near-field-related effects has been demonstrated.³⁴ This is caused by the feedback mechanism that, as well as allowing the tip to follow a sample structure, also causes unintentional and artificial features in the images. This is caused by the strong aperture-to-sample distance dependence of the intensity, where a very small change in this distance gives a large variation in the intensity. This makes the interpretation of the NSOM images and data of nonflat samples quite difficult. In fact, as was pointed out in Ref. 34, quite a few published articles have contained this type of misinterpretation of NSOM images. This kind of interpretation problem will not be discussed further in this review article.

There are two distinctly different modes of operating an a-NSOM: The collection mode, where the emission from the sample is collected by the aperture, and the excitation mode, where the excitation is transmitted through the aperture into the sample. In the excitation mode, the transmission of radiation through a subwavelength-sized aperture is only possible via the formation of wavelets, having extensions in the order of the aperture diameter. These wavelets can wind themselves around the aperture, although free-space propagation is forbidden. Consequently, radiative energy can be transmitted to the other side of the aperture and interact with the sample. The result of the interaction can be detected by some external mean, e.g., as luminescence as a result of recombination of the generated carriers or as a photoinduced current. This latter technique has been called near-field optical beam induced current (NOBIC) in terminological agreement with electron/optical beam induced current (EBIC/OBIC).³⁵

In the collection mode, the detected signal can be the evanescent waves from light that is undergoing total internal reflection at the sample-to-vacuum interface. These evanescent waves can be collected in the near field by the aperture

and then transmitted to an external detector. If the sample surface is modified with a thin, inhomogeneous layer, the intensity of the evanescent waves is spatially modulated. By recording the near field, the modification can be recorded. After the invention of the STM, the mode of tapping the evanescent waves from the total internal reflected geometry was one of the first applications to demonstrate the ability to collect the near-field emission. Due to its close relation with STM it was (and sometimes still is) denoted photon scanning tunneling microscopy (PSTM).^{36,37} The collection mode can also be used as a subwavelength-sized detector, scanning the sample surface. Although this is, strictly speaking, not detecting the near field from the individual emitters in the sample, the same considerations for aperture size and aperture-to-emitter distance have to be employed to obtain the optimum resolution.

The evanescent waves decay rapidly in free space, having a dependence of the transmitted energy as $\approx (a/d)^{3.7}$,²⁹ where a is the aperture diameter, and d is the distance from aperture apex. In practice, this rapid decay of the energy means that the near field can only excite on a length scale of a few aperture diameters away from the aperture. Figure 7 shows the lateral distribution of the electric-field intensity as a function of distance from the center of a circular aperture with a diameter of $\lambda/5$.³⁸ This demonstrates the rapid drop-off of the intensity with distance from the aperture, where the intensity is almost zero at a distance of $\lambda/5$.

One of the few experimental investigations of the resolution limits of NSOM in relation to the possibility to probe subsurface, low-dimensional structures, was a study³⁵ of the influence of the aperture size, sample-to-aperture distance, and wavelength of the excitation source. The sample consisted of three pairs of QWs, where the distance between the QWs in each pair was varied. The structure was grown in the intrinsic region of a $p-i-n$ structure. The NOBIC signal was recorded in cross section, as a function of vertical (sample-to-aperture distance), as well as lateral (on the sample) position. The wavelength of the excitation was chosen so that the absorption only took place in the QWs and not in the barriers. Since the distance between the QWs was varied it was possible to observe whether the excitation was due to evanescent waves or propagating waves, i.e., whether true near-field or far-field excitation occurred, as a function of aperture diameter and excitation wavelength. It was demonstrated that to have a true near-field excitation, the aperture diameter has to be smaller than half of the wavelength of the exciting light in the semiconductor material, i.e., $a < (\lambda/2n)$ (n =refractive index). In this case the wavelength of the excitation does not affect the resolution, and the resolution is given only by the aperture diameter (and the distance to the surface). The authors concluded that if the aperture diameter was larger than $\lambda/2n$, a coupling between evanescent and propagating waves was allowed in the sample. They found that the resolution deteriorated with an increase in the excitation wavelength. The higher spatial resolution was attributed to a stronger absorption (shorter penetration depth) for a shorter wavelength, as compared to a longer wavelength.³⁵ In another, more fundamental study of the relation between the aperture size and the tip-sample distance, the tapping of evanescent waves from the total internal reflected geometry was one of the first applications to demonstrate the ability to collect the near-field emission. Due to its close relation with STM it was (and sometimes still is) denoted photon scanning tunneling microscopy (PSTM).^{36,37} The collection mode can also be used as a subwavelength-sized detector, scanning the sample surface. Although this is, strictly speaking, not detecting the near field from the individual emitters in the sample, the same considerations for aperture size and aperture-to-emitter distance have to be employed to obtain the optimum resolution.

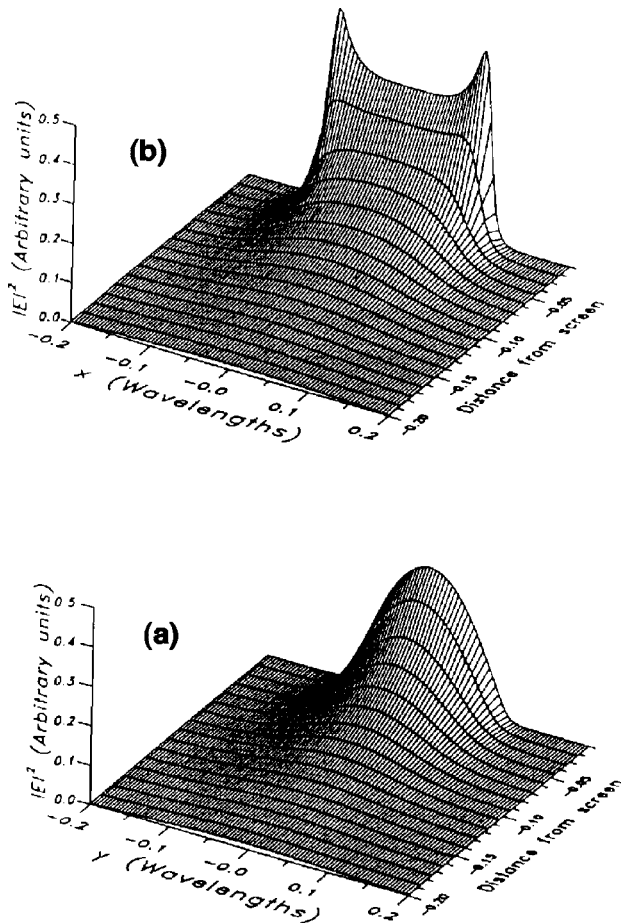


FIG. 7. Calculation of the intensity of the electrical field $|E|^2$ after a circular aperture, with a diameter of $\lambda/5$, where λ is the wavelength of the incident light. This is a typical aperture size that has been used in many NSOM studies of semiconductor structures. The intensity is plotted as a function of the lateral (parallel to the aperture plane) and vertical (perpendicular to the aperture plane) distances from the center of the aperture. The lateral direction is parallel to the electric-field vector in (a), and parallel to the magnetic-field vector in (b). The two plots illustrate the rapid decrease of intensity with distance to the aperture center. At a distance of the aperture diameter, very little of the intensity remains (Ref. 38).

nescent waves in the total internal reflected geometry was used.³⁹ The conclusion of this study was that an upper limit for the aperture sizes was 0.4λ , to obtain high resolution.

3. Cathodoluminescence

In CL, energetic electrons interact with the sample to generate the excitation. One of the most important factors for the generation mechanism in CL is the effective ionization energy (E_i), which is the average energy required to generate one e-h pair in the semiconductor. This is related to the band gap by $E_i = 2.8 E_g + M$, where M is a constant that only depends on the material and has a magnitude of 0–1 eV.⁴⁰ For a given acceleration voltage E_{acc} , the number of e-h pairs generated per primary electron (PE) G is given by⁴¹

$$G = (1 - \gamma) E_{acc} / E_i, \quad (4)$$

where γ is the fraction of the energy transmitted through and scattered back out of the semiconductor. In most cases, the semiconductor is thicker than the penetration of the PEs and

γ is, therefore, the fraction of the energy carried by the PEs that are scattered back out of the semiconductor.⁴²

The interaction between the PEs and the semiconductor can be either elastic or inelastic. The PE can be elastically scattered by the nuclei, or more correctly the screened nuclei of the atoms in the semiconductor. The main result is a change of the direction of the PE of up to 180° , whereas the change of the energy of the PE is negligible.⁴² PEs that are scattered back from the semiconductor are denoted backscattered primary electrons (BEs). In the SEM mode, they are used to form images that exhibit atomic number (Z) contrast caused by the monotone increase of the backscattering coefficient with atomic number. The PE can also interact inelastically with the electrons in the semiconductor.⁴² This interaction can take place with electrons of the inner electron shells of the atoms, as well as with the electrons in the conduction band and the valence band. This interaction generates free electrons, secondary electrons (SEs), within the semiconductor. Depending on the origin of the SE, the cross section, as well as the final energy of the SE differs.⁴² The interaction with the electrons of the conduction band and the valence band, which are loosely bound, has the largest cross section. This interaction leads to SEs with low energy (< 50 eV).⁴² These SEs travel a short distance before they are recaptured, in the order of 5 nm, and are referred to as slow SEs. Fast SEs can also be generated as a result of interaction with the more tightly bound electrons of the atomic levels in the semiconductor.⁴² These SEs are not as numerous as the slow SEs, but they have a larger kinetic energy and can travel much longer distances before they lose their kinetic energy and are recaptured. The energy of a fast SE can reach almost that of the PE it interacted with. Furthermore, the fast SEs are scattered at almost right angles from the trajectory of the PE. This, together with the larger traveling distance, contributes to the spreading of the excitation within the semiconductor, which deteriorates the resolution of the measurements. The PE interaction can also take place with the tightly bound electrons of the inner shells, e.g., the K and L shells in GaAs. Here, a large portion of the energy of the PE is transferred to the atom. This process also leads to the generation of SEs, but a more important consequence is the emission of characteristic x rays.

The SEs that are generated near the surface of the semiconductor have enough energy to overcome the work function of the semiconductor and thereby the possibility to escape from the surface, which results in SE emission. In the SEM mode, the detection of SEs is the most common detection mode. In this mode the topography of the surface is recorded, but additional information can be obtained regarding differences in compositions, doping levels, applied and induced voltages, and magnetic fields.^{42,43} In the case where the SE remains in the semiconductor it can combine with a hole to form an e-h pair, where the number of e-h pairs per PE is given by Eq. (4). One of the ways this e-h pair can recombine is via emission of a photon: CL. Other recombination ways are nonradiative and include emission of cascades of phonons and various Auger processes, where the energy of the interband transition is transferred to an intra-band transition.

The inelastic scattering includes two more processes, both relying on Coulomb interaction.⁴² When the PE passes through the Coulomb field of the atoms in the semiconductor, the PE velocity can be reduced and its direction can be changed. In this process the excess energy is emitted in the form of a photon. This phenomenon is known as bremsstrahlung (braking radiation). The energy of the emitted photon depends on the energy of the PE and the deviation from its original trajectory. The possible energy of the emitted photon ranges from virtually zero for very small deviations to the total energy of the PE. This radiation is mainly observed in the x-ray spectrum of a semiconductor, where it appears as a continuous background. It can also be observed in the CL spectrum of a semiconductor. This is particularly true in semiconductors with low luminescence yield, e.g., Si and Ge under high acceleration voltage and high probe current conditions. The second kind of Coulomb interaction is between the Coulomb field of the PE and the free-electron gas of the conduction band which induces plasmons, i.e., oscillations in the free-electron gas.⁴² These are almost exclusively emitted near to the direction of the trajectory of the PE, which afterwards is changed slightly. The cross section for this type of scattering is directly proportional to the density of free electrons and is, therefore, mainly significant for metals and not as important for semiconductors with their much lower free-electron density. For copper this is the largest of the inelastic cross sections.⁴² For details of the cross sections for the different scattering processes and for a discussion of the average energy loss per unit length the primary electron travels, we recommend, e.g., *Advanced Scanning Electron Microscopy and X-Ray Microanalysis*, by Newbury *et al.*⁴²

From the formulas of the energy loss, an average traveling distance can be computed. This is the average distance a PE will travel before it has lost all, or nearly all, of its energy. This distance is known as the Bethe range.⁴² However, this is in most cases not of any direct interest. The more interesting parameter is how far into the semiconductor the PEs can penetrate, usually referred to as the penetration depth. There are several formulas for calculating this. One of the first attempts was made by Grün⁴⁴ where the maximum penetration depth or electron range R_G is described as a function of density of the material and the acceleration voltage. In a later study⁴⁵ of the subject, the original expression by Grün was modified to

$$R_G = (0.0398/\rho)(E_{acc})^{1.75} \text{ (}\mu\text{m)}, \quad (5)$$

with ρ as the density in g/cm^3 . This expression was derived for a range of atomic numbers between 10 and 15 and E_{acc} between 5 and 25 keV. It was also concluded that 98% of the energy was dissipated within R_G .⁴⁵ A more general expression that accounts for most of the energy was derived by Kanaya and Okayama.⁴⁶ This expression gives a larger R_G as compared with Eq. (5), especially at lower E_{acc} , where the difference can be up to 2.5. However, Eq. (5) is often quoted as the more accurate for CL and EBIC work. Figure 8 shows the penetration depth as a function of E_{acc} for some semiconductors, calculated by Eq. (5).

Another factor that is more important for the ultimate resolution of the CL technique is the spreading of the PE

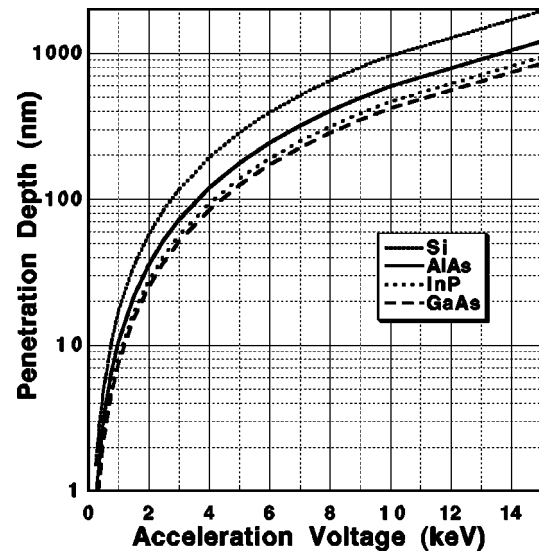


FIG. 8. Plot of the theoretical penetration depth, or electron range, as a function of acceleration voltage for Si and for some of the most usual III-V semiconductors AlAs, InP, and GaAs. The penetration range is calculated by using Eq. (5), suggested by Everhart and Hoff (Ref. 45). This equation is often considered to be the most accurate for CL work. About 98% of the energy is deposited within this range.

parallel to the surface of the semiconductor; the lateral spreading. This together with the penetration depth determines the volume of the semiconductor that is excited by the PEs. This volume is generally referred to as the generation volume.⁴¹ A good first approximation of the lateral extension of the excitation volume is a circle with the same diameter as the penetration depth. In fact, the first approximation of the excitation volume is a cylinder with a diameter of, as well as a length of R_G . The diameter of this cylinder is a first estimation of the resolution that can be obtained in CL.

In reality, the excitation density varies dramatically within this cylinder. There have been several attempts to make analytical expressions for the generation of e-h pairs in semiconductors. One of the more detailed expressions has been given in a series of articles by Donolato,^{47,48} where the normalized generation density $[g(z, r, R_G)]$ in a plane parallel to the electron beam is given by

$$g(z, r, R_G) \sim (\Lambda(z/R_G) \exp\{-[r^2/2\sigma^2(z, R_G)]\}) / [R_G \sigma^2(z, R_G)] \quad (6)$$

$$\Lambda(z/R_G) \sim \exp(-7.5\{[z/R_G - 0.3]^2\}) \quad (7)$$

$$\sigma^2(z, R_G) = 0.36d^2 + 0.11z^3/R_G, \quad (8)$$

where z is the distance from the surface in the direction of the electron beam, r is the radial distance from the original trajectory of the PE, $\Lambda(z/R_G)$ is the depth-dose function describing depth variation of the generation rate of e-h pairs, R_G is the penetration depth as given by Eq. (5), and $\sigma^2(z, R_G)$ describes the spreading of an electron beam of a Gaussian diameter d . Figure 9 shows an example of the normalized generation of e-h pairs in GaAs at 5 keV and a beam diameter of 50 nm. Although the shape of the genera-

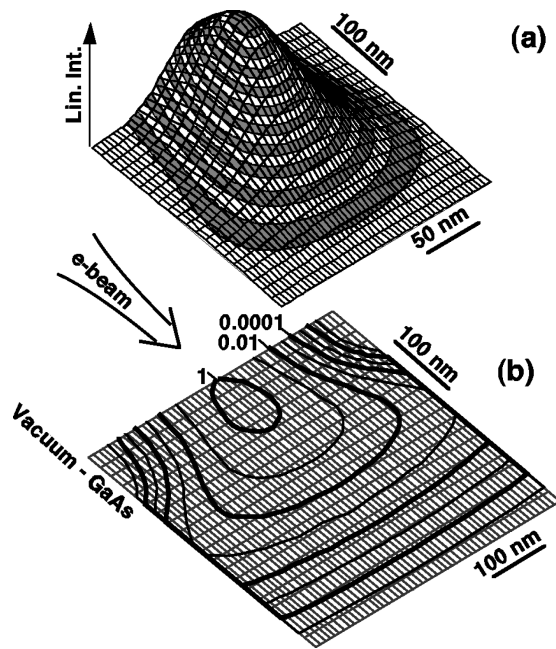


FIG. 9. Theoretical shape of the generation volume in GaAs using a 5 keV probe with a 50 nm diam. This plot is generated, using Eqs. (6)–(8), as described by Donolato and Venturi (Refs. 47 and 48). The plot demonstrates the spreading of the excitation in the sample. It also shows that the highest generation rate is actually below the surface. (a) shows a 3D plot, with linear excitation density, and (b) shows a contour plot, where each line trace represents a difference of 10 in excitation density. The sample surface is the top left in both figures.

tion volume is a very basic feature of electron microscopy and especially CL, work is still going on to refine the analytical models, see, e.g., Refs. 49 and 50.

Another approach to obtain the depth-dose function, as well as the shape of the excitation volume is to use the method of Monte Carlo (MC) calculations of the trajectories of the PEs.^{51–53} In this type of calculation, a continuous energy loss function for inelastic scattering is used together with the angle probability for elastic scattering. A PE of an energy E_{PE} has a mean-free path between elastic scattering events λ and loses approximately an energy $\Delta E = \lambda dE/dS$, due to inelastic scattering along this distance. The Monte Carlo approach is to calculate λ , move the electron the distance λ in a straight line, calculate ΔE for this distance, reduce E_{PE} , and obtain a scattering angle, using a random number. This is repeated in three dimensions until E_{PE} is below the limit, where the electron can no longer cause ionization.⁵⁴ From these calculations the depth-dose functions can be obtained, as well as the shape of the excitation volume. As a bonus, the backscatter coefficient can be obtained as the fraction of the PEs that are backscattered, as well as the fraction of the energy these PEs carry with them out of the sample.

Several methods have been used to verify the depth-dose function, as well as the shape of the excitation volume.^{45,55} One method is based on CL, where a focused electron beam is introduced into a gas of low pressure, which then emits light. The local emission intensity depends on the local excitation density.⁵⁶ The depth-dose function and the shape of the excitation volume can be obtained by recording the spa-

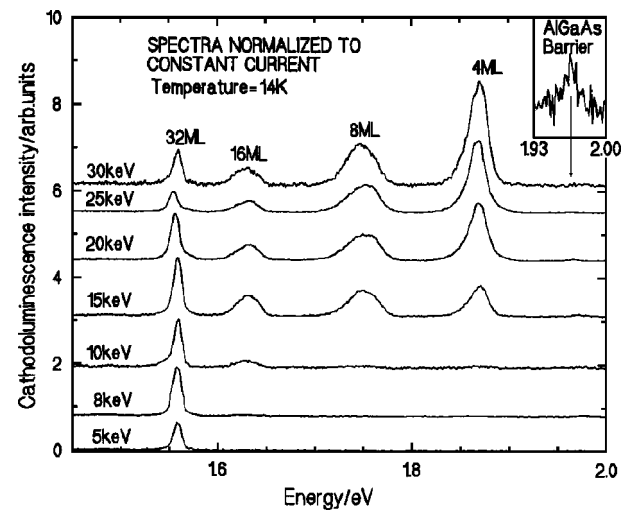


FIG. 10. Cathodoluminescence spectra from a GaAs/AlGaAs MQW using different acceleration voltages. The spectra were recorded with excitation in top view. The QWs are of different thicknesses, and therefore, different emission energies, so they can be separated spectrally. At 5 keV, only emission from the top QW can be observed. With increasing acceleration voltage, the emission from the deeper laying QWs starts to appear. At 30 keV, the deepest laying QW dominates the spectrum. (Courtesy of Trager-Cowan, Ref. 59.)

tial variations of the light intensity. Another method is based on the exposure of PMMA, normally used as an electron-beam resist for making patterns. After developing, all resist that has received a charge dose above a critical dose (i.e., current density times the exposure time) can be removed.⁵⁷ By looking at the profile of lines obtained with different doses (i.e., different exposure times rather than different currents), equidose lines of the excitation volume can be deduced and the shape of the excitation volume can be reconstructed. Yet another different method is to use the EBIC signal of a device where care has been taken to make sure that only direct generation of e–h pairs will be recorded rather than diffusion.^{55,58} This can be done by surrounding the depletion region with barriers of high band-gap material that block diffusion without reducing the collected current significantly.

There have been a couple of reports of measurements of the penetration depth and the lateral extension of the excitation volume using direct CL measurements of semiconductors. The penetration depth was measured for several acceleration voltages by the CL from a stack of QWs, where each QW has a different thickness, i.e., a different emission energy.⁵⁹ The main contribution to the excitation of the QWs comes from e–h pair generation in the barriers, but each QW only collects e–h pairs from the barriers immediately above and below it, and there is no transfer of excitation through the QWs. The penetration depth was obtained from the intensity ratios of the different peaks involved, with the electron beam normal to the plane of the QWs.⁵⁹ A good agreement was found with MC simulations of the depth-dose function. This is shown in Figs. 10 and 11.

A QW can also be used to determine the in-plane extension of the excitation volume, with the electron beam parallel to the plane of the QW.⁵⁸ In this case, a specially designed

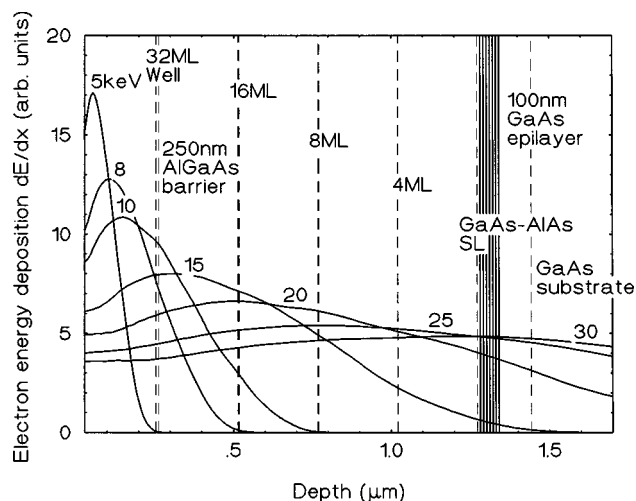


FIG. 11. Monte Carlo simulations of the depth-dose function for the acceleration voltages used in Fig. 10 with the positions of the QWs inserted. Using the integrated energy deposition in the barriers surrounding the individual QWs the intensity distribution of the spectra in Fig. 10 can be explained. (Courtesy of Trager-Cowan, Ref. 59.)

QW structure must be used to avoid effects of diffusion of carriers. The special design features a GaAs QW in between AlGaAs barriers. To prevent diffusion into the QW, an additional barrier in the form of an AlAs layer is put directly on each side of the QW. This gives a “detector” for the excitation with the width of the combined layers of the QW and the two AlAs barriers.⁵⁸ In this study, excitation profiles for different acceleration voltages were presented. By etching a wedge in the top layer of this type of structure, the penetration depth can be recorded as well, with the same precision.⁴⁹

The resolution obtainable in CL experiments is, as discussed above, determined by the acceleration voltage, where a lower acceleration voltage gives a smaller generation volume in the sample, and thereby a potentially higher spatial resolution. This is provided that the spot size of the electron beam can be made small enough. A very important factor for optimizing the resolution is the sample geometry. It is essential to choose an acceleration voltage that gives a substantial direct excitation of the structure of interest (cf. the peak of the electron-hole pair generation in Fig. 9). This can be exemplified by, e.g., top-view excitation of a buried QWR. If too low an acceleration voltage is used, all the excitation takes place above the QWR and the carriers have to diffuse (in 3D) to reach the QWR, which leads to a reduced resolution. If too high an acceleration voltage is used, there is a substantial direct excitation, but the generation volume is larger than necessary, and the resolution is lower than the optimum. Taking CL to the resolution limit usually means working in side view, where the structure of interest comes up to intersect the surface, and using the lowest practical acceleration voltage.

There have been a couple of reports of resolution in the 50 nm regime. This has involved deconvoluting the effect of diffusion in the barriers⁶⁰ and satisfying the Rayleigh criterion in a structure of known dimensions.⁶¹ Here, we present a monochromatic, side-view image of a stack of 15.2 nm thick $\text{Ga}_{0.85}\text{In}_{0.15}\text{N}$ QWs separated by 4 nm thick

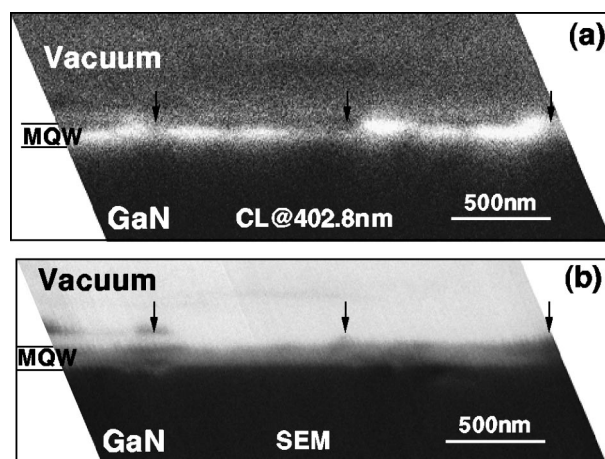


FIG. 12. (a) Monochromatic side-view CL image of a flat, 100 nm thick, $\text{Ga}_{0.85}\text{In}_{0.15}\text{N}/\text{Ga}_{0.5}\text{In}_{0.5}\text{N}$ MQW. This illustrates the resolving power of CL using a field-emission SEM operated at 1 keV and a sample temperature of 4.6 K. The MQW has thin (~ 50 nm), luminescent segments which appear to be displaced less than 50 nm with respect to each other in the growth direction and to the flat sample surface. (b) SEM image of the same area. The upper and lower interfaces of the MQW (lighter) are flat, with a few hillocks indicated by arrows in the images. Apart from the hillocks, there is no correlation between the variation in the luminescence intensities and variations in the MQW geometry. This illustrates an experimental resolution of better than 50 nm. [Images courtesy of Norman (Toshiba Cambridge), sample courtesy of Itaya (Toshiba RDC) (to be published).]

$\text{Ga}_{0.95}\text{In}_{0.05}\text{N}$ barriers (see Fig. 12). The multi-QW (MQW) structure (including the cap) has a total thickness of 100 nm and was grown on a GaN buffer. The SEM image reveals that the MQW layer is smooth, with flat upper and lower interfaces. The CL image exhibits strong CL from segments of the MQWs. These segments have a lateral extension of ~ 200 nm, and appear to be displaced with respect to each other in the growth direction. Since the surface is smooth in the corresponding SEM image, this displacement is not an effect of sample geometry. The magnitude of these displacements is less than 50 nm, effectively illustrating a spatial resolution of less than 50 nm.

4. Scanning tunneling luminescence

In STL,^{62,63} low-energy carriers are injected into the sample, similar to low-energy CL. One important distinction is that the injected carriers have energies in the order of eV (typically, 0.1–10 eV). In contrast to CL, it is possible to tune the energy of the carriers to be injected resonantly into specific states in the semiconductor. A significant difference is the inherent possibility to inject both electrons and/or holes, depending on the polarity and magnitude of the applied voltage.⁶⁴ The major advantage of STL over CL is considered to be the small excitation volume, due to the low energy of the injected carriers.

In this review article, we will present only the very basics of STM. The simplest case is when the tip, as well as the sample are metals, separated by vacuum. When the pointed tip approaches the surface, the electronic wave functions of the tip and the sample will start to overlap. If a voltage V_t is applied between the tip and the sample, the electrons can tunnel (quantum mechanically) through the vacuum

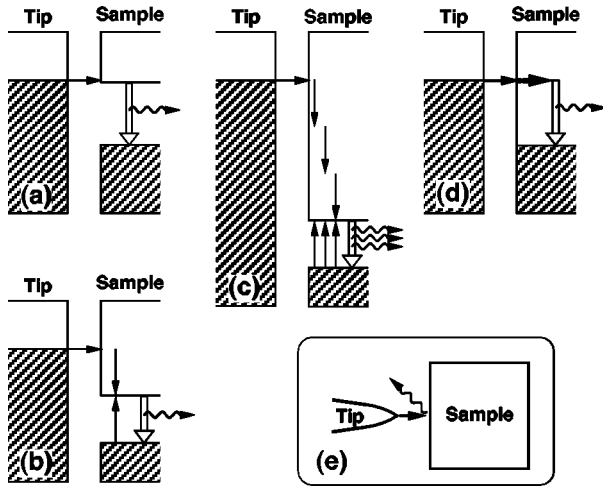


FIG. 13. Different tunneling injection conditions leading to radiative emission, using a metal tip. Thin arrows indicate flow of the carriers, hollow arrows indicate radiative recombination. (a) Resonant electron injection mediated emission in a semiconductor; (b) hot-electron Auger deexcitation mediated emission where the injected electron loses its energy in an Auger process, where a second electron is lifted from the valence band to the conduction band, creating an electron-hole pair; (c) high electron-energy (hot-electron) injection of carriers leading to impact ionization in a semiconductor; and (d) surface-plasmon decay mediated emission in a metal. (Adapted from Ref. 64).

barrier.⁶⁵ The distance between the tip and the surface will strongly affect the magnitude of the tunneling current I_t , and it is possible to show that the tunneling current has the following relation:^{66,67}

$$I_t \propto V_t r_s \exp(-ks\sqrt{\phi}), \quad (9)$$

where r_s is the local DOS of the sample surface, $k \approx 1 (\text{\AA} \sqrt{\text{eV}})^{-1}$, s is the tip-surface distance, and ϕ the average barrier height. With two metals in vacuum, the barrier height is an average of their respective work functions. When scanning a surface, the distance is usually adjusted to maintain a constant current, creating a constant-current surface map, which in turn is a map of the DOS of the surface. In metals this corresponds well with the actual atomic positions. This means that STM constant-current images give atomically resolved topographic information of the surface of the metal.

There are several different excitation or injection processes (see Fig. 13). The mechanisms described in the literature can be divided into two major groups: injection due to (i) elastic or (ii) inelastic scattering. In the case of inelastic scattering it is assumed that the energy of the tunneling carriers (electrons) can be converted into photons in the tunneling process. In one of the first studies of photon emission in a tunneling process, electrons from the tip recombined with a state in the sample and the excess energy was emitted as a photon.⁶⁸ This process is known as inverse photon emission. Following this initial report, there have been reports on plasmon mediated decay of the excess energy of the tunneling electrons,^{69,70} illustrated in Fig. 13(d).

For the case of elastic scattering processes, Figs. 13(a)–13(c), several studies^{62–64,67,71,72} have been reported where the carriers are injected into the semiconductor without any

loss of energy in the actual tunnel process. The subsequent emission is a result of recombination processes in the semiconductor. A carrier injected into a semiconductor from the STM tip can either: (i) recombine directly, either radiatively or nonradiatively; (ii) diffuse back to the surface depletion region and recombine nonradiatively at the surface; or (iii) diffuse inside the semiconductor, where it will be captured by a (non) radiative recombination center and experience a subsequent (non) radiative recombination. Since the carriers are injected close to the surface, these methods are very sensitive to nonradiative recombination processes, related to surface traps. It is, therefore, essential to minimize the number of surface traps to achieve enough intensity to be able to detect the radiative recombination of interest.

The most straightforward injection of (minority) carriers into a semiconductor is shown in Fig. 13(a), where a metal tip has been biased relative to the semiconductor, enabling electrons to tunnel into the conduction band of the semiconductor. Here, they can recombine with the hole distribution in the semiconductor. The minority-carrier injection of electrons into a p^+ -doped bulk InP sample was demonstrated in Refs. 63 and 72, where band-gap-related luminescence was detected. This resulted from the radiative recombination of the electrons with free holes in the p -type sample. The hole distribution was argued to be localized close to the surface depending on the actual band-bending situation. However, the actual band bending close to the tip was not known.

With increasing bias, the injected carriers will have increasing excess energy, and at some bias, carriers can even be injected directly into, e.g., a higher conduction band. This has been verified experimentally for injection into the Γ and L conduction bands in InP.⁶⁴ Above a specific bias there is also a possibility for injection of carriers having sufficient energies to create e-h pairs, i.e., impact ionization [illustrated in Fig. 13(c)]. Alvarado *et al.*⁷¹ have found (most probably) the onset of the impact ionization of e-h pairs. This was found to agree with the energy required for impact ionization of a GaAs surface covered by Cs_xO ,⁷³ as well as with the energy determined from measurements of the rate of e-h pair generation in GaAs.⁴⁰ In a different experiment using a bulk InP sample, the e-h pair generation was identified as a sudden increase in the detected luminescence when the bias was increased above 4.5 V. A different type of proof for e-h pair generation was an observation of a significant increase of the diffusion length, discussed in terms of a short electron diffusion length versus a long exciton diffusion length.^{74,75} This will be discussed in more detail in Sec. III C 6, on Stranski-Krastanow growth of quantum dots.

The injection of carriers from a metal tip to a semiconductor is pointlike. The carriers can be assumed to enter the semiconductor with the same trajectory as they left the tip. However, in Ref. 67 Pfister argues for a homogeneous angular distribution over the entire hemisphere in the semiconductor. This would be an effect of the relatively small effective mass of the carriers in the semiconductor, compared to the effective mass in the metal. The electrons (holes) can be assumed to be hot and ballistic when injected. In the semiconductor, the injected carriers will experience both elastic scattering events, which will change their direction, and in-

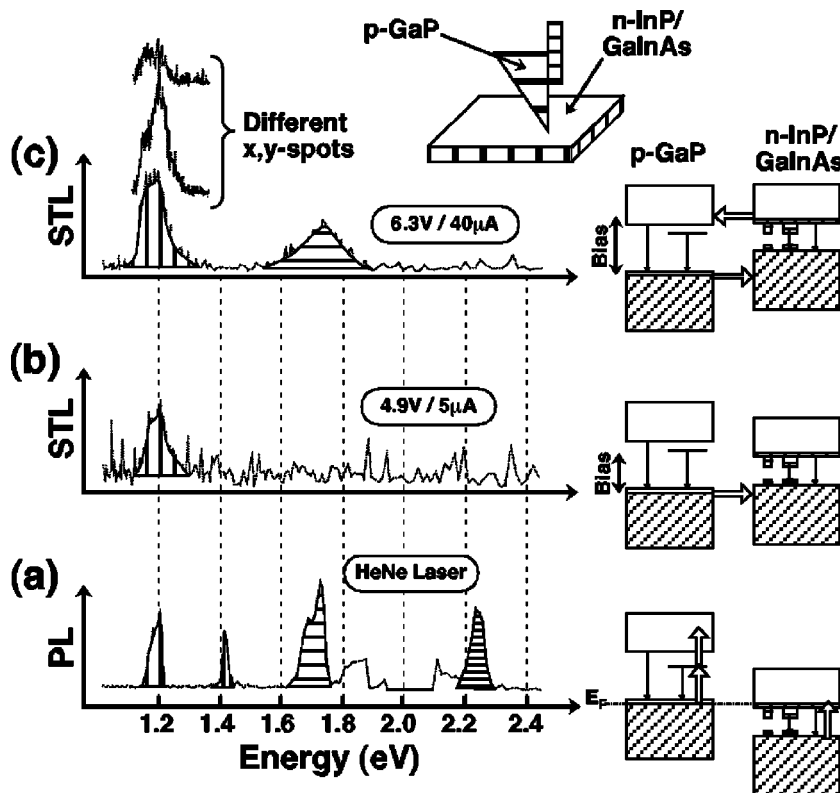


FIG. 14. 150 K STL and PL spectra of a *p*-type GaP tip and an *n*-type sample containing GaInAs QWs of different thicknesses in between InP barriers. Hollow arrows indicate excitation and thin arrows indicate recombination in the schematic illustrations to the right. (a) Typical PL spectrum, with a HeNe laser focused on the tip-sample region. The peak at 2.25 eV is the band-gap-related emission from the GaP tip and the peak at 1.7 eV originates from the Zn–O-related center in GaP. The 1.4 eV peak is related to the InP barrier and the 1.2 eV peak is related to one of the QWs. (b) When the tip is biased so that holes are injected from the valence band of the GaP into the valence band of the InP, only the emission from the QW can be detected. (c) At slightly higher bias electrons are injected from the conduction band of the sample into the conduction band of the tip. In addition to the QW luminescence, the Zn–O-related peak in GaP can be observed. Above the main spectrum of (c), two more QW-related spectra from different sample points are displayed, exhibiting a variation in the QW luminescence. This is most probably an effect of local thickness fluctuations in the QW. (Adapted from Ref. 64).

elastic scattering events, which will thermalize them. Scattering mechanisms include interaction with phonons, collision with impurities, carrier–carrier scattering, intervalley scattering, etc. The subject of how far carriers will travel before they are thermalized was also discussed in Ref. 67. To get an estimate of this thermalization length, LO-phonon scattering in the Γ valley of GaAs was considered as the only type of scattering event. This gave a theoretical thermalization length of ~ 130 nm for an excess electron energy of 100 meV. However, it was carefully pointed out⁶⁷ that this value should be an overestimation since only one type of scattering event was taken into account. This value can be compared with the tunneling hot electron transistor experiments by Heiblum *et al.*,⁷⁶ where 50% of electrons with an excess energy of 300 meV were shown to pass through a 31 nm GaAs layer with no detected energy loss, i.e., where the thermalization length was more than 31 nm.

A semiconductor tip and a semiconductor sample with different band gaps and doping types can be used to inject carriers either only into the sample, or into both the sample and the tip. By manipulating the voltages applied between the tip and the sample, it is possible to obtain information of the actual directions of carrier flow. This was shown experimentally⁶⁴ by employing a *p*-type GaP tip and an *n*-type InP/GaInAs multi-QW sample, where a series of spectra was recorded as a function of applied bias (see Fig. 14). For low biases, only injection of holes from the tip into the valence band of the sample was possible, leading to recombination in the MQW sample. With increasing bias, another channel was opened, leading to electron extraction from the sample into the conduction band of the GaP tip and the char-

acteristic GaP emission was detected, together with the InP QW emission.

B. Diffusion lengths, signal formation, and contrast

As mentioned above, the diffusion of the generated carriers and excitons plays a significant role in determining the resolution of most of the local luminescence probe techniques. Normally, diffusion is discussed in terms of diffusion of minority carriers, even though that is not always the case, especially not in CL experiments, where the excitation can create carriers in excess of the majority-carrier density. From a point source, the diffusion of carriers leads to a density profile $n(x) \sim \exp(-(x/L_{\text{diff}}))$ in 1D and $n(r) \sim (1/r) \times \exp(-(r/L_{\text{diff}}))$ in 3D.^{43,77} The diffusion length (L_{diff}) is a parameter that is complicated to model, and most data rely on empirical measurements. In principle, $L_{\text{diff}} = (k_B \mu T \tau / q_e)^{1/2} = (D \tau)^{1/2}$, in a macroscopic sense,⁷⁸ where k_B is Boltzmann's constant, μ is the mobility of the diffusing carriers, T is the sample temperature, q_e is the electron charge, D is the diffusion constant, and, finally, τ is the lifetime of the diffusing carrier.

Despite the reduction from three to two dimensions, the diffusion in QWs is more complicated than in bulk material. Most bulk materials can be assumed to be homogeneous, with only small fluctuations in the band gap. Carriers and e–h pairs can diffuse in any direction, virtually without any potential barriers. In both QWs and QWRs, a small variation in the thickness leads to a large variation in the confinement energy. This means that the diffusion can be limited by local variations in the thickness, i.e., local potential variations. De-

pending on the lateral extension and the magnitude of the energy variations, this can lead to a temperature-dependent diffusion length.⁷⁹ Furthermore, the nonradiative surface recombination can significantly reduce the effect of the diffusion for excitation close to the surface. A large portion of the carriers will diffuse to the surface and recombine there. This will act as a time window with a ps resolution, effectively freezing the shape of the excitation volume.⁶¹ This can, typically, be observed in CL images at very low acceleration voltage (1–3 keV) and even more apparent in STL images with a resolution governed mainly by the extension of the excitation spot. STL images with a sub-2-nm resolution have been presented.⁶²

As mentioned above, the dominating diffusion process in CL is not always the minority-carrier diffusion. This can be exemplified by a simple calculation of the density of created e–h pairs. The penetration depth in GaAs is $<0.5 \mu\text{m}$ at 5 keV, so in a simple model, the generated e–h pairs can be assumed to be confined to a cylinder with $0.5 \mu\text{m}$ diam and $0.5 \mu\text{m}$ long ($V \approx 0.1 \mu\text{m}^3$). This assumption actually underestimates the density of generated e–h pairs at the center of the excitation. Furthermore, a 5 keV electron will on average generate ~ 700 e–h pairs. This means that a probe current of 1 nA, which is a normal probe current used in CL, generates 4×10^{12} e–h pairs/s, which is 4×10^3 e–h pairs/ns. If we further assume a total lifetime of 1 ns, this gives a density of 4×10^4 e–h pairs/ μm^3 or 4×10^{16} e–h pairs/ cm^3 , which is one order of magnitude larger than the typical majority-carrier density in, e.g., undoped GaAs.

There are many ways to determine the diffusion length and here we will concentrate on measurements based on local luminescence probes. One of the most straight forward methods was introduced by Zarem and co-workers.^{80,81} The method is simply based on putting a mask, in the form of a 75 nm thick Al layer, on top of the semiconductor. This mask blocks the light from the sample, but does not significantly affect the penetration of the electron beam. In these types of experiments, the diffusion length is measured by scanning a focused electron beam perpendicular to the edge of the mask, simultaneously recording the position and the intensity. The slope of the intensity as a function of distance gives the diffusion length. Under high injection conditions ($\sim 10^{18}$ e–h pairs/ cm^3), the ambipolar diffusion lengths (the combined diffusion lengths of the generated electrons and holes) of various materials have been measured at 300 K.^{80–82} This method has also been used to determine the diffusion length in a GaAs/AlGaAs MQW structure at 8 K.⁸³ The data are presented in Table II, together with data obtained by other methods based on both PL, CL, and STL.

Spatially resolved PL can be used in several ways to measure the diffusion of excitons and minority carriers. In one of the earlier reports by Olsson *et al.*,⁸⁴ a pump–probe technique was used. Both the pump and the probe laser beams were focused on the sample. The time delay between the pump and the probe, as well as the spatial position of the pump and the probe were varied. It was thus possible to obtain an emission profile along a line as a function of time delay between the pump and the probe. Using this approach, it is necessary that the emission intensity depends nonlin-

TABLE II. Diffusion lengths for various semiconductors, obtained using local probe techniques.

QW/bulk	t (nm)	Barrier	L_{diff} (μm)	T (K)	Ref.
$\text{Al}_{0.50}\text{Ga}_{0.50}\text{As}$	...	...	$<0.5(\text{h})$	5–170	89
$\text{Al}_{0.38}\text{Ga}_{0.62}\text{As}$	...	...	2.5	300	81
$\text{Al}_{0.38}\text{Ga}_{0.62}\text{As}$	...	...	~ 1	300	71
$\text{Al}_{0.37}\text{Ga}_{0.63}\text{As}$	...	...	1.9	300	81
$\text{Al}_{0.37}\text{Ga}_{0.63}\text{As}$	75	AlAs	1.85	300	80
$\text{Al}_{0.25}\text{Ga}_{0.75}\text{As}$	...	...	0.7	300	81
$\text{Al}_{0.21}\text{Ga}_{0.79}\text{As}$	...	...	0.5	300	81
$\text{Al}_{0.21}\text{Ga}_{0.79}\text{As}$	75	AlAs	0.48	300	80
$\text{Al}_{0.09}\text{Ga}_{0.91}\text{As}$	...	...	0.7	300	81
GaAs	...	...	0.7	300	81
GaAs	...	...	100 (e)	30	86
GaAs	...	...	35 (h)	30	86
GaAs	75	AlAs	0.69	300	80
GaAs	74	$\text{Al}_{0.50}\text{Ga}_{0.50}\text{As}$	3.7	55	85
GaAs	4	$\text{Al}_{0.50}\text{Ga}_{0.50}\text{As}$	0.6	55	85
GaAs	2	$\text{Al}_{0.50}\text{Ga}_{0.50}\text{As}$	1.5(h)	5–170	88
GaAs	1300	$\text{Al}_{0.37}\text{Ga}_{0.63}\text{As}$	25 (h)	300	84
GaAs	10	$\text{Al}_{0.3}\text{Ga}_{0.7}\text{As}$	0.68	300	80
GaAs	5	$\text{Al}_{0.30}\text{Ga}_{0.70}\text{As}$	0.85	8	83
$\text{Ga}_{0.90}\text{In}_{0.10}\text{As}$	3.5	GaAs	0.9	16	91
$\text{Ga}_{0.80}\text{In}_{0.20}\text{As}$	3.5	GaAs	0.5	16	91
$\text{Ga}_{0.77}\text{In}_{0.23}\text{As}$	16–38	GaAs	0.4	6	92
$\text{Ga}_{0.77}\text{In}_{0.23}\text{As}$	16–38	GaAs	0.5	6	90
$\text{Ga}_{0.70}\text{In}_{0.30}\text{As}$	12.5	InP	10.5	300	82
$\text{Ga}_{0.59}\text{In}_{0.41}\text{As}$	20	InP	9	300	82
$\text{Ga}_{0.57}\text{In}_{0.43}\text{As}$	20	InP	7	300	82
$\text{Ga}_{0.47}\text{In}_{0.53}\text{As}$	9.5	InP	6	300	82
$\text{Ga}_{0.20}\text{In}_{0.80}\text{As}$	2.5	InP	3	300	82
GaN	...	...	<0.25	300	93
$\text{Ga}_{0.80}\text{In}_{0.20}\text{N}$	3	GaN	0.06	10	94
ZnSe	6	$\text{Zn}_{0.75}\text{Cd}_{0.25}\text{Se}$	0.5	300	87

early on the excitation power density. The sample was a double heterostructure of $\text{Al}_{0.5}\text{Ga}_{0.5}\text{As}/\text{GaAs}/\text{Al}_{0.5}\text{Ga}_{0.5}\text{As}$ where the GaAs thickness was $1.3 \mu\text{m}$. A hole drift mobility of $340 \text{ cm}^2/\text{Vs}$ was found. With a lifetime of 350 ns, this leads to a $L_{\text{diff}} = 25 \mu\text{m}$.

Another approach is to use a metal mask with micrometer-sized holes in close proximity to the sample. By measuring time-dependent PL through holes of different sizes it is possible to deduce the diffusion profiles. The carriers are generated immediately below the hole. For large holes (larger than the diffusion length of the carriers) the emission will have a certain lifetime. For small holes (comparable to or smaller than the diffusion length) some of the carriers will diffuse away from the hole and recombine under the mask. They will not contribute to the luminescence, i.e., a shorter lifetime is obtained, compared with larger holes. The method is somewhat indirect and requires modeling to get the emission profile. This approach has been used by Hillmer *et al.*,⁸⁵ who investigated exciton diffusion in GaAs/AlGaAs QWs. Different samples were used and the QW thicknesses were between 4 and 74 nm. The Al concentration of the barrier was, typically, but not exclusively, 0.5. At low temperatures, it was found that the diffusion of excitons was sharply reduced in narrow QWs compared to thick QWs. This was attributed to interface scattering. For narrow QWs, the amplitude of the exciton wave function is significant at the interfaces. Interface imperfections are expected to be im-

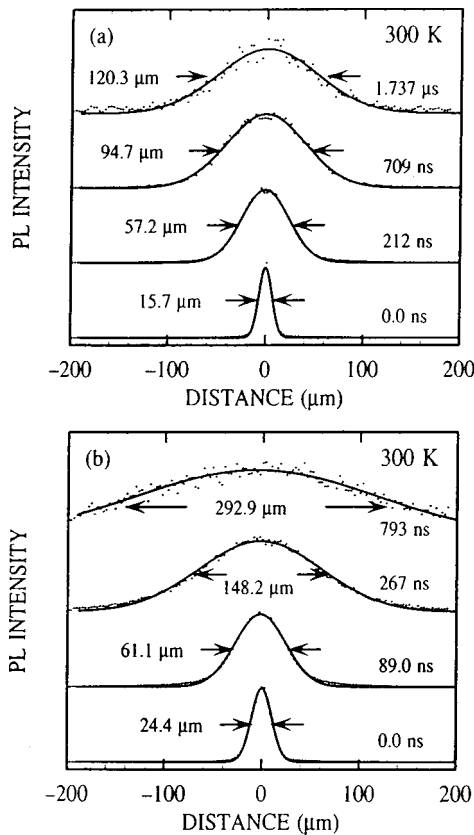


FIG. 15. The spatial distribution of the PL intensity as a function of time delay in a double heterostructure of AlGaAs/GaAs, viewed from top. The thickness of the double heterostructure was $9.82 \mu\text{m}$ in (a) and $0.30 \mu\text{m}$ in (b). At zero time delay (after the laser pulse) there is an initial width of the (Gaussian) distribution, which spreads out spatially with increasing time delay. The spatial distribution is Gaussian at all times, which indicates that the spatial spreading is due to diffusion (Ref. 86).

portant scattering channels. At higher temperatures it was concluded that other scattering mechanisms became important, such as phonon scattering and disorder scattering due to fluctuations in the alloy composition of the barrier. The mobility range was from $\mu = 300 \text{ cm}^2/\text{V s}$ and $L_{\text{diff}} = 0.6 \mu\text{m}$ for a QW of thickness 4 nm to $\mu = 30\,000 \text{ cm}^2/\text{V s}$ and $L_{\text{diff}} = 3.7 \mu\text{m}$ for a QW of thickness 74 nm , at a measurement temperature of 55 K . This is a difference of a factor 100. At a temperature of 200 K the difference in mobility was a factor of 4.

Wolford *et al.*⁸⁶ were able to follow the spatial diffusion of the minority carriers in double heterostructures of GaAs grown in between barriers of AlGaAs. They used time-delayed spatially resolved PL in a more direct way, where the emission from the samples was focused on the entrance slit of a monochromator as described in Sec. II C 1 on instrumentation, below. The GaAs thickness was varied between 0.118 to $9.82 \mu\text{m}$. Figure 15 shows the room-temperature time-resolved PL distributions. The spatial diffusion of the emission is clearly visible and the minority-carrier diffusion was found to be purely diffusive between 30 and 300 K . The room-temperature electron and hole mobilities were found to be 5000 and $250 \text{ cm}^2/\text{V s}$, respectively. The minority diffusion length was found to be $100 \mu\text{m}$ for electrons and $35 \mu\text{m}$ for holes. No dependence of the mobility on the GaAs layer

thickness was found, although the diffusion length became smaller for narrower heterostructures due to a decrease in the minority-carrier lifetime.

Perhaps the simplest way to measure carrier diffusion is to measure the spatial variation of the PL intensity using a microscope. It is necessary to have a tightly focused laser spot and to use modeling (solving the two-dimensional diffusion equation in the QW case). This approach has been used for studies of diffusion of carriers in $\text{Zn}_{0.75}\text{Cd}_{0.25}\text{Se}$ QW in between barriers of ZnSe. For a QW thickness of 6 nm , an ambipolar diffusion length of 500 nm was found (which also was the spatial resolution of the setup).⁸⁷

Another method for determining the diffusion lengths in QWs is to use local variations in the luminescence from a single QW (SQW), e.g., a dislocation or a locally occurring luminescence center. Using the CL from a decorated threading dislocation, a diffusion length of $1.5 \mu\text{m}$ was obtained for a GaAs QW, both for normal excitonic CL, as well as for donor-acceptor-pair CL.^{88,89} The former is related to the exciton diffusion length and the latter to the hole diffusion length (minority-carrier diffusion length). A different approach to determine the diffusion length is to study the quantum efficiency and the lifetime of the CL emission from partly relaxed QWs and make a correlation with the dislocation density. A systematic study of these parameters has lead to an ambipolar diffusion length of $0.5 \mu\text{m}$ in partly relaxed GaInAs/GaAs QWs.⁹⁰

A different approach for studying diffusion lengths of minority carriers is to use the pointlike injection of the STM.⁷¹ In this study, a sample containing two GaAs QWs, separated by a $1 \mu\text{m}$ AlGaAs barrier, was studied in cross section. By positioning the tip laterally over the structure, a significant decrease of luminescence as a function of injection distance from the QW was observed. Recording the intensity profile as a function of position, the authors were able to determine a minority-carrier diffusion length in the order of $1 \mu\text{m}$. The data in Table II show that the diffusion of e-h pairs will often limit the resolution in the local probe techniques.

The theoretical considerations for the signal formation in local luminescence probe imaging is a complicated subject.⁴¹ Most of the work has been concentrated on bulk material.⁹⁵⁻⁹⁹ We will just give a few fundamental ideas of the complex behavior. For this, we consider the simplified system of a bulk semiconductor with homogeneously distributed radiative centers having a constant lifetime (τ_{rr}). In addition, It has locally occurring nonradiative centers, in the form of dislocations with a lifetime (τ_{nr}). The excitation will create n e-h pairs per unit time. Here, we assume that they are equally spaced in the generation volume, staying where they were generated, i.e., neglecting the effect of diffusion. The general radiative recombination rate R_{rr} is given by

$$R_{rr} = Cn(\tau/\tau_{rr}), \quad (10)$$

where C is a constant and the total lifetime τ is a combination of the radiative and the nonradiative lifetimes:

$$\tau = (1/\tau_{rr} + 1/\tau_{nr})^{-1} = \tau_{rr}(1 + \tau_{rr}/\tau_{nr})^{-1} \leq \tau_{rr}. \quad (11)$$

In the simple case

$$R_{rr} = Cn \quad (\text{away from the dislocations}), \quad (12)$$

$$R_{rr} = Cn(1 + \tau_{rr}/\tau_{nr})^{-1} \quad (\text{at the dislocations}). \quad (13)$$

This is the recombination rate inside the semiconductor. The number of photons leaving the surface of the semiconductor has to be corrected for the internal absorption in the material. More significantly, only a small part of the total solid angle can be transmitted through the surface due to the high refractive index of most semiconductors. In general, $\sim 2\%$ – 3% of the total intensity is transmitted through the surface and the rest is reflected back into the semiconductor by total reflection.⁴¹ The orientation of microcrystals can, therefore, influence the intensity from a surface that is not flat. The detected signal intensity will further be influenced by the light collection geometry and the spectral response of the transmitting (lenses) and reflecting (mirrors and gratings) elements in the detection system. Finally, the detected signal depends on the spectral response of the detector.

To come closer to the experimental situation, the real extension of the generation volume has to be taken into account, together with the effect of the diffusion of the generated carriers, as well as the effect of nonradiative recombination at the surface of the semiconductors. For a more detailed analysis of the basic theory, see, e.g., Ref. 41. Due to the complexity of the contributions to, and influences on the signal formation, this is usually the limit of the theoretical work. Normally, the theories neglect the influence of a possible directionality of the diffusion. We have to bear in mind that, e.g., CL imaging only records the radiative recombination as a function of the position of the electron probe and does not record the actual location of the recombination center. Even in the simple case of a dislocation in bulk material, the strain field surrounding the dislocation can influence the band gap of the material. An increased band gap will lead to diffusion away from the center and more radiative recombination, whereas a decreased band gap will lead to diffusion into the center and more nonradiative recombination. In the more complicated case of a QW, the local variations in the thickness can introduce differences in the potential, which act as diffusion barriers. This can have an impact on the contrast in the monochromatic images of the different thicknesses of the QW. There is also a further possibility to saturate the radiative/nonradiative recombinations, which will also have an impact on the contrast. In principle, what is needed to model this is a complex Monte Carlo simulation, using the actual geometry and parameters of the sample.

C. Instrumentation

All the local probe techniques use similar detection systems and there is a number of possibilities, depending on the application, i.e., spectral range, signal level, and recording speed. The simplest version of the energy dispersion is the use of filters to select the emission of interest. The type of filter depends on the energy separation between the different peaks on the sample. A more common method is to use a monochromator for the peak selection. The main advantage

of this method is the option to record full spectra to determine the exact peak position, as well as to determine the optimum size of the detection windows. It is also possible to make studies of local variations in the spectra and effects of excitation density, as well as excitation energy. A third possibility is to use a Fourier transform spectrometer, usually used for low signal intensities in the infrared spectral region. This has the advantage of faster acquisition times for wide spectral ranges with high spectral resolution.

There is a wide range of different detectors available. The choice of detector depends on the application, but, in principle, there are two categories of detection. The first category is the single energy detectors, ranging from ordinary $p-i-n$ photodiodes, avalanche diodes to photomultiplier tubes (PMTs). The second category is the parallel detectors, which detect the whole spectrum, or parts of the spectrum, simultaneously. These detectors are, typically, diode arrays, charge-coupled device (CCD) arrays and streak cameras. The signal from these detectors can be detected in several modes, but there are basically two ways: current or voltage mode, where the signal is converted into an averaged current or voltage, which includes the noise of the detector, as well as the electronics. In the photon counting mode, pulses corresponding to single photons hitting the detector are detected. This virtually eliminates the contribution from the noise of the detector to the signal level, which results in a more true intensity level.

Both modes can be used for steady state, as well as pulsed measurements. In the steady-state mode, the data can be presented as spectra or intensity maps. In the pulsed mode either the rise time or the fall time of the emission can be studied. Such data can be presented as the time evolution of the spectrum or as rise/fall time maps. Using parallel detection, more data can be recorded in each point. Maps can be obtained where each point contains the information of the whole spectrum. In CL this is known as CL wavelength imaging (CLWI).¹⁰⁰ The data can be presented as either a normal intensity map for a certain energy or as a peak energy position map, where the energy position of the most intense energy in the spectrum is plotted. Further, the use of a streak camera could add the time dependence to the data. However, the difficulty with the CLWI, and similar methods, is the amount of data it generates. Just an example: A 1024 element CCD array with 16 bit resolution (65 536 intensity levels) generates one spectrum containing 2 kb of information per pixel. A 128×128 pixel image then uses about 32 Mb.

For studies of semiconductors, it is generally invaluable to work at low temperatures using a cryostat cooled with liquid nitrogen (LN_2 , 77 K) or liquid helium (lHe, 4.2 K). The reason is mainly related to the emission. Reducing the temperature, generally, increases the quantum efficiency of the emission and suppresses the nonradiative recombination paths. The reduced temperature also means that the average energy of the carriers is lower and the energy spread of the carriers is lower, leading to sharper and more intense emission peaks. There are, however, technical challenges in implementing the cooling without deteriorating the resolution.

1. Microphotoluminescence/microphotoluminescence excitation

One of the most direct ways to obtain local luminescence in the form of spot-mode spectra and intensity maps is by spatially resolved PL. This technique is simple and straight forward, as well as nondestructive. The resolution, however, is limited to $\sim 1\text{--}2\ \mu\text{m}$ in III-Vs or even less ($10\text{--}100\ \mu\text{m}$) in silicon.¹⁰¹ The resolution is determined by the size of the excitation spot on the sample and by minority-carrier (or exciton) diffusion, which may be very long in, e.g., high-quality silicon. The low excitation power density used in these types of measurements means that the diffusion is governed by minority carriers. This is in contrast to the high excitation power densities, which are generally used in, e.g., CL. There are two dominating ways of obtaining maps of the luminescence, either by direct imaging or by scanning either the excitation spot or the sample.

In direct imaging, the area of interest of the sample (which may be in a cryostat) is excited by a light source, usually a defocused laser. A magnified image of the sample is focused on a video or CCD camera, which gives a comparatively short measurement time.¹⁰² To obtain monochromatic images, the interesting spectral range can be selected using filters. A disadvantage of using direct imaging is that the bandpass of the filters is usually quite large, which means that some spectral information may be sacrificed. However, if the technique is used for the mapping of the spatial origin of a defect that has a broad luminescence peak, there is a tremendous improvement in measurement time without any loss of spectral information. The spatial resolution is limited by the optics but can with suitable lenses (i.e., corrected for the spherical aberration introduced by the cryostat windows) be as good as the diffraction allows [$\sim \lambda/2$, see Eq. (3)].

If a higher spectral resolution is needed a monochromator must be used. This complicates matters as it is not possible to obtain direct intensity maps using this technique. There are several other options to obtain these maps. The simplest way is to focus the excitation source on the sample and to record the intensity at a specific wavelength (or a complete spectrum) as a function of excitation position.¹⁰³ The positioning can be done by either scanning the excitation over the sample or keeping the excitation position fixed and scanning the sample. It is easier to keep the focus of the excitation beam if the sample is stepped. However, if the sample is kept in a cryostat, this will usually require that the whole cryostat is moved. A second option is to use the analog of direct imaging, where the whole area of interest of the sample is excited and a magnified image of the sample is focused on the entrance slit of the monochromator. This means that only the emission from one point of the sample is detected. If an array-type detector or CCD/video camera is used, a line can be detected. Using this type of detection, it is also necessary to either move the sample or the image on the entrance slit to obtain the maps.

Recently, fast imaging streak cameras, with a time resolution down to 5 ps have come on the market. Usually, the output slit of a monochromator is replaced with a streak camera, which will record the emission spectrum as a function of time. If, instead of a monochromator a filter is used to

select an interesting spectral feature, it is possible to image the spatial distribution of the emission as a function of time. Of course, only the spatial behavior along one direction can be followed during one measurement.

In a combination of the two scanning techniques, a laser is focused on the sample and a magnified image of the sample is focused on the entrance slit of the monochromator. If the image of the sample is shifted on the entrance slit, the emission from different parts of the sample can be investigated. By detecting at a suitable spatial offset from the excitation spot, it is possible to detect the emission from carriers which have diffused a certain distance (determined by the offset).⁸⁶ This technique is particularly valuable if the exciting laser is pulsed and the detection takes place at certain time delay after the exciting pulse (analogously to a box-car system).

Combined systems, which are able to not only record PL maps but also to record maps of photoreflectance are commercially available.¹⁰³ Photoreflectance is a two-beam experiment where a strong, modulated, laser beam causes a change in the reflectivity which is probed by a second (weak) laser beam. The strong beam causes saturation of surface states, which changes the band bending at the surface, this in turn will change the reflectivity due to the small changes in the electric field. The change in reflectivity is linked to the band structure, which can, therefore, be mapped.¹⁰⁴ One strong advantage of this technique is that it is possible to obtain information about excited states. It is also a derivative technique with quite high spectral resolution, compared to the related technique of absorption measurements.

In many cases, spatially resolved PL is used for the inspection of wafers to evaluate the quality¹⁰⁵ and to optimize the growth conditions both in bulk using the Czochralski technique¹⁰⁶ and in epitaxial growth.¹⁰⁷ One of the most common applications is to inspect the wafer for the distribution of impurities. It is clearly an advantage to have a (known) uniform distribution of impurities to make workable devices from the whole wafer. This screening of wafers can reduce the failure rate of subsequently grown devices. Not only radiative impurities are studied in bulk samples. The reduction of the emission lifetime due to the presence of nonradiative impurities or defects can be used to gain information of the distribution of nonradiative defects if a map of the emission lifetime is made. In this review we concentrate on low-dimensional semiconductor structures and we recommend Refs. 41, 54, 103, 105, and 107 for properties of bulk and bulklike materials.

μ -PL is used widely in the literature, though the definition is not totally clear. Originally, the term was used to describe PL excitation through an optical microscope, where the size of the laser spot on the sample is near the diffraction limit, $\sim 1\ \mu\text{m}$. In a wider sense, the term is used to describe any type of PL setup, where the spatial resolution is of the same order, regardless of how it is obtained. The system used in the authors' laboratory is a typical μ -PL setup¹⁰⁸ (see Fig. 16). The system differs from a conventional PL setup on two points: the cryostat and the optical microscope. The cryostat is a commercial IHe-flow cryostat, with a temperature range of 4–300 K. The sample is mounted very close to the

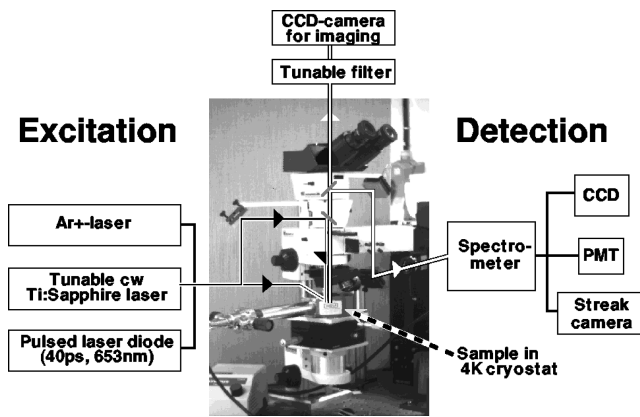


FIG. 16. In our μ -PL setup, the sample is mounted in a He-flow cryostat, with the sample very close (<10 mm) to the optical access window. The sample can be observed through the eye pieces of the microscope and by a CCD camera. A tunable filter is put in front of the CCD to record monochromatic images. The microscope has a section of parallel light, where access to the beam path can be made by beam splitters and mirrors. The excitation sources, in the form of various lasers, can be entered through a beam splitter and focused on the sample, or entered below the objective at grazing incidence on the sample. The emission from the sample can be taken out via a mirror and dispersed through a monochromator. The detection is either made by a PMT, a CCD, or a streak camera. (Courtesy of Hessman, Ref. 108.)

optical access window. This is important, since a short distance between the objective of the microscope and the sample is essential for the spatial resolution. The spatial resolution of a high-quality objective in air is $\approx 0.6\lambda/\sin\theta$ [cf. Eq. (3)], where θ is the half angle of the solid angle collected by the objective. With the relatively large objective-to-sample distance, the objectives are generally restricted to $\sin\theta \leq 0.5$. However, due to the aberration of the optical window of the cryostat, the resolution is limited to $\sim 1.5\ \mu\text{m}$, i.e., $\sim 2\lambda$.

The objective is infinity corrected, which means that light from a spot in the focal plane results in parallel light. In the other direction, parallel light from, e.g., a laser will be focused on the same focal plane. The excitation source is generally a laser that can be introduced in two ways. It can be introduced at a grazing incidence for global excitation. It can also be introduced into the parallel beam path of the microscope by a beam splitter for local excitation. The latter case gives a considerable amount of stray light in the form of reflections of the laser beam. The parallel light from the objective is focused on an intermediate image plane by a lens system. This image can be observed either by ordinary eye pieces or by a CCD camera. A tunable filter is put in front of the CCD to obtain monochromatic images. For more detailed spectroscopy, a mirror can be inserted in the parallel beam path and the light can be dispersed through a monochromator. At the moment, there are several excitation sources: An Ar^+ laser, a tunable Ti:sapphire laser (cw and pulsed), and a pulsed laser diode. The emission is detected by either a PMT, a CCD camera, or a streak camera.

2. Near-field scanning optical microscopy

The possibilities to apply near-field microscopy concepts, in combination with the advances in scanning probe

technology, have in recent years enabled a new spectroscopy technique to be developed.^{18–20,29–31,109–111} The use of a tapered single-mode optical fiber for NSOM was proposed by Betzig and co-workers.²⁰ The tapered end of the fiber forms the small aperture (aperture diameter $<$ wavelength of the light) necessary for near-field work. This type of aperture can be scanned laterally with high precision and at a constant vertical distance from the surface of the sample. Usually, the vertical distance is maintained at a constant height by using a force detection scheme, sensing the shear force on the optical fiber probe.^{111,112} There are three major modes of operating a NSOM: (i) In the simplest case of NSOM, a far-field illumination is incident on the sample inducing radiative recombination (identical to conventional PL). The small aperture will collect the local emission from the sample, with a lateral detection resolution given by the scanning system and the aperture size. (ii) In the reverse case, the excitation is done through the aperture and the detection is done in the far field. This is the analog of the focused-laser PL, but with a smaller spot size. (iii) In the most advanced case, both the excitation and the detection is done through the same aperture. This yields the highest spatial resolution and most complex experimental setup, as well as the lowest signal intensities.

One of the key technology steps in designing an a-NSOM system is the definition of the small aperture at the fiber apex. Usually, this is done in a fiber pulling machine using a high-power CO_2 laser that melts a point on the single-mode fiber while the fiber is pulled. The melted area becomes thinner until it breaks, forming the tapered end. The tapering can also be formed by chemical etching,^{113–115} a technique initially used for atomic force microscope (AFM) tips.¹¹⁶ The diameter of the resulting aperture can be controlled down to approximately 20 nm.²⁰ After this step, the fiber has to be coated with a reflecting surface to trap the light in the fiber. This coating is usually aluminum, evaporated onto the tip. An alternative way for tip fabrication was demonstrated in Ref. 117, where a focused ion beam was used to define the taper of the tip. After a complete metallization, the focused ion beam was used to define the aperture in the tip. Recently, uncoated tapered fibers have also been used,¹¹² thereby avoiding the critical metallization stage.

There are many alternatives to the use of tapered fibers to create the NSOM probe, such as other aperture definition schemes and using near-field scattering mechanisms or luminescent effects.^{29,30,118} The problems associated with the critical aperture formation technique have triggered the realization of apertureless concepts.^{119,120} However, for probing the luminescence properties of low-dimensional structures, the preferred method seems to be the tapered-fiber definition of the aperture. The advantages are: the large versatility of optical contrast formation; the independence of the employed samples; and last, but not least, the proven reliability. All these advantages have been considered by many workers in the field, who nearly all use the tapered-fiber approach. Maybe the best argument for this approach is that the commercially available instruments utilize the tapered-fiber approach.

With the tapered-fiber approach, the resolution has been shown to reach down to 12 nm ($\sim \lambda/40$) with visible light.²⁰

A problem with a-NSOM is that the photons have to pass through a small aperture, smaller than the wavelength of the light. As an example, the transmission efficiency of a pulled fiber tip, with an aperture of size $\sim \lambda/10$ is in the order of 10^{-5} – 10^{-6} , where the aperture size is proposed to dramatically reduce the fiber transmission efficiency. The situation is better for the etched fiber tips, where the transmission efficiency is $\sim 10^{-2}$ – 10^{-3} for similar aperture sizes.^{115,121} The transmission efficiency is suggested to decrease as the fourth power of the aperture radius.¹²² This together with the low signal levels involved suggest that the aperture might have to be increased and thereby sacrificing some of the spatial resolution in order to obtain a sufficient signal intensity. This is why most of the spectroscopic NSOM work reported has been obtained with a probe that has an aperture diameter of 150–250 nm ($\lambda/5$ – $\lambda/3$). Even though this is far from $\lambda/40$,²⁰ the advantage over conventional far-field optics [limited by diffraction to $\lambda/2$ as discussed in Eq. (3) above] is large.

A serious technical challenge with NSOM is the regulation of the probe-to-sample distance that has to be maintained shorter than the aperture size in order to obtain maximum spatial resolution.¹¹² This is, as already mentioned, often done by optical detection of the shear forces acting on the tip.^{112,122,123} Usually, this means that the tip is vibrated in a motion parallel to the sample surface at one of its mechanical resonances, and when the tip approaches the surface perpendicularly to a distance of some few tens of nm the amplitude of the vibration will be decreased. The typical amplitude of the vibration is 3–5 nm, which is small enough not to reduce the resolution. The decrease in amplitude can, in turn, be monitored by the lock-in technique in combination with detection of the intensity of the specular reflection of the laser beam focused on the tip,^{112,122} or by differential interferometry.¹²⁴ To date, the nature of these shear forces is still unclear. Recently, a new purely electrical scheme detection of the vibration was proposed by Karrai and Grober,¹²⁵ thereby avoiding an unwanted optical background level in the cases where the photons should be limited to those around the tip aperture. The authors also claim that it should have advantages over the optical shear-force detection scheme at low-temperature and high-vacuum conditions. The method is based on a piezoelectric tuning fork held at resonance frequency and, in a way similar to the vibrated fiber tip, the amplitude changes when the sample surface is approached. Height regulation of the types used in atomic force microscopy¹¹⁸ and STM¹²⁶ have also been reported.

The group that has made the largest impact in NSOM studies of low-dimensional semiconductor structures, is the group around Betzig.^{20,27,111} They have developed a IHe cooled version of the NSOM. Their system,¹²² consists of: (i) *in situ* translation stages permitting course positioning of the fiber tip relative to the sample; (ii) fine scanning and positioning with piezoceramics; (iii) *in situ* far-field optics for imaging of the sample–tip region; (iv) a tip–sample distance regulation method based on the shear-force sensing concept; and (v) direct contact of the sample with the cryogenic liquid to deal with the high incident power (mW distributed over a few μm). The microscope is designed as an insert into a IHe

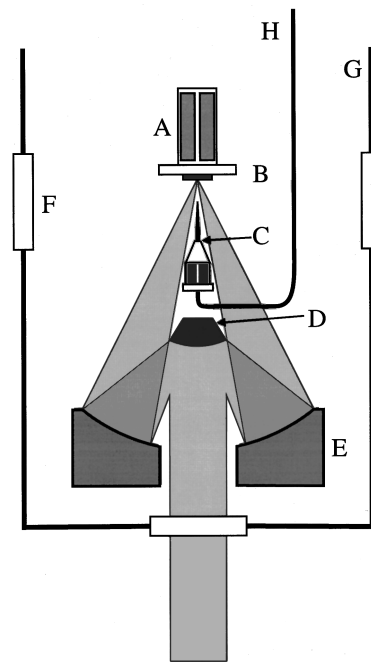


FIG. 17. Schematic showing the design of the optical system of a typical low-temperature NSOM. This system can operate down to a temperature of 1.5 K and be used in both collection mode and excitation mode. The sample is mounted on a platform (B) and scanned by a piezotube (A) with respect to the excitation/detection system. The far field is focused on the sample using two mirrors (D and E). The near-field probe is the tip (C) of a fiber (H), which is placed in the solid angle that is blocked by the primary mirror (D) of the far-field system. The whole assembly is mounted in a cryostat (G) with optical access through windows (F). (Courtesy of Harris, Ref. 122.)

bath cryostat, with the possibility to use the microscope in the temperature range between 1.5 and 300 K. The fiber probe is positioned on the optical axis of a reflecting microscope objective (see Fig. 17), permitting the system to be used either in collection mode or excitation mode. The reflecting microscope objective has some advantages over a conventional microscope such as a high numerical aperture (≈ 0.4) and a long working distance (15 mm). The disadvantage is the loss of viewing field in the center of the objective, where the second, central, mirror of the objective is located. This is not very serious, as this part of the solid angle is blocked by the fiber tip anyway (see Fig. 17).

3. Cathodoluminescence

The excitation source in a CL system is an electron gun, where the electrons have been accelerated by an electric field ranging from a couple of hundred volts to several kilovolts. The special case of extremely low-voltage CL, where the excitation source is the tip of a STM, is discussed in a separate section below: II C 4 on scanning tunneling luminescence. The high excitation energy of conventional CL makes it possible to study radiative processes in most semiconductors and insulators. One of the strong points with CL imaging and spectroscopy is the ability to excite semiconductors with large band gaps, which require lasers in the UV, or even deep UV, as an excitation source in PL. This typically applies to materials like diamond,¹²⁷ GaN,^{128–131} GaInN/GaN QWs,^{94,132} and BN.¹³³ The excitation source used is normally

the electron beam of a SEM or a STEM.⁴¹ This means that the beam is well defined in energy and it is also focused to a small spot, with the possibility to scan the beam over the sample to obtain a map of the intensity variations of the emission.

The use of a SEM or a STEM means that the wealth of information obtained from the ordinary detection system can be used as a complement to the CL signal. The normal SEM signals, typically, include topographical information obtained by detection of secondary electrons, atomic number contrast obtained by detection of backscattered primary electrons, and compositional information in the form of characteristic x rays,⁴² detected as energy or wavelength dispersive analysis of x rays (EDAX/WDAX). More complicated signals can be detected as well, e.g., scanning deep-level transient spectroscopy¹³⁴ and EBIC.⁴³ In the STEM, the normal transmission electron microscopy (TEM) facilities, such as identification of dislocations and determining their Burger's vectors.¹³⁵ Electron energy loss spectroscopy (EELS) and direct lattice imaging are also available in the STEM. Together, this gives the advantage of locating the interesting areas, as well as identifying irregularities and artifacts.

As the electron beam is easily guided and focused by electrical and magnetic fields, it is possible to scan an electron probe with a diameter of, typically, 1–100 nm over large areas of the sample with high precision. With an electric field near the electron source, it is also possible to deflect the beam and turn the excitation on and off. Rise and fall times, as well as pulse, with a duration in the sub-ns range can be obtained to study transient behavior of the emission.¹⁰⁰ Furthermore, the SEM/STEM provides high-precision mechanical movement of the sample relative to the exciting electron beam. This is very useful for studies of samples that are significantly larger than the field of view of the detection system. The sample can be moved to access different areas or to make a series of overlapping images.

The nature of the very local excitation means that, like in all local probe techniques, the total intensity of the generated luminescence can be very limited. This puts very high demands on the efficiency of the light collection and detection systems. Many of the systems are of different design, even though the basic features are the same in most cases. In its simplest version, the light collection system consists of a lens with the sample at one focal point, with a detector after the lens. This type of system will only provide panchromatic images. In a slightly more advanced system a filter can be placed between the lens and the detector for monochromatic imaging. One special application of this is to use red, green, and blue filters and present the images in three colors, a mode that is known as real color CL.¹³⁶ This type of system can often be supplied with the SEM itself as optional equipment. A simple alternative is to place a detector directly under the sample in a mode that is known as transmission CL.¹³⁷ Another alternative is to place a large-area detector, with a hole for the incoming electron beam, above the sample.

For more efficient light collection, a mirror is usually employed. Previously, the most commonly used systems were based on an ellipsoidal mirror with the specimen in one

focal point and a fiber bundle in the other. This configuration allows a very large solid angle collection, where a full 2π of solid angle can be collected. This is, however, limited by the numerical aperture of the fiber system. The use of a fiber makes the system very flexible and it is easy to switch between detection systems by just attaching the other end of the fiber bundle to different systems. The same signal can be shared by detection systems simply by parting the fiber bundle into 2. In this configuration even simultaneous PL/PLE measurements can be combined with the CL measurements, since the area on the specimen that the CL signal is collected from is also the area that will be excited, as well as collected in the PL/PLE measurements.¹³⁸ The second feature is a geometric factor of the ellipsoidal mirror itself; the distance from one focal point, via the mirror to the second focal point is constant. This means that there is no time delay depending on the point where the photon hits the mirror. A big drawback with the fiber-based systems is the high loss of intensity in the fiber bundle. This is caused by the fact that a large portion of the area of the fiber bundle is actually cladding material, as well as the loss from scattering at the ends of the fiber bundle.

The loss can be overcome by a carefully designed system, where the specimen is put in one focal point and the detection system is put in the other. This usually means that the entrance slit of the monochromator is placed in the second focal point. This can cause some problems in designing an optimized system as the f number (the useful solid angle admitted into the monochromator) of the monochromator should match the f number of the mirror. This reduces the flexibility of the system, as any type of detection system will have to be placed in a fixed position in space (the second focal point of the mirror). This problem can be overcome by using a parabolic mirror. This type of mirror features parallel light from the mirror, which makes the f -number fit independent of the mirror. In the case of low signal intensity, this arrangement can also be used with various types of interference filters. A very special application is to study the polarization behavior of the emission. This can be performed by putting a polarizer after the collection mirror.^{139,140}

Many CL systems for use with semiconductors have a cold stage, operated with either liquid nitrogen or liquid helium, where the temperature range between 300 and 4 K can be covered. The most common systems have a flow cryostat, where the liquid helium or nitrogen is flowed through the cryostat. The specimen is usually mounted directly on the cryostat. Care has to be taken to prevent vibrations caused by the boiling liquid from ruining the potential resolution of the measurements. A solution to the vibration problem is the way it was done previously in the authors' laboratory; the cryostat was connected to the specimen holder via a flexible braid.¹⁴¹

There are commercial CL systems available,¹⁴² but as an example of a typical CL system, we will give a description of the system used in the authors' laboratory (see Fig. 18). The system is based on a conventional SEM with a few modifications to the SEM itself. The vacuum system is improved by fitting a turbomolecular pump, which has magnetic bearings which make the pump virtually vibration free.

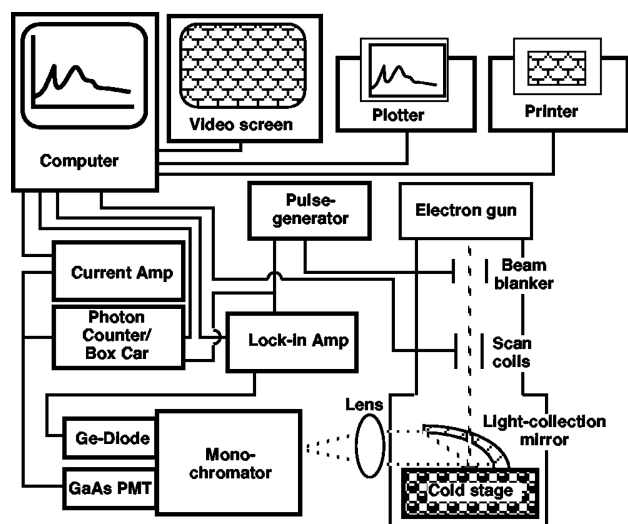


FIG. 18. Illustration of a typical CL setup, used in the authors' laboratory. The system is based around a conventional SEM, where the scan of the beam is computer controlled. The original specimen stage has been replaced by a liquid He cold stage. Both the CL and SE signals are recorded by the computer. The electron beam can be switched on and off with a beam blanker. The CL emission is dispersed through a monochromator and detected by either a GaAs PMT or a Ge diode. The PMT is used either in current mode or in photon counting mode. The diode is used in lock-in mode together with the beam blanker. The beam blanker together with the PMT can give time-resolved measurements using a box car.

A fast (<1 ns) electrostatic beam blanker is installed just under the anode of the electron gun to provide either pulsed excitation for detection in lock-in mode, or pulses with sub-ns rise and fall times for time-resolved measurements. The useful range of accelerating voltages is 1–30 keV and the electron source is a pointed tungsten filament. However, for work at low accelerating voltages, a LaB₆ emitter or a field-emission gun is preferable.

The cryostat is a commercial IHe/IN₂ flow cryostat, which gives a stable temperature range from <5 to 300 K. The light collector consists of a parabolic mirror, with a focal length of 4 mm. The mirror is made of Al and the surface is uncoated for good performance over a large spectral range. The collection efficiency (solid angle coverage) is $>60\%$. The useful field of view is strongly dependent on the focusing on the detection system, but the minimum field of view is $\sim 20 \times 20 \mu\text{m}^2$, with the system focused for highest local collection. Reducing the local collection by moving the focusing lens off the optical axis or by moving the sample out of the focal point of the mirror leads to a larger field of view of up to $500 \times 500 \mu\text{m}^2$, but the intensity can be reduced by up to two orders of magnitude.

The normal mode of operation is to focus the parallel light out of the mirror onto the entrance slit of a monochromator. The dual exit monochromator normally uses two different detectors, with overlapping spectral ranges. The normal combination is a Peltier cooled GaAs PMT for the spectral range 200–900 nm and a nitrogen cooled Ge pin diode for the range 800–1800 nm. The PMT is either used in current or photon counting mode. The diode is used in lock-in mode only. In the photon counting mode the PMT can be used in the time-resolved mode with a box car. This

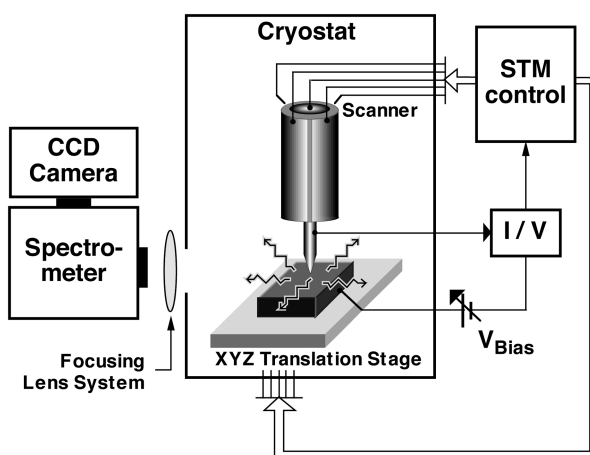


FIG. 19. Complete STL setup with the different subsystems used in the authors' laboratory. The STM is immersed in cryogenic liquid inside a cryostat. An optical detection system using a focusing lens is placed close to an optical window in the cryostat. The lens projects an image of the sample–tip region onto the entrance slit of a monochromator equipped with a CCD camera. Normal STM control electronics is used and the electrical feedthrough consists of voltages to the piezoscanner; the coarse XYZ-translation stage, bias to the sample and current lead to the tip. (Courtesy of Lindahl, unpublished).

way, a time resolution of ~ 1 ns can be reached, currently limited by the boxcar. As an option, we can put a fiber at the entrance slit of the monochromator to transfer the light to a streak camera for better time resolution and for parallel detection as well. The monochromator, as well as the scan of the microscope, are fully computer controlled. This means that spectra, as well as intensity line scans, can be recorded and images with different resolutions (1024×1024 pixels down to 16×16 pixels) can be recorded.

4. Scanning tunneling luminescence

The basis of a STL setup is a STM and a good optical collection and detection system. In some cases, a cryostat is incorporated in the system. The STM should provide the possibility to move the sample with respect to the tip both on a macroscopic and on a very fine scale, i.e., a coarse and a fine positioning system is required. Standard STM accessories such as software and hardware for scanning the tip and monitoring the current and/or the feedback signal are also required (see Fig. 19). A detailed description of the principles for tunneling and the different possibilities to employ the STM for obtaining information about the electronic structure of the sample goes beyond the scope for this article and the reader is referred to articles and textbooks which cover the STM methods beautifully.^{143–145}

The STL technique puts some very special requirements on the STM instrument itself. The most obvious is related to the light collection, where there must be a clear line of sight from the detector to the tip sample region. This is to be able to both monitor the tip with respect to the sample structures, as well as to make it possible to detect the (often) weak luminescence signal. The most crucial part in STL is the detection system. The low emission intensity is a result of both the very local nature of excitation and the low excitation power levels.

In most cases, the emitted radiation is collected by a lens, which has the tip-sample region in its focal point.^{62,146} In order to collect as large a solid angle as possible, the lens should be placed as close to the tip as possible. In other systems, a parabolic mirror⁷² or an elliptical mirror,¹⁴⁶ similar to CL systems, has been used. Although such an approach should be ideal, the practical problems with such a small mirror are severe. It has to be mounted with precision to have the tip-sample region in its focal point. The comparatively large tip will effectively block a large portion of the solid angle of the collected and reflected light since it will be in the line of sight from the mirror to the detector. Furthermore, the limited space around the tip sample restricts the actual size of the mirror even though the size has to be large enough to allow either the sample top surface or the sample edge (the cross section) to be approached by the STM tip. In many cases, the wavelength selection to obtain monochromatic images has been done by filters,^{62,71,146} but in some cases a monochromator has been employed.^{63,64,72,146,147}

STM equipments that have been used to obtain STL data are with few exceptions ultrahigh vacuum (UHV) systems operated at room temperature. It is, however, desirable to incorporate a cooling stage. Apart from the general efficiency and linewidth considerations, as discussed above, the major reason for cooling the sample is the possibility to increase the tunneling current levels without destroying the delicate sample surface. Our system is based on putting a small STM inside lHe or lN₂ inside the cryostat.^{63,72} In this system, the sample surface will in most cases be contaminated already before the sample is introduced into the cryostat and it will be nearly impossible to obtain atomic resolution in the STM mode. However, even though there is a contaminated layer on the surface, it is in most cases possible to tunnel through this layer. We also have the possibility to cleave the sample either in the inert gas environment or inside the actual liquid. In this case, it is possible to obtain a surface quality that enables atomic resolution. The demands on the equipment increase considerably with the inclusion of a cryostat system, since the cryoliquids introduce increased noise problems. There are now a few commercial low-temperature STMs available, although none of them has an efficient luminescence collection.

Recently, an instrument combining a STM for carrier injection with NSOM detection has been demonstrated.¹⁴⁸ The tip of the STM is an etched fiber, coated with a transparent as well as conductive layer. This works as a conventional STM tip. The tapered fiber is also used for collection of the emission from the sample, like in a NSOM system. The authors' claim a higher collection yield than in conventional STL using this setup. This system has potentially a very high resolution, since it combines the very local excitation of STL with the very local detection of collection-mode NSOM. The STM itself is mounted on a lHe cold stage in UHV, capable of cooling the system to 10 K.

III. APPLICATIONS OF LOW-DIMENSIONAL LUMINESCENCE PROBES

In this review we will concentrate on properties of low-dimensional structures. For properties of bulk material, as

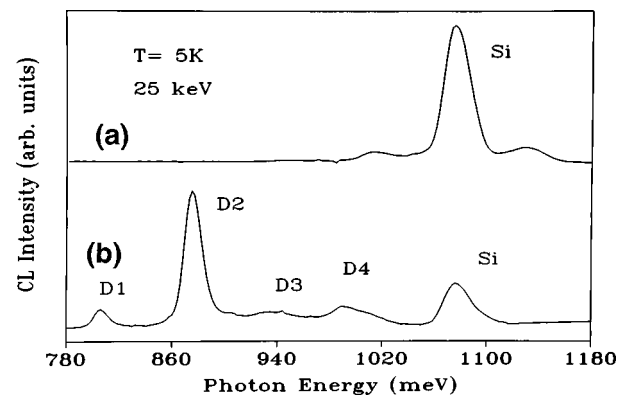


FIG. 20. CL spectra from (a) as-grown Si/Si_xGe_{1-x}/Si epilayer with ~ 2 μm thickness and $x=0.01-0.02$. (b) the same layer after transition-metal (Ni) contamination and subsequent annealing. The (a) spectrum is dominated by a bound exciton in Si and its phonon replica. The (b) spectrum shows the same Si-related features and, in addition, four peaks related to dislocations, D_1 , D_2 , D_3 , and D_4 . The appearance of the D peaks does not seem to depend on the actual transition-metal contamination, as Ni, Cu, and Mn all give rise to the same peaks. (Courtesy of Higgs, Ref. 154.)

well as bulklike epitaxial layers, e.g., dislocations and defects, we recommend two review articles of the CL technique by Petrov¹⁴⁹ and by Yacobi and Holt⁴¹ and a monograph by Yacobi and Holt.⁵⁴ We also recommend an article about CL by Holt,¹⁵⁰ extending to properties of related materials, such as ceramics and minerals. For spatially resolved PL studies of bulk properties we recommend several reviews on the subject.^{103,105,107}

A. Group IV compounds

1. SiGe

As discussed above, the bulk properties of most group IV semiconductors are reviewed in Refs. 41, 54, and 149, and are therefore, left out of this review. We are making an exception for the luminescence properties of epitaxial Si_xGe_{1-x}, an alloy system with up to 4% mismatch of the lattice constants. The exception is made because of recent advances in the quality of the grown layers and for the progress in the field of CL imaging, as well as spectroscopy of thick epitaxial layers.^{151,152} The growth of high-quality strained or partially relaxed epitaxial Si_xGe_{1-x} results in low-temperature luminescence with several well-defined peaks. The typical features are an excitonic line and its phonon replicas. The exciton is either the free exciton or excitons bound to one or more impurities and the phonon replicas are the LA and TO phonons. Additional features in the luminescence spectrum of relaxed or partly relaxed Si_xGe_{1-x} are usually four peaks, related to the misfit dislocations. These peaks are denoted D_1 , D_2 , D_3 , and D_4 , where a higher index corresponds to a higher peak energy position. These features can be enhanced by decorating the dislocations by low concentrations of transition metals, e.g., Cu, Fe, and Ni.¹⁵³ A typical CL spectrum of a partly relaxed layer of decorated Si_xGe_{1-x} is shown in Fig. 20.

Using monochromatic CL imaging in top view, the positions of the misfit dislocations were first determined by imaging the exciton luminescence from the Si substrate, Fig.

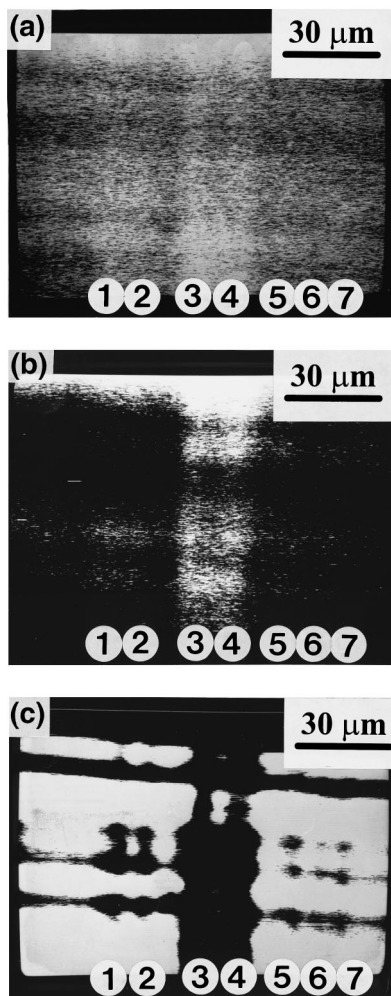


FIG. 21. Monochromatic, top-view CL images of a Si/SiGe/Si epilayer, similar to the one in Fig. 20, but in this case Cu was used as the contamination. (a) Detection of the D_2 peak, (b) detection of the D_4 peak, and (c) detection of a peak originating in the Si substrate. The dark lines in the (c) image correspond to the positions of misfit dislocations. The D_4 emission coincides with the lateral positions of the misfit dislocations, but there is a considerable intensity between the dislocations as well. The image of the D_2 emission reveals a higher intensity at the dislocations, but there is a considerable intensity between the dislocations as well. The interpretation is that D_4 and its phonon replica D_3 are related to the core of the dislocations and that D_2 and D_1 are associated with point defects trapped in the strain field of the dislocations. (Courtesy of Higgs, Ref. 154.)

21(c), where the dislocations showed up as dark lines.^{153,154} The positions of the dark lines were compared with the monochromatic images of each of the D_n peaks. This comparison showed that the D_3 and D_4 emissions originated in the same lateral position as the dislocations, Fig. 21(b), whereas the D_1 and D_2 emissions were found over the whole area, but slightly more intense at the dislocations, Fig. 21(a). It was also established that the D_3 and D_4 peaks originated in the same spatial position. This conclusion was reached since the difference in peak energy position, as well as the intensity ratio of the peaks was found to be constant in spot-mode excitation. The energy difference was identical to that of the TO phonon. D_1 and D_2 , however, appeared independent of each other, as well as of D_4 (and, thereby, D_3 as well) and were found not to be related to each other. The interpretation was that D_1 and D_2 originate in transitions

involving impurities that are trapped in the strain field of the misfit dislocations, whereas D_4 and its phonon replica D_3 are related to transitions at the dislocation core.

Monochromatic CL imaging has also been used to study the formation of dislocations, as well as composition fluctuations in seeded-growth crystals of $\text{Si}_x\text{Ge}_{1-x}$.^{155,156} The crystals were grown by a technique called heteroepitaxial lateral overgrowth, where the SiGe was grown on a Si substrate covered with a SiO_2 mask, with small (typically, 20 by 100 μm^2) growth windows in it. The initial growth of the $\text{Si}_x\text{Ge}_{1-x}$ only took place in the growth window and the thickness of the mask was chosen so that the SiGe was perfectly relaxed when the growth was level with the top of the mask. This provided a perfectly relaxed buffer for further growth, which took place both on top of the buffer, as well as sideways, over the mask. The monochromatic CL imaging revealed a high density of threading dislocations in the entire microcrystallite straight above the window. These threading dislocations were found to originate in the interface between the Si and the SiGe crystals. The density of dislocations was lower, and in some cases virtually dislocation free in the areas that had grown out over the mask. These dislocations originated in the differences in thermal expansion coefficients between the SiGe and Si substrates. These dislocations were introduced when the specimen was cooling down from the growth temperature ($\sim 800^\circ\text{C}$).

Monochromatic CL imaging has also been used to study the quality and behavior of SiGe QWs in between Si barriers,¹⁵⁷ where the dominating spectral features were an exciton line and its TO phonon replica. In thin QWs, the only intensity variations in the images were dark spotlike features, with typical extensions of $\sim 10\text{--}50\ \mu\text{m}$, whereas thicker QWs exhibited dark line defects, corresponding to misfit dislocations. The formation of misfit dislocations will be discussed further in Sec. III B 1 on misfit dislocations and strain in III–V heterostructures. An interesting discovery was made about the dark line defects: The dark line contrast disappeared after annealing, though the misfit dislocations were still present. A possible explanation given in the article was that the nonradiative centers at the dislocations could be passivated by hydrogen during the annealing process.¹⁵⁷

2. Porous Si

One of the major drawbacks of silicon and germanium is the indirect band gap. A way to overcome this has been predicted to be a superlattice of Si and Ge, which should result in a direct band gap (or nearly direct band gap),¹⁵⁸ something that still needs to be experimentally verified. A different approach to obtain luminescence from Si is to use porous silicon.¹⁵⁹ For more details of the properties of porous silicon, we recommend a Ref. 160. Porous silicon is generally formed by anodization of single-crystal Si in a HF solution. This leads to the formation of pores in the surface, where each pore spreads in a tree-rootlike fashion below the surface, to form a porous layer. In PL and electroluminescence (EL) the porous silicon emits light in the visible range from red to blue, depending on the etching conditions. Several explanations of the phenomenon have been suggested, mainly, a quantum size effect versus some sort of hydride or

oxide formation. However, more and more is pointing to the involvement of a quantum size effect in some way.^{160,161} Although the type of emission seems very suitable for CL investigations, very little work has been done on this. This is explained by the decaying luminescence intensity with electron-beam irradiation, where the loss of intensity is irreversible.^{162–164} However, recently it has been demonstrated that it is possible to fabricate porous silicon that is stable even under electron-beam irradiation.¹⁶⁵

In a study of a porous silicon layer, the global PL showed a broad luminescence peak and the surface appeared orange red and quite homogeneous.¹⁶⁶ Top-view CL investigations of the surface revealed brighter spots on a homogeneous background. These spots generally had luminescence with a different energy position than that of the background. The spots appeared white, but also blue, green, and red spots were found. A trend in the emission peak position and the morphology was found, where the blue spots were located at more visibly damaged areas. In different cross-section experiments,^{162,163} the major luminescence intensity was found to originate in the surface layer, rather than from the major part of the porous layer. It was also found that there were two contributions to the luminescence:^{162,167} near the surface, emitting at 660 nm and at the porous silicon-to-Si interface, emitting at 420 nm.

The origin of the CL emission observed from porous silicon is not necessarily the Si nanostructures.¹⁶⁰ The high energy of the primary electrons can induce emission from, e.g., the SiO_x that is generally present on these samples. This was concluded by comparison of known emission bands from SiO_x and by an increase in emission intensity with electron bombardment, interpreted as an increase in the density of active defects in the oxide.¹⁶⁸ A severe reduction in the emission intensity was the result of an oxide removal, linking the emission with the oxide.¹⁶⁹ A quenching of the emission bands (290, 440, and 620 nm) by H-plasma treatment was also observed, which was interpreted as a reduction in the defects in the SiO_x, which give rise to the various emission bands.¹⁷⁰ It has been demonstrated that it is possible to observe emission from the true porous Si, generally around 750 nm.¹⁷¹

STL has also been used to study porous silicon. In the first reports, grains of 1–5 nm in size were found to be the origin of the emission.^{172,173} These were interpreted as individual QWRs of porous silicon. In other STL studies, the optical properties of porous silicon as a function of the surface preparation process was investigated. A correlation between the reduced size of the surface features and the increased blueshift of the luminescence was revealed.^{174–176} One of the problems with STM/STL investigations of the porous silicon layer was the large surface resistivity, approaching levels around 10⁹ Ω/cm. This only allows investigations of very thin layers of porous silicon. Furthermore, it was discovered that the porous silicon surface could be altered by increasing the level of the bias voltage of the STM, thus leading to reproducible local surface modification.¹⁷⁴

However, when operating the STM at bias levels between 0 and 110 V, it was possible to excite EL, increasing monotonously with applied bias. This was found to be in

clear correspondence with the current increase. The porous silicon layer was assumed to consist of small and randomly oriented individual crystalline particles, separated by electrically passivated regions. The luminescence was attributed to e–h-pair recombination in a single Si crystalline particle. In the EL case, thicker *n*-type samples were employed and the generation mechanism proposed was impact ionization, where hot electrons were injected in a specific Si particle.

In more recent studies, wavelength-resolved STL experiments have been performed, providing insight into the spectral signature for individual nanoparticles. In this study,¹⁷⁵ the tip was placed on top of three different sized structures: 3, 6, and 9 nm, respectively. The spectra exhibited a clear blueshift with decreasing feature size. The behavior of porous Si can be modeled, based on the local density approximation.¹⁷⁷ In this case, good agreement between predicted and measured blueshift was obtained, clearly pointing to a quantum confinement effect in the Si particles.¹⁷⁵ Similar results were obtained in a study employing NSOM,¹⁷⁸ although the authors did not succeed in the imaging of individual Si particles. Also, in this case the results pointed towards a quantum confinement model.

B. III–Vs and II–VIs: Quantum wells, superlattices, and barriers

Over the last decade there has been a large interest in studies of heterostructures, mainly QWs in III–V semiconductors. There are mainly three areas that have been investigated by luminescence-based imaging: (i) generation of misfit dislocations in heterostructures; (ii) interface quality of QWs, formation of ML-flat terraces or islands; and (iii) quality of the barrier material in terms of ordering, phase separation, and compositional variations.

1. Misfit dislocations and strain in III–V and II–VI heterostructures

A field where the local probe techniques have been used extensively is for the studies of dislocations in heterostructures, mainly, III–V heterostructures. A necessary prerequisite for these types of studies is the knowledge of the behavior of, and the luminescence properties of, dislocations in general. In an important paper, Petroff *et al.*¹³⁵ concluded that most types of dislocations act as nonradiative centers. A dislocation will, therefore, show up as a dark contrast in an image, either as a dark spot, if the dislocation is oriented perpendicular to the plane of the image, and otherwise as a dark line. It is commonly considered that the core of the dislocation is involved in the nonradiative process. In some cases, especially in II–VI semiconductors, the dislocations can be associated with radiative recombination at a significantly lower energy than the band-edge luminescence.^{179–181} The dislocation-related recombination competes with the band-edge recombination and in an image of the band-edge emission the dislocations appear dark, whereas they appear bright in an image of the dislocation-related emission. Lately, evidence has been presented that the nonradiative processes could be associated with the impurities that are trapped in the strain field surrounding the dislocations. These impurities are generally known as the Cottrell atmosphere.¹⁸²

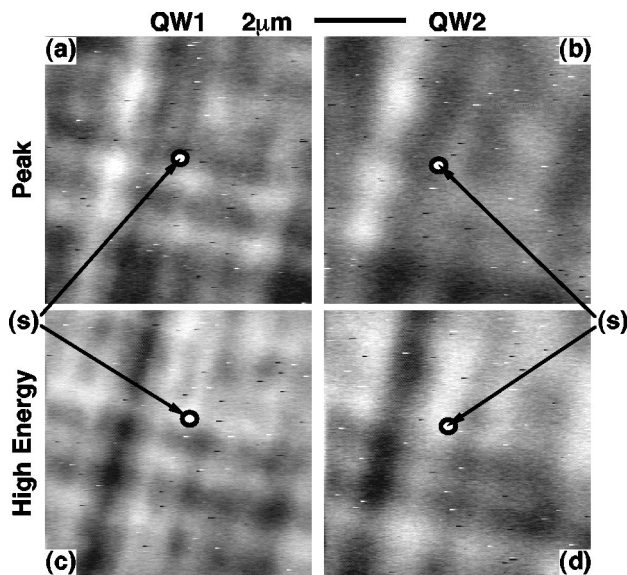


FIG. 22. 25 K monochromatic, top-view CL images of two different GaInAs/InP QWs, QW 1 and QW 2, grown after a thick GaInAs buffer layer containing a network of misfit dislocations. With the detection on the respective average peaks, (a) and (b), a network of dark lines appears, corresponding to the positions of the dislocations in the buffer layer. This was revealed by imaging the buffer layer at a higher acceleration voltage. With detection on the high-energy side of the respective peaks, (c) and (d), the contrast is reversed. The strain field of the misfit dislocations in the buffer layer introduces local variations in the band gap of the QWs, which are responsible for the dark/bright contrast.

The structure, on which this report was based was a partly relaxed superlattice, where a TEM study revealed that the misfit dislocations were confined to the first and last layers. CL imaging clearly showed that the emissions from all the layers in the superlattice (SLS) were reduced and not only in the first and last layers. This was the case for the nominal emission from the SLS, as well as for the impurities-related emission from the SLS.

The Cotrell atmosphere around dislocations mentioned above have been studied by spatially resolved PL.¹⁸³ The sample in this case was Si-doped GaAs grown from melt by the liquid encapsulated Czochralski method. The interest in this study was focused on grown-in dislocations moved by thermal stresses during postgrowth cooling. The Cotrell atmosphere around the grown-in dislocations was found to consist of Si donors on a Ga site. In addition, an increase of the luminescence intensity close to the dislocation was found, in contrast to the argumentation above for a decrease of the intensity caused by the Cotrell atmosphere.¹⁸² This increase of the intensity was interpreted as a decrease of the concentration of deep levels normally associated with dislocation cores.

Care has to be taken in interpreting images of dark line defects to distinguish between the effect of nonradiative recombination and the effect of the strain field of a dislocation.^{184–186} Figure 22 shows a series of CL images, illustrating the two effects. In this case, the sample consisted of three GaInAs QWs, with a low mismatch to InP barriers. The two thinner QWs (QW 1 and QW 2) were grown after the thicker QW (QW 3). The QWs were separated by 30 nm thick InP barriers. The images of QW 3 showed a cross-

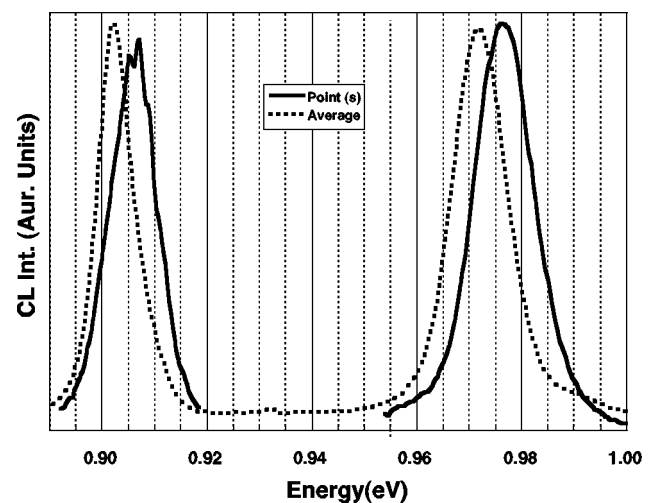


FIG. 23. Average spectrum of the area in Fig. 22 and a point spectrum of point (s), where two orthogonal misfit dislocations in the buffer layer meet. At point (s), where the strain fields of two misfit dislocations interact, the introduced shift is ~ 5 meV, compared to the average spectrum.

hatched pattern of dark lines, identified as a network of dislocations, lying at the interfaces between QW 3 and the barriers. Figure 22 shows four monochromatic top-view images of the same area of the QW, but obtained at different energies. Figures 22(a) and 22(b) show the images corresponding to the average peak energy position of the emission from QW 1 and QW 2, respectively. The images reveal the same cross-hatched pattern as in the image of QW 3. Detecting at ~ 5 meV higher energy, Figs. 22(c) and 22(d) lead to a complete contrast reversal; the same cross-hatched pattern, but with bright lines on a dark background. Furthermore, shifting the detection to the low-energy side resulted in an enhanced contrast of the dark line pattern. This can be attributed to an energy shift introduced by the strain field of the dislocations in QW 3. This was confirmed by recording spectra in spot mode along and beside the lines. Compared to the spectra obtained away from the lines, the spectra along the line were shifted ~ 3 – 5 meV (see Fig. 23). The largest shift was found where two perpendicular lines meet, i.e., where two strain fields were added.

Strained III–V semiconductors often give polarized emission and it is, therefore, possible to obtain a map of the strain field by imaging the degree of polarization across an interesting region such as a dislocation¹⁸⁷ or a laser diode.¹⁸⁸ The strain resolution can be as good as 10^{-5} , i.e., $\Delta a/a = 10^{-5}$ where a is the lattice parameter and Δa is the change in the lattice parameter. A polarization beamsplitter cube and two Si detectors were used to analyze the polarization state of the luminescence signal and the sample was scanned to obtain the images.¹⁸⁷ Figure 24 shows a map of the degree of polarization around an edge dislocation, as well as a simulated image. The effect of the strain field could be observed as far as $15 \mu\text{m}$ away from the dislocation, which agrees very well with the calculated pattern.

A subject of recent interest is the temperature dependence of the contrast of misfit dislocations in QWs. The variations in contrast with temperature have been studied in detail by EBIC,¹⁸⁹ but little attention has been focused on the

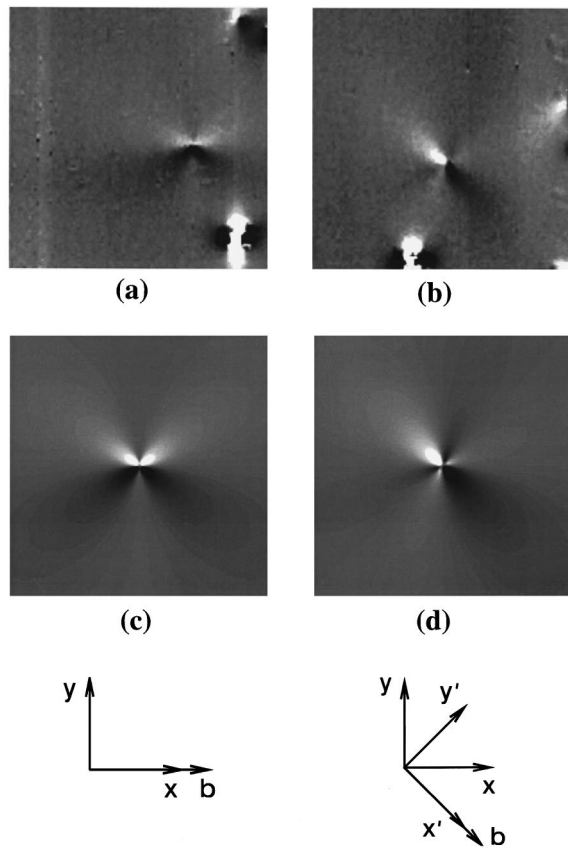


FIG. 24. Strained III–V semiconductors give rise to polarized emission. The figure displays the measured degree of (linear) polarization introduced by an edge dislocation in InP, viewed along the dislocation line. The degree of polarization is defined as $(L_x - L_y)/(L_x + L_y)$, where L_x and L_y are the detected intensities polarized in the x and y directions, respectively. The sample area is $50 \times 50 \mu\text{m}$. The sample has been rotated 45° clockwise between (a) and (b). In (c) and (d) the calculated degree of polarization is shown for the same sample orientations. (Courtesy of Cassidy, Ref. 187.)

CL contrast. It is possible to obtain an activation energy for reemission out of a QW from a plot of the CL intensity from a lattice-matched QW as a function of temperature. In a partly relaxed lattice-mismatched QW, there is an additional effect of the activation of nonradiative centers with temperature. In a study¹⁹⁰ of the temperature dependence of the CL intensity from partly relaxed GaInAs/GaAs MQW structures, the spatially integrated emission displayed two different activation energies. In the low-temperature range (80–150 K), an activation energy of ~ 29 meV was found and in the high-temperature range (150–250 K) it was ~ 79 meV. By comparing monochromatic images, pixel by pixel, in the two temperature ranges, it was possible to determine the local activation energy. It was concluded that the activation energy at the dislocations was higher than the surrounding material in the high-temperature range. The behavior in the high-temperature range was explained by a lower barrier for thermal emission out of the QW in the fully strained areas (without dislocations) than in the relaxed areas (with dislocations). This is caused by the strain influence on the positions of the valence and conduction bands in the QW and in the barrier. In the low-temperature range, the behavior was governed by the barrier thickness. For a thick barrier (40 nm), the activa-

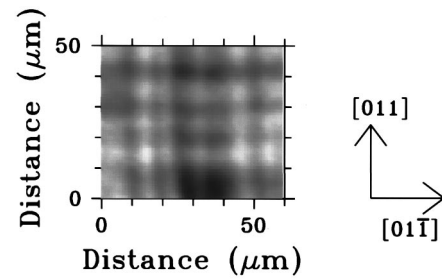


FIG. 25. Imaging PL is very suitable for the (indirect) observation of misfit dislocations in strained layers. The image shows a cross-hatched pattern of misfit dislocations of GaInAs grown on GaAs. The detection energy corresponds to the band gap of the GaInAs and the misfit dislocations cause a decrease of the emission intensity. Very low densities of dislocations can be observed. However, it is not possible to resolve dislocations that are very close ($< 0.5 \mu\text{m}$) to each other. (Courtesy of Giling, Ref. 191.)

tion energy was lower at the dislocations than the surrounding areas, whereas the opposite effect was found for thin barriers (< 12 nm). The behavior in the low-temperature range was explained by thermally activated nonradiative recombination and by a difference in the distribution of defects with barrier thickness.¹⁹⁰

With the knowledge that most misfit dislocations cause dark contrast, related to nonradiative recombination, the effect of dislocations in terms of radiative efficiency and lifetimes can be studied. Spatially resolved PL is, along with CL and x-ray topography, very suitable for the study of very low densities of misfit dislocations.¹⁹¹ Very low densities of misfit dislocations can be detected, where segments $< 20 \mu\text{m}$ long can easily be observed and spacings in excess of $100 \mu\text{m}$ can be observed.¹⁹² The density of these misfit dislocations is initially very low and they are difficult to detect using techniques such as x-ray diffraction¹⁹³ and TEM.¹⁹⁴ However, if the density of dislocations is high, with a spacing of less than $0.5 \mu\text{m}$, the individual dislocation lines cannot be resolved. Using CL or PL imaging, the formation of misfit dislocations in mismatched layers can be studied in a nondestructive way.^{92,191,192,195–199} The onset of the formation of misfit dislocations as a function of layer thickness and mismatch can be determined with high accuracy. Figure 25 shows a PL intensity map of a compressively strained GaInAs layer on GaAs. The misfit dislocation lines reduce the band-edge emission and appear dark in the image. It is clear that very low misfit dislocation densities can be detected. The detection of low dislocation densities is only limited by measurement time, and in principle, one misfit dislocation per wafer can be detected. This technique is very suitable for detailed investigations of the critical thickness (referred to as h_c , i.e., the thickness to which a strained layer may be grown without the formation of misfit dislocations). Techniques where a high density of misfit dislocations is needed tend to overestimate the critical thickness.¹⁹² A comparison between the images of the emission of the layer and the barrier can further help to determine if the misfit dislocations are located uniquely in one of the materials. In combination with etching (a wedge or steps),²⁰⁰ an even better determination of the position in the structure can be made.

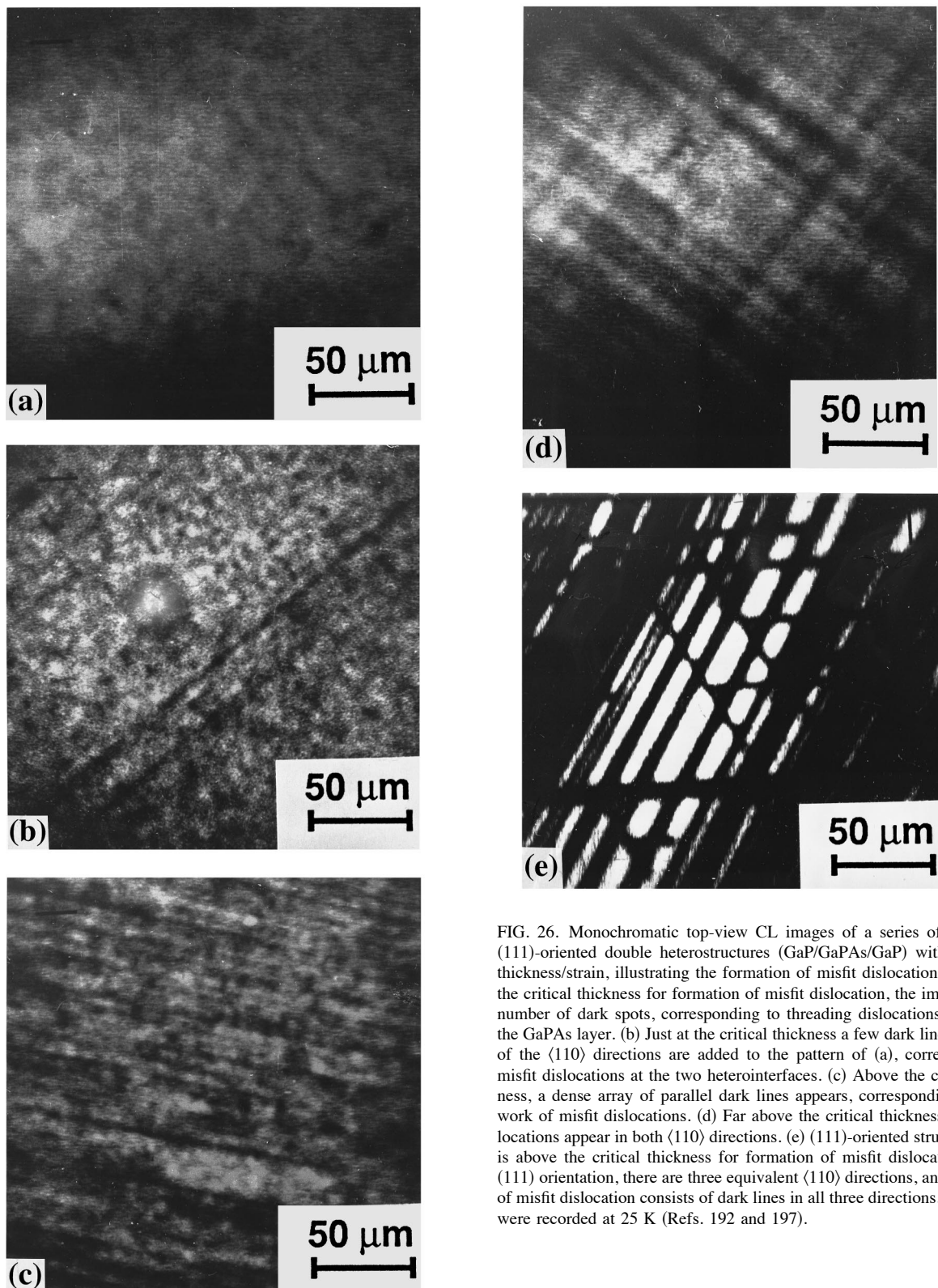


FIG. 26. Monochromatic top-view CL images of a series of (001)- and (111)-oriented double heterostructures (GaP/GaPAs/GaP) with increasing thickness/strain, illustrating the formation of misfit dislocations. (a) Below the critical thickness for formation of misfit dislocation, the images show a number of dark spots, corresponding to threading dislocations penetrating the GaPAs layer. (b) Just at the critical thickness a few dark lines along one of the $\langle 110 \rangle$ directions are added to the pattern of (a), corresponding to misfit dislocations at the two heterointerfaces. (c) Above the critical thickness, a dense array of parallel dark lines appears, corresponding to a network of misfit dislocations. (d) Far above the critical thickness, misfit dislocations appear in both $\langle 110 \rangle$ directions. (e) (111)-oriented structure, which is above the critical thickness for formation of misfit dislocations. In the (111) orientation, there are three equivalent $\langle 110 \rangle$ directions, and the pattern of misfit dislocation consists of dark lines in all three directions. The images were recorded at 25 K (Refs. 192 and 197).

In a series of publications, top-view CL imaging was used to study the formation of misfit dislocations in GaP/GaP_xAs_{1-x}/GaP heterostructures.^{192,197} The dislocation density was studied as a function of layer thickness, as well as composition x . All the monochromatic images of the emission from the different GaP_xAs_{1-x} layers could be divided into four categories: (1) Figure 26(a): for thin layers,

below the critical thickness the images exhibited a pattern of dark spots with a separation of $\sim 20\text{--}50\text{ }\mu\text{m}$, corresponding to threading dislocations penetrating the GaP_xAs_{1-x} layer. (2) Figure 26(b): for layers just above the critical thickness, the images exhibited the same pattern of dark spots, as well as a set of parallel dark stripes along one of the two $\langle 110 \rangle$ directions in the (001) growth plane. These dark stripes were

attributed to single misfit dislocations and are separated by 50–100 μm . (3) Figure 26(c): further above the critical thickness, this network of dark lines developed further and the lines were separated by 1–5 μm , and were attributed to bunching of several misfit dislocations rather than single ones. (4) Figure 26(d): even further above the critical thickness, a pattern of dark lines appeared in the orthogonal $\langle 110 \rangle$ direction, forming a cross-hatched pattern of dark lines. From the analysis of these images, a very accurate determination of the critical thickness could be made over a range of compositions ($0.55 < x < 0.9$). The effect of substrate orientation was also made in this study, where a similar behavior could be observed for growth on (111) GaP substrates: (1) Dark spots of threading dislocations (below the critical thickness). (2) Short dark lines of misfit dislocations in all three $\langle 110 \rangle$ directions in the (111) growth plane (just above the critical thickness). (3) Figure 26(e): a dense pattern of parallel dark lines in all three $\langle 110 \rangle$ directions in the (111) growth plane (well above the critical thickness). It was found that the critical thickness was much higher ($\sim \sqrt{2}$) in the (111) case. In both cases, the values of the critical thicknesses were found to agree with the values predicted by the mechanical equilibrium theory of Matthews and Blakeslee.^{201–203} This theory assumes that the first misfit dislocations originate in preexisting threading dislocations. This theory balances the force exerted on a threading dislocation in the mismatched layer and the energy per unit length (force) for the formation of the misfit dislocation(s). In a single heterostructure, the misfit dislocations are formed at the interface, whereas in a double heterostructure, the misfit dislocations are formed at both interfaces. When the force on the threading dislocation is larger than the force needed to form the misfit dislocation(s), the threading dislocation starts to glide within the layer and the misfit dislocation(s) is formed.

In another study, misfit dislocations were studied in the time-resolved mode of top-view CL imaging. Here, the total lifetime was studied as a function of lateral position.^{90,92,198} A map of the total lifetime of the emission showed that the lifetime around the dislocation lines was reduced from 1.5 to 1.0 ns. This was a result of the additional effect of the competing nonradiative recombination. The total lifetime τ is related to the lifetime of the radiative τ_{rr} and nonradiative τ_{nr} recombinations via: $\tau = 1/(1/\tau_{rr} + 1/\tau_{nr})$ as discussed above. Images were recorded with different time windows, at different time delays after the excitation pulse. Immediately after the excitation pulse (0–3 ns), dark lines appeared at the lateral positions of the misfit dislocations, which is shown in Fig. 27(a). After a further delay (3–4 ns), the center of the dark lines appeared even darker compared to the surrounding areas, shown in Fig. 27(b). This increase in the intensity difference was related to the decreased total lifetime at the dislocation lines. Another observed effect was that the width of the dark lines increased with delay. This was attributed to diffusion of carriers or excitons into the areas of the dislocations.

For high dislocation densities it is impossible to image individual dislocations by spatially resolved PL or CL. In this case it is still possible to get an estimate of the dislocation density by using luminescence lifetime mapping as done

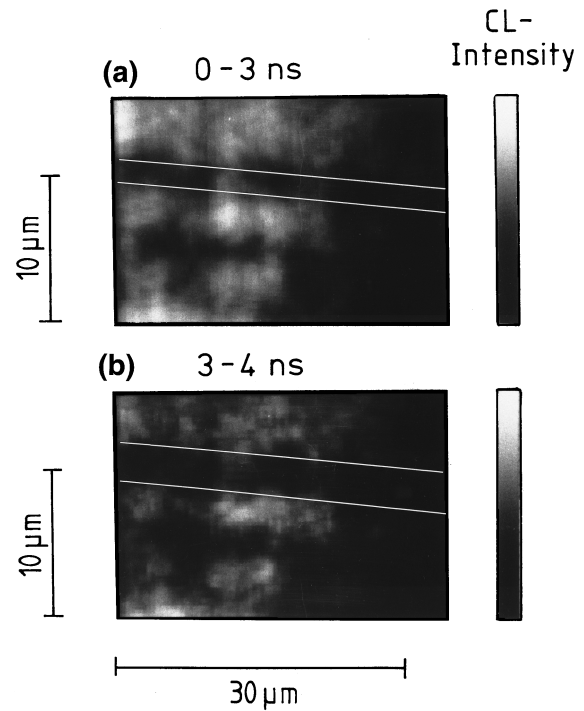


FIG. 27. Pair of time-resolved monochromatic top-view CL images of a GaInAs/GaAs QW containing two parallel misfit dislocations, indicated with two pairs of white lines. (a) Recorded during 0–3 ns, and (b) recorded 3–4 ns after the excitation pulse. In (b), the contrast of the dislocations is higher and the dark bands are wider. This illustrates the diffusion of carriers into the dislocations, where they will recombine mainly nonradiatively. The different scales in the x and the y directions are an effect of sample tilt in the experiment. The sample temperature was 6 K during the experiments. (Courtesy of Grundmann, Ref. 92).

by Nishimura *et al.*²⁰⁴ for the case of GaAs grown on Si. The minority-carrier lifetime is reduced in the presence of dislocations. A simple model for the lifetime is described in Ref. 205. This model assumes that the excess minority-carrier concentration is zero at the dislocation and increases to a steady-state value away from the dislocation. By solving the 1D continuity equation,

$$\frac{\partial n}{\partial t} = D \frac{\partial^2 n}{\partial x^2}, \quad (14)$$

where n is the excess carrier concentration and D is the diffusion constant, it was found that the minority-carrier lifetime is

$$\tau = \frac{L_{\text{diff}}^2}{D}, \quad (15)$$

where L_{diff} is given by

$$\frac{1}{L_{\text{diff}}^2} = \frac{1}{L_0^2} + \frac{\pi^3 N_d}{4}, \quad (16)$$

where L_0 is the diffusion length without any dislocations present and N_d is the dislocation density. It was concluded from a comparison of the lifetime with the etch-pit density that the lifetime adequately represented the dislocation density for a dislocation density higher than 10^{-5} cm^{-2} . It must, however, be remembered that the etch-pit density measures

the surface density of dislocations while the lifetime averages the dislocation density over a finite depth (depending on the diffusion length of the carriers). It is, therefore, difficult to get high accuracy with this method.

A topic that is related to dislocations is the effect of strain on the band gap of the semiconductor or QW. As discussed above, strain will shift the peak energy position of the emission. This is an effect suitable to study by CLWI in top view, as this can be used to make a map of the peak energy position for each pixel of the recorded image. Two examples of this have been reported. One is where a thin foil (110 nm in thickness) containing a GaAs QW embedded between AlGaAs barriers was grafted onto a narrow ridge ($\sim 2 \mu\text{m}$ wide) sticking up ($\sim 1.5 \mu\text{m}$ high) from a Si substrate. The foil stuck to the ridge, as well as to the substrate.²⁰⁶ Using the CLWI technique, maps of the local stress introduced by the bending were made. By comparing with theoretical models of the stresses, it was possible to determine whether the foil was flat against the top of the ridge or if the foil was bending and only touching the edges of the ridge, where both cases were found. Using the approach of grafting, it has been possible to make strain-induced QWRs in the original QW. A second case where CLWI has been used is in a side-view study to visualize the strain relaxation of InP grown on nonplanar Si substrates. The (001) substrates were patterned with V grooves along the [110] direction with, typically, $8 \mu\text{m}$ widths ($\sim 3 \mu\text{m}$ depths) of the grooves, with {111}A sidewalls. Here, it was found that there was a difference in the relaxation process on the two planes involved, observed as a difference in the residual strain.²⁰⁷ After 200 nm of InP on the (001) plane, the peak energy position was found to be nearly constant, corresponding to the position of the emission from a fully relaxed layer. On the {111} planes of the sidewalls of the grooves, the peak started off at a lower-energy position than on the (001) plane and was constant over $1 \mu\text{m}$ and then jumped to the same position as on the (001) plane. This corresponded to a slower relaxation in the grooves, in line with the study of the critical thickness discussed above. This study concluded that the growth of strained InP on grooved substrates resulted in a significant reduction of defects on the planar regions between the grooves, with a quality close to that of homoepitaxial InP. A similar study was made with InP mesas grown by LPE on a masked Si substrate, where CL imaging was used to establish that defect-free stripes as long as $300 \mu\text{m}$ could be produced. Also, in this case CL imaging was used to map the strain in the InP.²⁰⁸

2. Imaging of transport in quantum wells

Recently, Nagamune *et al.*²⁰⁹ have used $\mu\text{-PL}$ in a very novel way, which bridges transport studies and PL. By top-view imaging of the emission of a GaInAs QW to which a point contact was made, they could (indirectly) observe the drift of the holes and the electrons due to applied voltage over the point contact. The effect most directly seen was the hole drift. The electron drift was rather seen as a decrease of the PL due to the high k vector of the current carrying electrons. The interpretation was that electrons with a high k vector are unable to recombine with holes with a low k vec-

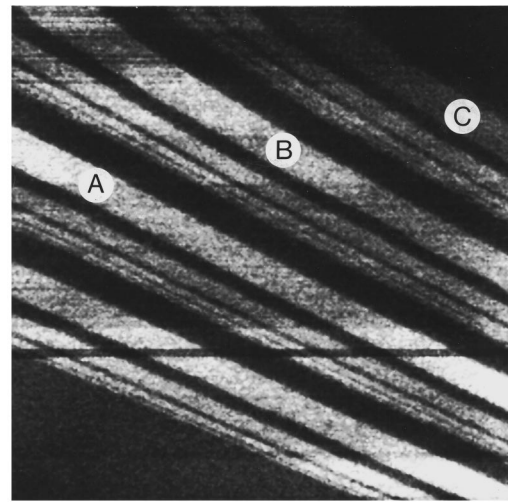


FIG. 28. 300 K, panchromatic, cross-sectional STL image of a MQW structure consisting of four sets of different GaAs/Al_{0.38}Ga_{0.62}As QWs with varying doping density. Each series consists of 2/2, 5/5, 10/10, 20/20, 50/50, and 100/100 nm wide QW/barrier pairs. The labels A, B, and C mark the 100 nm QWs in three of the sets, with the corresponding doping densities $p = 10^{19}$, 3×10^{18} , and 10^{18} cm^{-3} , respectively. As can be seen in the image, the luminescence intensity scales with the doping concentration, reflecting the number of available holes in the valence band. It was possible to image emission from QWs down to 10 nm in width. Scan size $1.73 \times 1.73 \mu\text{m}^2$. (Courtesy of Alvarado, Ref. 71).

tor. As pointed out in Ref. 209, this technique of observing transport through imaging PL holds strong potential for studies of edge states, electron focusing, and even electron interference phenomena.

3. Cross-sectional imaging of quantum wells

It is of fundamental interest to obtain conditions where it is possible to image the low-dimensional structures and to study how the sample geometry influences the possibility to resolve and to image individual structures. An example of the influence of sample geometry was demonstrated in a cross-sectional STL study of a GaAs/Al_{0.33}Ga_{0.67}As QW sample.⁶⁷ In this study, minority carriers were injected into the barrier, 50 nm away from the GaAs QW, and STL from both the thick barrier and the QW was detected. The barrier STL was significantly decreased when moving the tip closer to the QW. Using this approach, it should be possible to determine the diffusion length of the injected carriers.

In a different study, the authors were able to image QWs down to 10 nm widths with cross-sectional STL⁷¹ (see Fig. 28). In a trace of the intensity, it was even possible to detect the signal from the 5 and 2 nm QWs. The sample used was a MBE-grown, GaAs/Al_{0.38}Ga_{0.62}As MQW sample. The obtained contrast was attributed to the lower band bending at the surface in the GaAs QWs, as compared with the AlGaAs barriers. The stronger band bending results in a reduced STL intensity in the barrier region due to a decreased number of available hole states in the valence band, or to an increased nonradiative near-surface recombination process (as argued by Renaud and Alvarado²¹⁰). It was also shown⁷¹ that the STL intensity was related to the p -doping concentration. It was argued that the intensity reflected the number of avail-

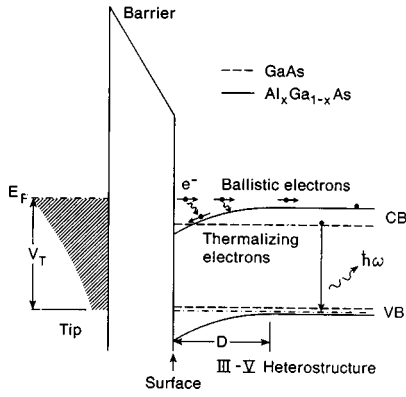


FIG. 29. Schematic view of the tunnel injection process of electrons into GaAs and AlGaAs. The strong band bending of AlGaAs was suggested to induce quantum-mechanical electron reflection and carrier trapping near the surface, therefore, reducing the fraction of electrons injected into the bulk. This means that the luminescence intensity from the barrier will be lower than from the QW for the same carrier injection density. In the semiconductor, the solid (dashed) lines indicate the band edges of the AlGaAs (GaAs) and the dash-dotted line indicates the Fermi level. The figure also shows depletion region D in the semiconductor and applied bias V_t . The band bending δ is the energy position of the conduction-band edge on the surface relative its bulk energy position (δ is negative for AlGaAs in the figure). (Courtesy of Alvarado, Ref. 71.)

able holes at the Γ point in the QW. The onset of luminescence as a function of bias was not found to depend on the quality of the surface.

One of the fundamental parameters of heterostructure semiconductors is how the band edges line up. This is especially important for QWs, where the height of the barriers for the electrons and holes is determined by the band offsets. The threshold voltage for generation of luminescence has been used as a tool for determining the conduction-band profiles in a QW structure.^{71,210} In this case minority carriers (electrons) were injected into the freshly cleaved edge of p -type MBE-grown single and MQW structures. The structures used in this study were AlGaAs or GaAs QWs in between AlGaAs barriers. The authors assumed the surface Fermi level to be pinned due to surface contaminants or other surface perturbations. Using that assumption, it is straightforward to model the injection of electrons into the p -type semiconductor by a simple vacuum tunneling model. The energy (E_e) of the electrons when entering into the semiconductor can then be given by

$$E_e = qV_t - E_{fc} - \delta, \quad (17)$$

where V_t is the tip voltage and E_{fc} is the energy difference between the Fermi level and the conduction-band edge, and δ is the band bending, which is introduced by Fermi-level pinning. The magnitude of δ is the energy of the conduction-band edge on the surface with respect to the bulk conduction-band-edge position. In Fig. 29, $\delta < 0$ for AlGaAs and $\delta = 0$ for GaAs.^{71,211} Figure 29 also shows the depletion region D in the AlGaAs layer. Note that flatbands are assumed for the GaAs layer. In the bias region $E_{fc} + \delta < qV_t$, only a fraction of the injected electrons may ballistically cross the vacuum barrier into the depletion region. In this case, the electrons are confined to the depletion region. However, when $qV_t > E_{fc}$ the electrons can tunnel into the

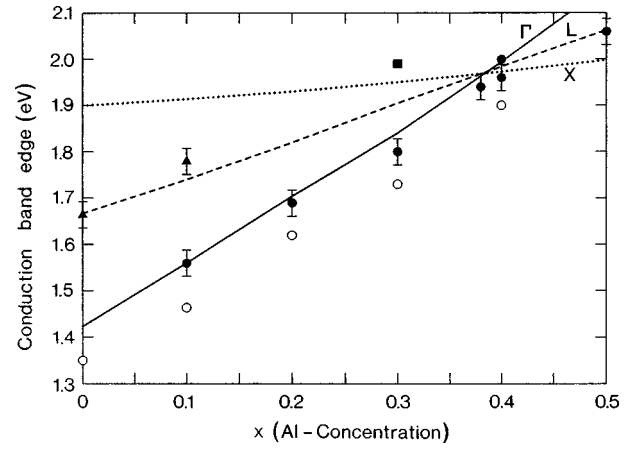


FIG. 30. The position of the conduction band relative the tip Fermi energy, as determined from luminescence intensity versus bias voltage. The figure illustrates the case for a series of Be-doped $\text{Al}_x\text{Ga}_{1-x}\text{As}$ samples, with $p = 10^{19} \text{ cm}^{-3}$. The heavy p doping means that the Fermi energy can be assumed to be equal to the valence-band edge and that the relative position of the conduction band is equal to the band gap. The solid circles represent the bulk (Γ -) band edge. The error bars indicate the scatter in measurements performed on different samples. The solid line is the theoretical Γ -conduction-band-edge energy. The triangles and the square show features related to the injection of electrons into the L and X bands, represented by the dashed and dotted lines. The open circles are measured when the radiative recombination at the surface was strong due to a lower band bending, resulting in a higher density of holes in the valence band available for radiative recombination. (Courtesy of Alvarado, Ref. 210.)

semiconductor and give rise to luminescence due to recombination of carriers behind the depletion region. The carriers will mainly recombine in the QWs since the diffusion length of thermalized electrons is longer than the thickest barrier width in this sample. As shown in Fig. 30, the authors were able to determine the position of the conduction band as a function of Al concentration in the barriers.²¹⁰ This was done by detecting the threshold of the STL, with the tip positioned at the barrier. The heavy doping means that the Fermi level is practically at the valence-band edge, and the threshold is, therefore, identical to the band gap.

In a different experiment, the STM tip was moved across a MQW stack. The profile of the conduction band was traced, including the band bending around the barrier-to-QW interface. Since the measurements were performed in UHV on an *in situ* cleaved surface, it was possible to make a detailed investigation over a long period of time under stable conditions. However, a decrease of surface radiative recombination was found as a function of increased surface contamination. This was argued by the authors to originate in an increased band bending, which caused the nonradiative near-surface processes to become dominant.

4. Monolayer flat islands in quantum wells

The optical quality of a QW is influenced by several factors. One of these factors, the importance of dislocation-free material, has been discussed above. A second very important factor is the state of the interfaces between the barriers and the QW.^{212,213} This is a subject that has been studied extensively over the last decade. The field picked up interest when the growth of SQWs resulted in PL emission

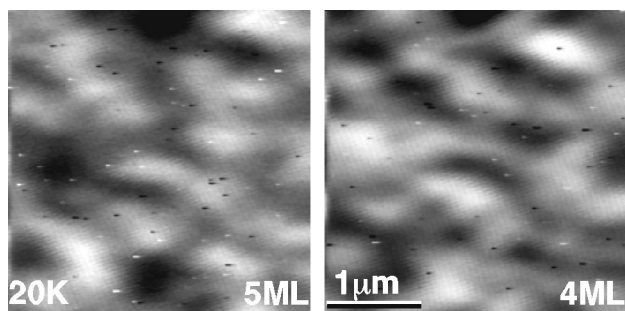


FIG. 31. Pair of monochromatic top-view CL images, obtained by using the two peaks originating in one single GaInAs/InP QW at 20 K. The peaks correspond to excitonic recombination in areas of the QW that are 4 and 5 ML thick. The two images show a clear complementary behavior, where a bright spot in one image corresponds to a dark spot in the other image, and vice versa (Ref. 227).

that was split into several narrow peaks.²¹⁴ The difference in energy of two adjacent peaks corresponded to the difference in confinement energy of a difference in QW width of one ML.²¹⁴ One ML is defined as a complete layer of the semiconductor, e.g., in GaAs the shortest periodicity of sheets containing both Ga and As. In the $\langle 001 \rangle$ directions this is $a/2$ (GaAs: 1 ML = 0.283 nm) and in the $\langle 111 \rangle$ direction this is $a/\sqrt{3}$ (GaAs: 1 ML = 0.326 nm). To exhibit several sharp peaks, the QW must have areas that in some sense are flat and have an in-plane extension of the QW, which is larger than the 20–30 nm diam of the excitons in the QW.²¹³ This is the area of the QW that the exciton samples when it recombines. These areas are generally referred to as ML-flat islands or terraces.^{215,216} The first reports of top-view CL imaging of extended ML-flat islands were published by two group independently in Refs. 217 and 218. Both samples were SQWs of GaAs in between AlGaAs barriers, grown by MBE. Since then, there have been reports on island formation in both MOVPE, as well as LPE-grown QWs. Observations of ML-flat areas have been made in the GaAs/AlGaAs,^{215,219–222} GaInAs/InP,^{216,223,224}

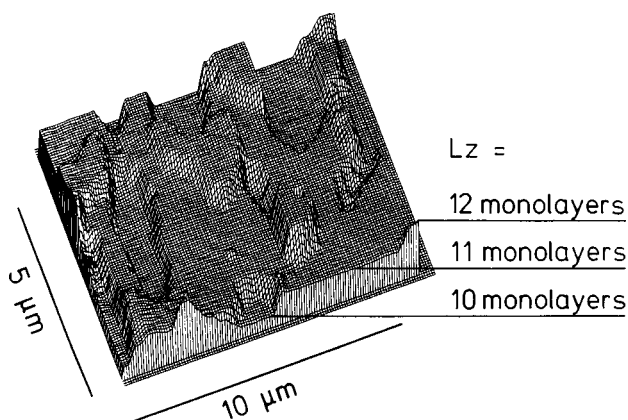


FIG. 32. Thickness map of a GaAs/AlGaAs QW, constructed from a CLWI map of the QW. The image is generated by plotting the peak position of the strongest peak for each pixel (spectrum in the CLWI map). The thickness scale of the plot is in reality the emission wavelength, which corresponds to a thickness. In addition to the extension of the ML-flat terraces, this illustrates how little the wavelength varies within an area of a certain thickness. (Courtesy of J. Christen, Ref. 100.)

GaInAs/GaInPAs,²²⁴ CdSe/ZnSe,²²⁵ and ZnCdSe/ZnSe²²⁶ material systems. An example of images of ML-flat islands is shown in Fig. 31, where the emission from a thin GaInAs/InP SQW was split into two peaks originating in areas of 4 and 5 MLs of GaInAs.²²⁷ In these images the sizes of the islands were found to be 0.5–1 μm , with a high degree of complementary behavior between the two images: a bright area in one image is dark in the second image and vice versa.

Using the technique of CLWI it is possible to get more data out of the images.¹⁰⁰ Extracting the energy position of the most intense emission in each pixel, a map of the quasi-thicknesses can be created, illustrated in Fig. 32. In this case the sample is a single GaAs QW between AlGaAs barriers, grown by MBE. The emission from this QW is split into three peaks, corresponding to 10, 11, and 12 MLs.¹⁰⁰ In this thickness map, the islands are in the range of several μm . It is also worth pointing out the high degree of uniformity of the peak position within a single thickness. This gives a further indication of the high quality of the interfaces in this sample. In this material system an island size of up to 8–10 μm has been reported²¹⁸ for samples grown by MBE with long growth interruptions (120 s).

Recently, there has been a discussion of the nature of these extended islands. Based on results from TEM, PLE, PL, and side-view CL imaging investigations, it has been claimed that even state-of-the-art QWs, which exhibit large islands in top-view CL images, are in some sense uneven.^{228,229} This unevenness, or roughness, exists on all scales, from the atomic level and upwards. The main question is which scale the analysis method senses. In the case of CL imaging, the lower limit of the analysis scale is 0.1 μm . This limit is set by the three factors governing the resolution of the CL imaging: spot size, generation volume, and diffusion, as discussed above. The upper scale is limited by the field of view of the collection system, i.e., 10–500 μm .

The effect of diffusion of the carriers or excitons can play an important role in the contrast formation in the images of QWs. It was argued that the contrast in the images is a function of not only the extension of the ML-flat islands but also affected by diffusion of excitons.²³⁰ Reports on this subject have shown that a magnetic field can be used to increase the spatial resolution in terms of observing smaller features with higher contrast.²³¹ This covers the part of the diffusion of minority carriers. However, there is still a smearing of the details due to diffusion of excitons, which are not restricted by the magnetic field.²³² It has also been demonstrated that the sample temperature can influence the resolution in the CL images of the QWs.²³³ Here, it was shown that on increasing the temperature the peak ratio of the low-to-high-energy peak was reduced due to Boltzmann-like thermalization of excitons from low-to-high-energy areas of the QW. It was argued that at equilibrium between the diffusion of excitons from low-to-high-energy areas and from high-to-low-energy areas, the true island size would be revealed. In recent publications,^{227,234} the influence of the temperature has been studied. It was found that the contrast of the images of the ML-flat islands was reduced with increasing temperature, making the images more diffuse. As the images became more diffuse, the finer features were lost and the apparent

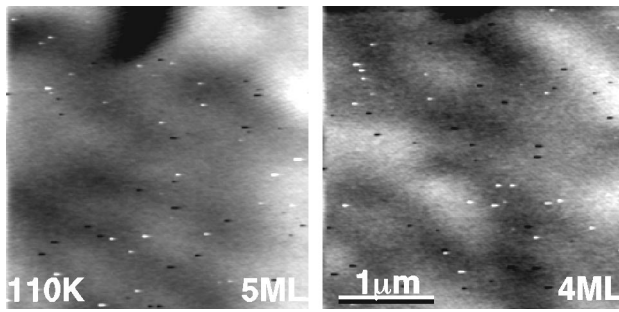


FIG. 33. Pair of monochromatic top-view CL images obtained by using the two peaks originating in one single GaInAs/InP QW at 110 K, recorded from the same area as Fig. 31. It is tempting to interpret the fluctuations in this type of image as a direct map of the thickness fluctuations of the QW. The fluctuations of the same area at 110 K, are on a larger scale than at 20 K. Most of the features are present at both temperatures, but with less contrast at the higher temperature. This is caused by increased mobility of the excitons in the QW. The conclusion is that Fig. 31 is not likely to be a map of the areas of different thicknesses. Instead, the extensions of the intensity variations in the pair of images give an indication of the *upper limit* of the size of the ML-flat islands (Ref. 227).

sizes of the islands increased, as can be seen in Fig. 33. An important observation was that there was a high degree of complementary behavior of the contrast, but not the total intensity: at 20 K the contrast variations were 80% of the intensity, whereas at 110 K this was reduced to 30%. This was illustrated in Ref. 227 by two sets of line scans in the area of Fig. 31, where the line scans were recorded in the temperature range 20–110 K. This led the authors to model the ML-flat islands in terms of agglomerates of microislands, which appear to be extended solid islands due to carrier, or exciton diffusion, mainly from high-energy areas to low-energy areas at low temperatures.²²⁷ Since this, there have been reports that it should be possible to deconvolute the actual extension of the ML-flat islands, using the apparent extension in the images, as well as the absolute contrast of the intensity variations.²³⁵

The high spatial resolution of NSOM at lHe temperature has also been used to investigate the spectral variations in the emission from QWs in top view.²³⁶ The sample used was a

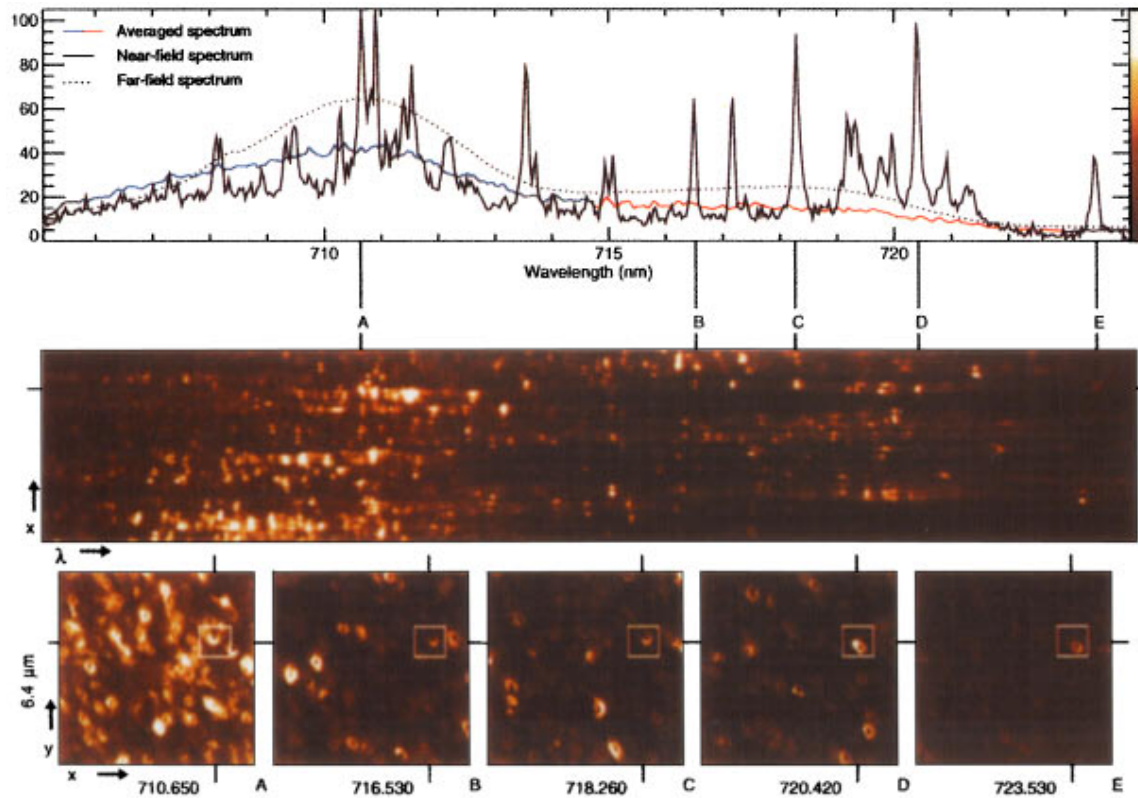


FIG. 34. A complete top-view NSOM study at 2 K of a 2.3 nm thick QW GaAs SQWs in between AlGaAs barriers. The excitation was done below the band gap of the barrier. Far-field excitation and near-field collection was employed in the NSOM (top). The three spectra show a spot-mode near field, far field, and spatially averaged near-field PL. The spot-mode near-field spectrum exhibits a number of sharp emission lines, which when spatially averaged resembles the two broad peaks of the far-field spectrum (middle). A spatially resolved intensity map showing the spatial and spectral variations in the QW emission along a $6.4 \mu\text{m}$ distance (bottom, A–E) Real-space near-field images of a $6.4 \times 6.4 \mu\text{m}^2$ area with detection at the five wavelengths indicated in the spectra above. The luminescence was found to originate from only certain spatially and spectrally discrete sites, which appeared as rings. These rings are caused by the excitation/detection geometry. The fiber tip blocks the total solid angle of the far-field excitation when the luminescence center is directly below the tip, so the center appears dark. The tick marks outside each image indicate the position where the near-field spectrum at the top was recorded. The ticks also indicate the position of the line scans along which the middle spectra were obtained. By comparing images A, C, and D in the bottom part of the figure, it is evident that the different spectral lines shown in the (top) near-field spectrum do not originate from the same site in the sample. However, the spatial origin of emission lines B and C is more or less identical. In the latter case, different quantum states from the same site might be responsible for the luminescence. The emission lines might also originate in sites that are spatially separated, but closer than the limit of the spatial resolution. The overall message from the figure is that the potential energy of the QW is corrugated, leading to a trapping of the diffusing excitons into potential minima, where they recombine radiatively. The intensities of the bright rings reflect the depth of the local minima, as well as the local diffusion lengths to reach the minima. The potential fluctuations were attributed to variations in the QW thickness. (Courtesy of Harris, Ref. 236.)

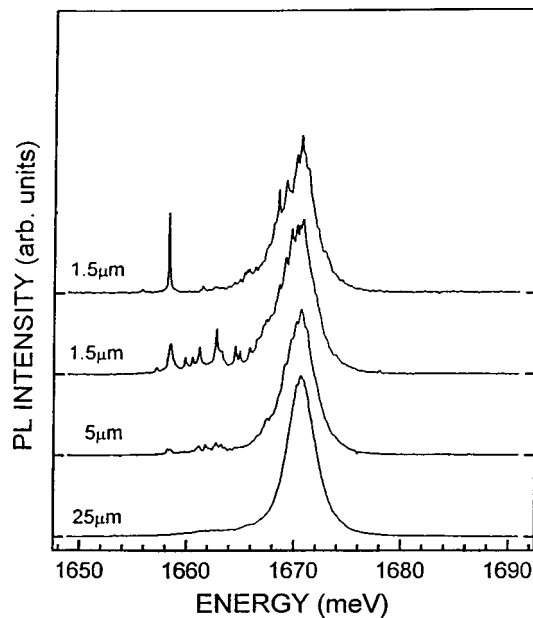


FIG. 35. The PL emission from a QW of GaAs in between AlGaAs barriers using different sizes of the probe. The emission was collected through a microscope and a pinhole in one of the image planes was used to select the probed area. With increasing spatial resolution, the emission from the QW splits into sharp emission lines. These lines vary on a local scale as illustrated by the top two spectra taken with the same spatial resolution, but at different positions on the sample (Ref. 237).

stack of four GaAs QWs of different widths: 2.3, 3.1, 4.5, and 9.0 nm, separated by 25 nm AlGaAs barriers. The sample was excited in the far field and the emission was detected in the near field. In each spatial position, a complete spectrum was recorded. In this way a 3D luminescence map $I(\lambda, x, y)$ was created, making it possible to study, e.g., the luminescence intensity as a function of a line coordinate, exhibiting clear rapid variations with position. As shown in Fig. 34, very narrow emission lines were reported in this study. When all the individual spectra of an area were added, the resulting spectrum was a replica of the far-field spectrum. In this study, it was found that the intensity at different wavelength features changed considerably, but the total density of luminescent states did not change.

The sharp spectral lines were interpreted as individual luminescence centers in the SQWs. Magnetic-field dependence of the near-field spectra suggested that these centers are mainly local thickness variations. These variations were deduced to be on a much smaller scale than the ML-flat areas first reported by CL imaging,^{217,218} but in agreement with the picture described above.

Since the SQWs exhibit local thickness variation and the excitation is near resonant, each locally created exciton only diffuses a short distance before being captured. In the intensity map of Fig. 34 the resulting luminescence appears as rings instead of dots. The reason for this is simply the obscuring tip at the center of the NSOM setup. Since the tip is larger than the diffusion length in the QW, none of the far-field excitation reaches the luminescence center when the tip is positioned above it, which results in the dark spot at the center. The brightness of each ring reflects not only the oscillator strength of the associated energy minimum in the

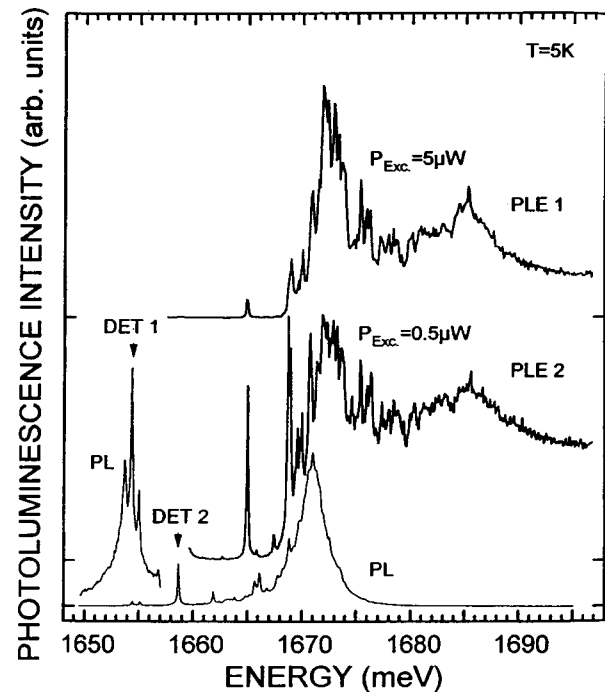


FIG. 36. PL and PLE spectra of excitons localized to interface fluctuations of a QW. The spatial resolution was obtained using an optical microscope. PLE spectra taken at two different detection energies (indicated by DET1 and DET2) are shown. The PLE spectra are quite similar showing that the two peaks DET1 and DET2 share the same excited states (Ref. 239).

QW, but also the local diffusion length. This local diffusion length should scale with the depth and density of the potential-energy minima, as well as with temperature. The depth of the energy minima can be controlled by varying the QW thickness, which was also reported. The wider the QW is, the longer the possible diffusion length. Since the variation in the potential is smaller in the thicker QWs, there is a reduction in the exciton scattering and trapping.²³⁶

In a very interesting and recent development in the field of spatially resolved PL it has been possible to obtain very sharp emission lines from narrow QWs of GaAs in between AlGaAs.²³⁷ The techniques used in these cases have been conventional, using an optical microscope for excitation and detection. These sharp emission lines have been interpreted as emission from laterally localized states within the QW resulting from fluctuations in the width of the QW.²³⁷ It has, however, also been observed that composition fluctuations in the barrier, and not only thickness fluctuations can have significant influence on localized states in QWs.²³⁸ Figure 35 shows the emission spectra from a 3.2 nm thick QW using different spot sizes of the laser excitation. It was observed that there was an emission peak with a linewidth of ~ 100 μeV , for a spot size of 1.5 μm . The emission was found to be localized to areas that are smaller than the experimental spatial resolution. It is interesting to notice that for thicker QWs (5.5 and 7.0 nm) it was not possible to obtain any sharp line emission. This situation is very similar to a corresponding experiment using NSOM.²³⁶

It is very gratifying that narrow-line spectroscopy can be performed using relatively simple techniques (compared with NSOM). It has also been possible to perform excitation

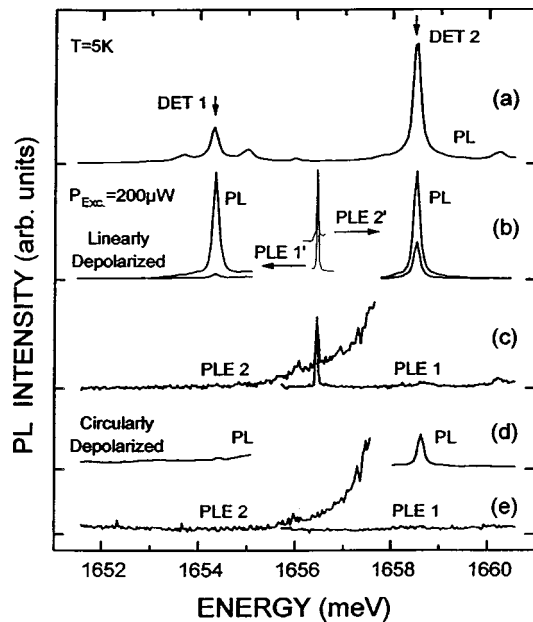


FIG. 37. (a) part of the PL spectrum of Fig. 36 at higher spectral resolution. In the PLE spectra (c) of both peaks a resonance appears at an energy of 1.6564 eV. (b) displays the PL spectra of peaks DET1 and DET2, when exciting in the PLE peak of the resonance (high intensity) and with the excitation detuned 0.2 meV (low intensity). Peak DET1 was attributed to a biexciton and peak DET2 to a single exciton. Traces (d) and (e) show the same experiments but with circularly depolarized light and no resonance is seen. The resonance was attributed to the creation of a biexciton through absorption of two photons (Ref. 239).

spectroscopy on these narrow lines and sharp line PLE spectra are obtained as shown in Fig. 36,^{237,239} exhibiting narrow linewidths down to 0.1 meV. PLE spectra have been obtained both from the exciton in the localized site, as well as from the biexciton in the same localized site.²³⁹ Figure 37 shows the two-photon absorption of the biexciton, as well as the exciton which has a linewidth of 30 μ eV. This linewidth is smaller than that obtained in one-photon absorption. This was attributed to the fact that no excited states are populated in this experiment. Polarization-resolved PLE spectra from excitons localized by interface fluctuations in the GaAs/AlGaAs system have been performed by Gammon *et al.*²⁴⁰ Also, in these experiments the emission linewidths are extremely narrow, down to 32 μ eV. Clear polarization effects were found, which were attributed to the elongated shape of the islands which bind the excitons in the QW. No emissions attributed to biexcitons were reported in this study.

It is not only high-quality flat QWs that have attracted the interest of local luminescence investigations. The growth of thin QWs by LPE results in small, separated, islands with diameters of 10–30 nm. CL imaging of these islands gave the results that islands of the same thickness tend to agglomerate forming areas of equithick islands.²⁴¹ This effect was attributed to ML steps in the substrate. CL imaging has been employed to study the island formation in InAs QWs in between InP barriers, where peaks corresponding to 1–8 ML in thickness were identified.²⁴² It is obvious that the sizes of these islands (10–100 nm in diameter) were well below the resolution that can be obtained with CL imaging. However, some useful information about the thickness distribution and

upper lateral size of the islands could be obtained. The bright areas in the images, corresponding to single islands, tended to overestimate the lateral extensions of these islands, as a normal island size was found to be ~ 10 nm in high-resolution TEM imaging,²⁴³ whereas the normal size of the bright spots was 500 nm, given by the resolution of the measurements. The degree of overlap between bright spots in the images of the different thicknesses was used as an indication of the real separation of the individual islands of various thicknesses, as well as an indication of the upper size limit of the lateral extension of the islands.

Nonuniform QWs have also attracted attention lately, as these provide a possibility for direct growth of QDs, with a high density. These nonuniform QWs are normally highly mismatched, e.g., InAs/InP, InP/GaInP, and GaInAs/GaAs. A long growth interruption applied to the top interface leads to formation of 3D islands (discus, or truncated, or full pyramid shaped) and these islands are significantly thicker than the original flat QW. The interest in these islands is twofold: A study of strain relaxation of highly strained material systems and a quest for the growth conditions, where all the islands are uniform in thickness, as well as in lateral extension: the ideal QDs. The growth of thin heavily strained layers is an efficient way to fabricate QDs and will be discussed further in Sec. III C 6 on Stranski–Krastanow growth of quantum dots.

Other areas of variations in QW thicknesses/compositions have also been investigated, especially by CL. An example is where a mask was placed a 5–10 μ m above the substrate during growth of a QW structure, a technique known as shadow-masked growth.²⁴⁴ This leads to a variation in both composition and thickness of a GaInAsP/InP QW. The deviation from planar growth depends on the width of the mask. This technique was used to incorporate a tapered waveguide at the end of a laser directly in the MOCVD growth step.²⁴⁵ The tapering was intended to be a mode-size converter, to reduce the transverse far field of the lasers. It was found that the full width at half maximum of the far field could be reduced from 47° to 30°, without any significant increase in the threshold current of the device. A reduction down to 21° could be achieved, with the penalty of an increased threshold current. Along the length of the tapering it was found that the band gap of the light guide increased by 130 meV from the laser side to the output facet of the device. The tapering in shadow-masked growth has also been shown in the case of MBE growth.²⁴⁶ Here, the directions of the beams were used to reach under the shadow mask at different parts of the sample. The idea was to displace the edges of a stack of QWs with respect to each other. This was done in order to be able to make electrical connections to single QWs without a complicated post- or pre-growth procedure. In this case, CL was used to determine the behavior of the QWs at the edges of the masks. With a mask displacement of 1 mm above the surface, the CL measurements showed a 800 μ m tapering distance with a linear decrease of the QW thickness towards the mask.

Other areas of intentional variations of the QW composition address, e.g., the incorporation of In in GaInAs/GaAs QWs. An example of this is when growing a QW by MBE

on a substrate containing wide ridges. Typical In migration lengths of up to 20 μm were reported.²⁴⁷

5. Barrier material: Compositional variations, strain, phase separation, and ordering

For the fabrication of high-quality QWs, it is important to have homogeneous and good quality barrier material. Several effects can influence the behavior of the barrier. In tertiary barriers like AlGaAs and, more specifically GaInP, the composition can vary on a very small scale, submicron scale, or on a long scale. The latter is known as phase separation, also known as spinodal decomposition.²⁴⁸ The typical variation in composition is $\pm 2\%$. In the case of $\text{Ga}_{0.5}\text{In}_{0.5}\text{P}$ this leads to a band-gap variation of ~ 10 meV, including the effect of strain.²⁴⁹ More dramatic effects can be introduced if the layer separates in an ordered fashion. This ordering means that alternate group III layers are more or less pure, giving e.g., a $\cdots\text{Ga}-\text{P}-\text{In}-\text{P}-\text{Ga}\cdots$ stacking sequence in GaInP.²⁵⁰ Depending on which plane is involved the ordering is denoted in a different way. If the alternating layers are on $\{111\}$ planes, the ordering is denoted CuPt, which is the most common type of ordering in III-V semiconductors.²⁵⁰ A perfect CuPt ordering can lower the band gap by as much as 100 meV in GaInP.²⁵¹ For the fabrication of high-quality QWs it is important to have a homogeneous barrier, as a shift in the barrier height will introduce a shift in the confinement energy. Because of the larger variation in energy, it is especially important to control the ordering/disordering of the barrier.

GaInP grown on a planar GaAs substrate can exhibit a pattern on a submicron scale, which shows complementary behavior between the high- and low-energy sides of the averaged CL peak. This behavior has been shown to be related to phase separation.²⁴⁹ This was concluded from simultaneous EDAX and CL measurements on thinned epilayers, using a STEM equipped with a CL system. The influence of misorientation of substrates on ordering, has also attracted the attention of local probes. A study of ordered areas, using TEM, AFM, and monochromatic CL imaging, showed a correlation between the pattern of steps on the surface, direction of ordering, and orientation of bright bands in the CL images.²⁵² The authors also found nonradiative recombination at the boundaries between ordered and disordered domains. There have been some reports on imaging of ordered and disordered phases of GaInP, grown on a nonplanar substrate.^{251,253} This resulted in ordered areas, which had an extension of several μm . This has been confirmed by both dark-field TEM imaging using the superlattice spot of the diffraction and micro-Raman spectroscopy. In this set of samples, an averaged PL spectrum showed three distinct peaks, separated by ~ 100 meV. In the CL study, the low-energy peak was found to originate in the ordered region of the GaInP layer, whereas the two high-energy peaks originated in the disordered regions. Similar results were found in a NSOM study of the same sample.²⁵⁴ This confirms the link between ordering and the reduction in the band gap.

There are a few recent articles on CL investigations of II-VI epilayers. An example is where thick mismatched layers have been studied by depth-resolved CL, where the strain

has been monitored as a function of depth. In an 8 μm thick ZnSe layer both light-hole and heavy-hole exciton peaks were observed.²⁵⁵ With increasing penetration depth, the split between the peaks increased, which was attributed to a higher degree of strain nearer the interface to the GaAs substrate. Most of the investigations, however, address the inhomogeneous incorporation of impurities in the epilayers. In one study of incorporation of impurities, it was found that illumination of a sample during MBE growth could reduce the donor concentration by one order of magnitude, as well as reduce the growth rate of ZnSe.²⁵⁶ These effects were shown by illuminating the sample with a first- and second-order optical grating during growth. This resulted in height variations due to the first-order grating, with a $\sim 4\%$ reduction of the growth rate. A minor effect was also found by the second-order grating. A more interesting effect was that both gratings resulted in a much lower iodine incorporation. This was identified in the CL images of the donor-acceptor-pair luminescence, where this was observed as dark bands of reduced intensity. The authors predicted that in the ultimate limit, structures with submicron dimensions could be produced.

The growth on patterned substrates does not only introduce ordering effects, but can also influence the composition of the grown layers. An important application of this is AlGaAs grown on a substrate with V grooves.²⁵⁷ This is a technique which is used for the fabrication of QWRs and will be discussed further in Sec. III C 5 on growth in grooves and recesses, a study of the compositional variations of the AlGaAs layers grown on V grooves with a planar section at the bottom. (001)-oriented GaAs substrates were used with near $\{111\}$ sides of the V grooves. The width of the groove was varied from 10 to 53 μm . With a 10 μm etch depth, the planar section in the groove was varied between 0 and 40 μm . The LP-MOCVD growth of an AlGaAs layer with 28% Al, resulted in a structure defined by (001) planar areas on the unetched areas, as well as at the bottom of the grooves for the wider grooves. In the wider grooves, the area between the (100) bottom and the $\{111\}$ sides developed a $\{112\}$ plane, resulting in four different areas of the AlGaAs layer. Using spectrally resolved PL imaging, with a 1 μm spatial resolution, it was found that the $\{111\}$ layers had a slightly higher Al content of 30%. In the wider grooves the $\{112\}$ had a lower Al content of 20%. The Al content of the planar area at the bottom of the grooves depended on the width and the position in this area. Close to the sidewalls the Al content was $\sim 25\%$ and the value increased towards the center of the groove. This increase was found to approach the nominal value for the wider grooves. The lower Al content in the bottom of the V grooves is consistent with general observations of the GaAs/AlGaAs QWR structures grown on V-grooved substrates, where a sheet of Ga-rich material is found at the bottom of the groove. This forms a vertical QW (VQW), which has several points of interest. It can be used as a simple way of forming a QW, where no complicated switching sequence of the growth apparatus is needed. Lasers, based on this type of structure have been demonstrated.²⁵⁸ The role of the VQWs in the capture of carriers and subsequent feeding into QWRs grown on V-

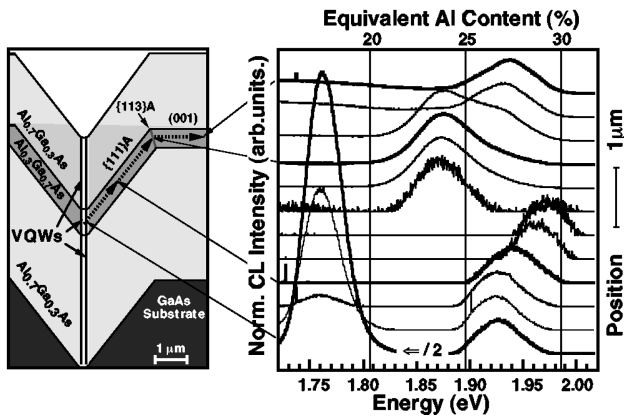


FIG. 38. (Left) schematic of a triple-layer VQW structure of AlGaAs grown on a V-grooved GaAs substrate. The active layer is the 300 nm thin $\text{Al}_{0.3}\text{Ga}_{0.7}\text{As}$, which is surrounded by thick layers of $\text{Al}_{0.7}\text{Ga}_{0.3}\text{As}$. During growth there is an accumulation of Ga at the bottom of the groove. This forms a sheet of Ga-rich material, which is a VQW. (Right) series of spot-mode spectra along the $\text{Al}_{0.3}\text{Ga}_{0.7}\text{As}$ layer from the bottom of the groove to the planar region. The arrows indicate where the spectra were recorded. At the bottom of the groove, the spectrum is dominated by the VQW peak with an energy position corresponding to $\sim 18\%$ Al (uncorrected for any quantum confinement effect). Along the quasi- $\{111\}$ side the Al content increases from $\sim 27\%$ near the VQW to $\sim 30\%$ on the verge of the groove. Between the $\{111\}$ side of the groove and the planar region, the structure is determined by a short $\{113\}$ area. This has a lower Al content of $\sim 23\%$, and the emission dominates the spectra in this region. Outside the groove, the Al content is $\sim 27.5\%$.

grooved substrates has been discussed from a theoretical view point²⁵⁹ and it has been demonstrated experimentally.²⁶⁰

The variations of the Al content of AlGaAs layers, grown under different conditions on a V-grooved, GaAs substrate have been the subject of side-view CL investigations, where a higher spatial resolution was achieved.^{261–263} In one study²⁶¹ it was concluded that LP-MOCVD growth of $\text{Al}_{0.33}\text{Ga}_{0.67}\text{As}$ on a V-grooved substrate resulted in four regions of distinctly different Al content, resulting in four different emission peaks. The different regions of the $\text{Al}_x\text{Ga}_{1-x}\text{As}$ layer that were identified were: (i) the $\{111\}$ sidewalls of the groove with $x=0.33$; (ii) the $\{001\}$ regions between the grooves with $x=0.31$; (iii) the region where the $\{001\}$ and $\{111\}$ facets meet, identified as a $\{225\}$ facet, with $x=0.27$; and finally, (iv) the VQW at the bottom of the groove with $x=0.23$.

In a more detailed study, the compositional variations of an AlGaAs layer grown on a V-grooved substrate were studied. CL spectra were recorded in spot mode and the composition was determined from the energy position of the emission. These spot-mode spectra and a schematic picture of the structure are shown in Fig. 38. The structure consisted of a 300 nm, 30% Al layer in between thicker layers of 70% Al grown by AP-MOCVD. The spot-mode spectra were recorded in side view with the spot moving from the bottom of the groove to the planar region, along the 30% Al layer. At the bottom of the groove the emission is dominated by the VQW, which has an Al content of $\sim 18\%$ Al. This is uncorrected for any quantum confinement effect in the VQW, so the actual Al content is probably slightly lower. Along the 2 μm of the $\{111\}$ A side of the groove, the Al content gradu-

ally increases from $\sim 27\%$ to $\sim 30\%$. At the verge of the groove, the structure is determined by a short $\{113\}$ plane, which has a lower Al content of $\sim 23\%$. Due to its lower-energy position, this region dominates the spectra in this area. Finally, on the planar region next to the groove the layer has a 27.5% Al content.

The situation is even more complicated, for growth of AlGaAs in a tetrahedral-shaped recess. The substrate is generally a $\{111\}$ GaAs substrate with $\{111\}$ sidewalls. The growth leads to the formation of several types of facets with different Al incorporation and intersections between these facets forming VQW-type features.^{264,265} Also, in this case CL has been useful in identifying the various facets.²⁶⁵

C. Quantum wires and quantum dots

1. Etched quantum wires and quantum dots

An area where local probe techniques are extremely useful is studies of nanostructures, such as QWRs and QDs. The CL technique really comes to the fore in this field. Many of the fabrication techniques only cover a very limited area of the sample (typically, $100 \times 100 \mu\text{m}^2$) and, e.g., in CL, the normal SE mode of the SEM can be used to locate the structures. Once a structure of limited area coverage has been located it can be studied spectroscopically. The ability to observe the topography (SE mode) and the luminescence (CL mode) simultaneously can identify several causes for irregularities in the emission from these structures. An example is the case of etched out and free-standing QD columns. Even for high-quality QDs with an area coverage of $<1\%$ the intensity from an area containing the QDs is reduced by roughly the same amount, as compared to the original QW. This means that an unintentional mesa of a few μm^2 in size can dominate the luminescence properties in an area of several thousands of μm^2 . CL imaging can easily reveal these mesas and turn this from a negative effect to a positive effect. These unintentional mesas can be used as references for peak energy positions and intensity.²⁶⁶ Unless the original QW structure is very homogeneous in terms of peak positions and intensity, this form of reference is more favorable than a different piece of the original QW structure. There could, e.g., be a difference in peak position between the two pieces, already before processing.

The ability to find the patterns of etched QWRs and QDs using the SEM mode and the possibility to control the probe current means that it is ideal for assessing the intensity of the etched structures compared with the unprocessed material. The most common way to describe this is to normalize the intensity to the area coverage of the remaining material.^{267,268} This has a tendency to overestimate the intensity from the QWRs/QDs. In the case where the original QW is grown on a thick layer of barrier material, the excitation density can be significantly higher in the QWRs/QDs than in the original QW at the same excitation density. Most of the carriers generated in the barriers diffuse into the QWs, in fact, most of the excitation of the QW emission is generated in the barriers. Removing a large portion of the upper barrier and the QW, but not the lower barrier, leaves a proportionally larger barrier volume to supply the remaining QW ma-

terial with excitation. This leads to a higher excitation density in the QWRs/QDs as compared with the reference QWs, for the same global excitation density. A simple way to avoid this is to have a thick layer of the QW material just below the actual QW(s). This provides a diffusion sink, which prevents most of the diffusion from the substrate/lower barrier into the QWRs/QDs. This has been demonstrated in a 450 nm pitch (200 nm width) etched QWR structure with a diffusion sink in the form of a 50 nm thick GaInAs layer below the GaInAs/InP QW structure. In the CL images of the structure, the single QWRs were clearly resolved.²⁶⁹

In several studies of etch damages in QWR/QD structures, the normalized intensity was studied as a function of width of the structures. Using these data it was possible to determine the width, for which the intensity of the structures vanished. From this, the damaged, or dead, layer on the sides of the structures could be determined for various material systems and etch processes. A typical extension of the etch damage for reactive ion etching of GaAs/AlGaAs QDs using BCl_3/Ar was found to be 20–50 nm,²⁶⁷ and it was also concluded that subsequent overgrowth of the etched structures could result in a reduction of this effect.²⁷⁰ In a series of more recent articles,^{271,272} CL has been used to study the influence of etching away a part of the top layer of a MQW stack. The effect of etching time (etch depth), and thereby the penetration depth of the damage was studied. In these studies, the influence of the QW thickness was also studied, by reversing the order of the QWs in the stack. The authors' conclusion from the studies was that thinner QWs are more sensitive to etch damage and that the etch damage can penetrate up to 80–100 nm. It was speculated that the thinner QW was more sensitive to damage as the wave function of a thinner QW penetrates further into the barriers than that of a thicker QW. This behavior was further taken as an indication that the damage is spread in the barrier, as well as in the QWs, rather than being confined to the QW. One problem in determining the damage introduced by etching is to obtain several parameters on the same sample. Recently, a way to determine both the diffusion length and the surface recombination velocity at the etched sidewalls of a sample containing a GaAs/AlGaAs QW structure was used.⁸³ The technique is based on the method for determining the diffusion length, introduced by Zarem and co-workers^{80,81} as discussed above. Instead of a single metal layer, two layers were used. The first thin layer protects the sample from the etching process and the second layer acts as the mask. The intensity profile of a line scan perpendicular to the edge of the mask is recorded. From this profile, the diffusion length can be determined. If the first metal layer is missed out, the area outside the mask will be etched. By making the same line scan across the edge of the mask, the recorded intensity profile will contain the additional effect of the sidewall surface recombination velocity. A result of this study was that the dead layer was determined to be 30 nm.²⁷³ $\mu\text{-PL}$ has also been used for the investigation of etched out and subsequently overgrown QDs.²⁷⁴ This investigation showed that the PL efficiency of these disks was unaffected by the fabrication process, for disk diameters down to 0.1 μm .

The introduction of etch damage and the reduction of the damage by surface treatment is a very complicated subject. This has been shown in an extensive study of the effect of several parameters on the CL intensity from bulk GaAs with the appropriate title: “*The bewildering surface attributes of gallium arsenide: A cathodoluminescence perspective.*”²⁷⁵ This study showed that there are many effects to take into account. One of the major influences in the CL intensity was found to be the surface states, where an increase in the CL intensity was introduced by surface passivation using ammonium sulphide. It was also observed that the CL intensity increased under e-beam exposure, which was related to charging effects rather than the deposition of a carbon-based film generally deposited in a SEM during operation. It was also observed that the band bending at the surface, and more importantly, the reduction of the band bending also influences the CL intensity, where it was concluded that CL investigations are generally performed under nonflatband conditions. A high excitation density can lead to flatband conditions. A superlinear intensity response can be the consequence of going from nonflatband to flatband conditions.

One of the signatures of a QWR as compared to a wider QW stripe is the shift of the emission to a higher energy (blueshift) due to the quantization in the QW plane perpendicular to the QWRs. Over the last decade there have been numerous reports of unexpectedly large and unexplained blueshifts in etched out QWR/QD structures.^{269,276,277} A possible explanation for this was shown in a study of etched QWRs where the original QW structure consisted of three QWs of different thicknesses.²⁷⁸ The most interesting feature was that the thinnest QW exhibited a clear (55 meV) split between the peaks of the 2 and 3 ML thick areas of the QW, the only one of the QWs in the structure that exhibited this clear ML split of the emission. In a 100 nm wide QWR the two thicker QWs exhibited different blueshifts, 4 and 8 meV, respectively, the smaller shift for the thicker QW. The thinnest QW did not show any blueshift, but rather a change in the intensity ratio between the peaks, in favor of the high-energy peak in the QWR: 1:5 in the QW and 3:1 in the mesas. This effect was attributed to a reduced diffusion of excitons from high-energy to low-energy areas of the QW in the QWRs. The reduced diffusion from high-to-low-energy areas rather than an increased diffusion in the opposite direction was determined from a simultaneous PL and absorption measurement on the same type of QW structures.²⁷⁹ The conclusion of the CL study was that a reduced diffusion of excitons resulted in a ratio shift in the CL from the QWR. If the peaks were resolved, this could be observed directly, and if the peaks were not resolved, this could be observed as a blueshift. The magnitude of the blueshift was related to the difference in confinement energy of a thickness difference of 1 ML in the QW.

2. Ion beam and heat induced quantum wires and quantum dots

One of the first types of structures to be studied by a local probe technique was fabricated by ion implantation and subsequent annealing, starting from a substrate containing a QW.^{280,281} This makes the implanted areas of the QW and

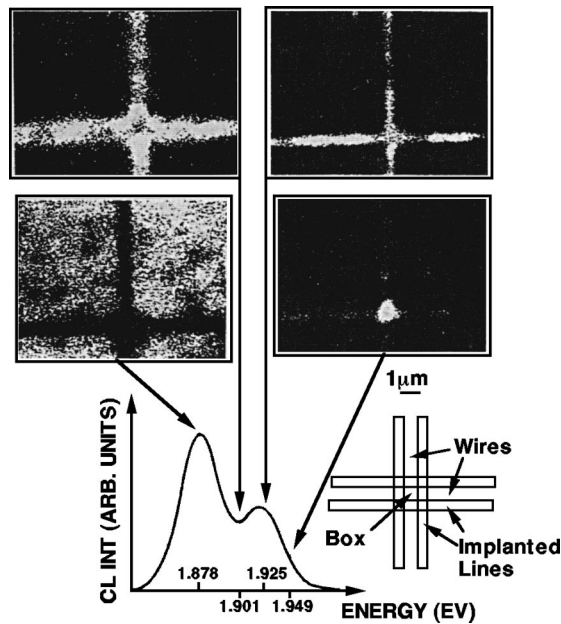


FIG. 39. Illustration of the technique of FIB-induced QWRs and QDs starting from an existing GaAs/AlGaAs QW. A cross was defined by the FIB using two pairs of parallel lines. The area outside the cross represents the unprocessed QW with an emission peak at 1.878 eV, at 14 K. The area between the parallel lines is blueshifted to 1.925 eV, corresponding to the formation of a QWR. At the point where the two sets of lines meet, the emission is further blueshifted to 1.949 eV. This is illustrated by a series of monochromatic top-view CL images detected at four different energies, indicated in the average spectrum. The figure also illustrates the principle of the implantation lines. (Courtesy of Merz, Ref. 283.)

the barriers interdiffuse to increase the band gap in the QW. This introduces a potential barrier in the QW plane, which can lead to the formation of QWRs or QDs. The ions can be implanted with either a simple mask to stop the ions from penetrating the QW, or by writing directly on the sample by a focused ion beam.^{138,282} An example of the latter is shown in Fig. 39. The original QW was patterned with a cross of two parallel lines.²⁸³ The two lines were separated by 200 nm. The monochromatic CL images revealed three areas: (i) the original QW emission outside the cross; (ii) an emission at slightly higher energy between the parallel lines of the cross, where QWRs were formed, and finally, (iii) in the center of the cross, the emission was shifted to even higher energy, where a QD was formed. These structures together with the etched structures are fairly simple structures to analyze, as they have mainly one emission peak, or one peak per QW, and the original structure is well characterized, so the original peak position is well known.

In a slightly different fabrication technique, the local interdiffusion in the QW is caused by local heating of a focused laser beam, instead of an ion beam. QDs fabricated with this technique (in the GaAs/AlGaAs system) have been studied by μ -PL using a microscope for light collection. Figure 40 shows the PL from a QD with a lateral size of 450 nm, although the size cannot be specified by a single number since the interdiffusion causes a slowly varying lateral potential. In contrast to the original QW, the PL emission consists of several lines due to additional the lateral quantization. It was concluded in Ref. 284 that the relaxation of the

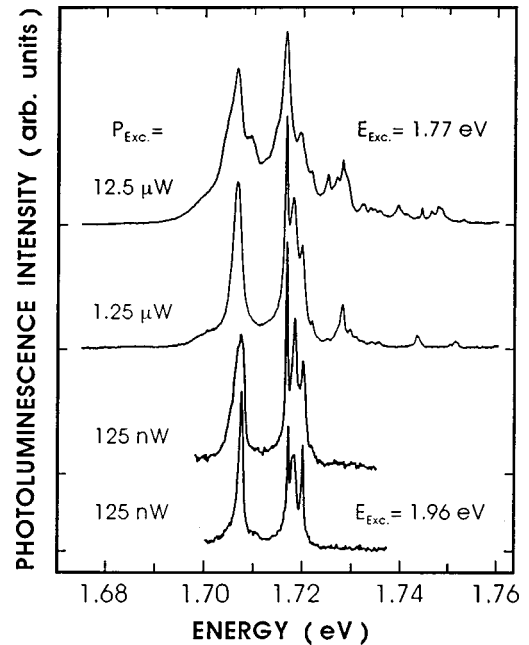


FIG. 40. PL from one single QD fabricated with a technique based on heat-induced interdiffusion of a QW. The emission consists of several sharp lines in contrast to the original QW, which is dominated by a single emission peak. At increasing excitation power density the lines broaden and new peaks appear, particularly at higher energy. The appearance of several peaks in emission was attributed to a reduced relaxation rate of electrons in the QD (Ref. 284).

electron to the ground state was reduced due to a reduced phonon scattering rate. At increasing excitation power intensity the lines broaden due to the presence of several excitons in the QD simultaneously.

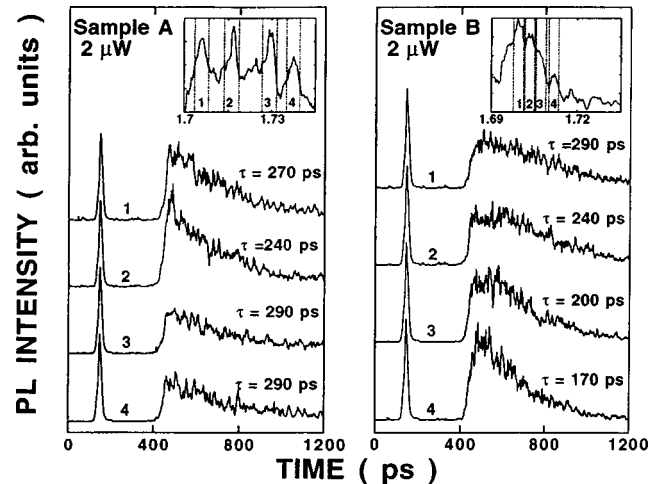


FIG. 41. The inset panels show the emission spectra of two single QDs, having level spacings of 10 meV (sample A) and 4 meV (sample B). The time dependences of the emission from the ground and excited states are shown. For sample A, the higher-energy states have almost the same or even slower time decay than the low-energy states, whereas, for sample B, the higher-energy states have a faster time decay than the low-energy states. From the rapid onset of the emission from the ground state it was concluded that the relaxation of the carrier is caused by Coulomb scattering. The authors argued that the difference in decay times between the two samples could be explained in terms of screening of the Coulomb scattering (Ref. 285).

Bockelmann *et al.*²⁸⁵ have studied the time evolution of individual QDs fabricated by laser-induced interdiffusion in the GaAs/AlGaAs system. Figure 41 shows the temporal evolution of emission lines for two QDs (samples A and B). The energy spacing of the states in sample A was 10 meV and 4 meV in sample B. For sample A, it was observed that the decay time for the high-energy states was identical or slower than for the low-energy states. For sample B, the opposite behavior was seen, i.e., the high-energy states had a faster decay time than the low-energy states. The rapid onset of the emission from the lower laying states demonstrates that the relaxation of carriers in QDs is not limited by phonon scattering, as suggested in Ref. 286. Instead, it was concluded that the carrier relaxation proceeds via Coulomb scattering.²⁸⁷ The difference in temporal behavior of the excited states was discussed in terms of screening of the Coulomb scattering.²⁸⁵

Similar studies have also been performed at different temperatures²⁸⁸ and with applied magnetic fields.^{289,290} In the magnetic-field experiments a diamagnetic shift was observed, which was interpreted to be due to the excitonic origin of the PL.²⁹⁰ The authors also observed a Zeeman splitting of the lines in PL and PLE.²⁸⁹

3. Growth on vicinal substrates and cleaved edges

One of the first methods for direct growth of low-dimensional structures that was investigated was growth on vicinal surfaces.²⁹¹ Initially, the interest was focused on the self-ordering of a vertical SLS while growing thick layers of AlGaAs. This later developed into formation of QWRs when the angle of the SLS was intentionally varied during growth. This led to the formation of a serpentine SLS, forming a SLS of QWRs.²⁹² It is not only lattice-matched AlGaAs that can form vertical SLS and serpentine SLS. Growing a sequence of (InP)₂/(GaP)₂ bilayers in between Ga_{0.51}In_{0.49}P barriers on a vicinal GaAs substrate (3° off [001]), leads to a strain-induced lateral ordering.²⁹³ The composition varies up to 30% in the bilayer region. The domains are ~40 nm wide by ~20 nm in height (for MOVPE growth) and ~10 nm by ~5 nm (for MBE growth). These domains are extended along the direction of the steps in the substrate. These QWRs exhibit polarization anisotropy in the CL emission, as expected.^{293–296} They also show band-filling effects with increasing excitation density and carrier relaxation effects in time-resolved measurements.^{294–296}

Lately, the growth on vicinal surfaces has taken a new direction, where the growth of a thin QW shows thickness variations oriented along the steps on the surface of the vicinal surface.^{297–299} The alignment of thickness variations along steps on the surface is a well-known phenomenon, and this has been demonstrated on near-(001)-oriented substrates in the past.^{231,300} However, it was only with the growth on (110) and higher index surfaces that the thickness variations of vicinal QWs were discussed in the context of QWRs. The main reason for the formation of the QWRs is a difference between the growth of AlGaAs and GaAs on a vicinal surface. The AlGaAs tend to follow the angle of the substrates, whereas the GaAs tries to form low-index facets. This has

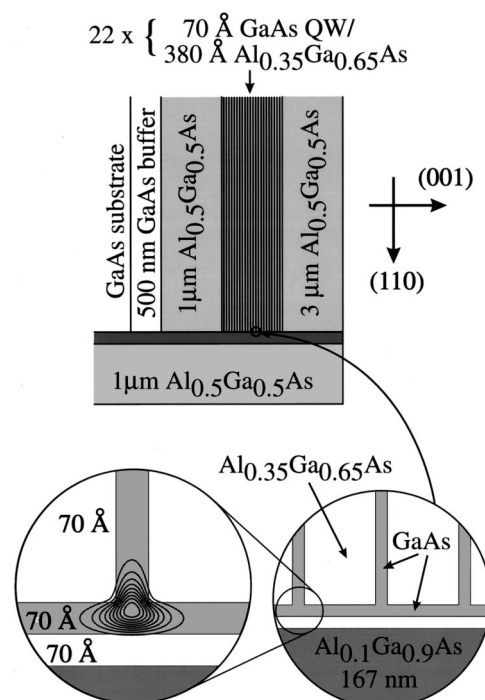


FIG. 42. Schematic showing the structure of a multistack of T-shaped QWRs grown by cleaved-edge overgrowth. The original structure was grown on a (001) GaAs substrate and consisted of a stack of identical GaAs QWs in between AlGaAs barriers. The subsequent cleaved-edge overgrowth consists of a single GaAs QW grown directly on the (110) side of the substrate. The first inset shows the intersection in more detail and the second inset shows a contour plot of the QWR wave function. (Courtesy of Harris, Ref. 312.)

been shown both on high-index surfaces³⁰¹ and on the side-walls in growth in V-grooved substrates.³⁰²

In a different category of samples, the QWR structures are formed by direct growth on the side of a single or MQW or SLS.³⁰³ By growth of a QW on the cleaved edge or etched sides of a strained QW/SLS, the QWRs are formed by the strain-induced variations in the band gap of the QW.^{303,304} Another way to form the QWRs is to grow the QW directly on the cleaved edge, to form a T of the two QWs: T-shaped QWRs.^{305,306} The latter type has attracted some interest for local probe investigations, mainly because of the possibility to achieve a significant difference in quantum confinement between the T region and the nominal QW, where the T region has the lower energy.³⁰⁷ A difference between the two regions of 54 meV has been reported.³⁰⁸

If instead of a QW, the cleaved edge contains a QWR perpendicular to the surface, a QD is formed where the QWR meets the QW of the overgrowth.³⁰⁹ QDs can also be formed in the first overgrowth step, under growth conditions that result in 3D growth, rather than layer-by-layer growth,³¹⁰ this will be discussed in Sec. III C 6 on Stranski–Krastanow growth of quantum dots. In the case of CL, monochromatic imaging has been used to identify the spectral feature of the T-shaped QWR, where a SQW was grown on the side of an etched multi-QW. In monochromatic images, the spectral positions of the MQW, as well as the T-shaped QWRs were identified. In spot mode, spectra were recorded of the same area, using the emissions from the side QW, top QW, as well

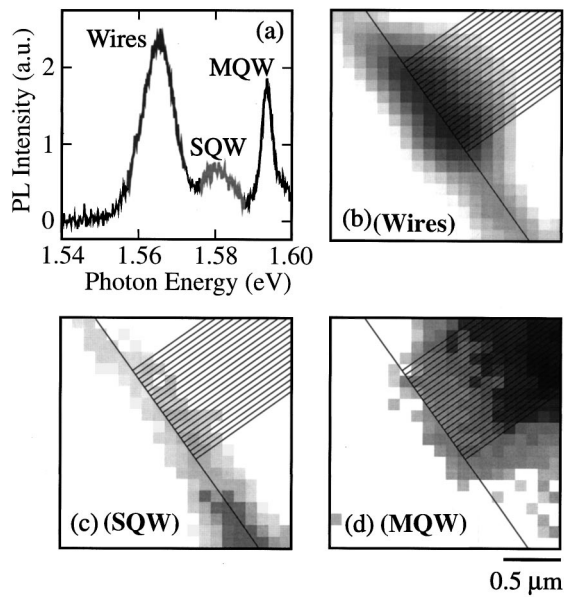


FIG. 43. (a) Typical far-field PL spectrum. (b)–(d) Monochromatic collection-mode images obtained by monitoring the emission from the low-, middle-, and high-energy peaks of the spectrum in (a). In these images white represents low intensity and black high intensity. As a guide, a schematic indicating the orientation of the sample is superimposed on each image. The images positively identify the spatial origin of the three peaks. The QWRs [(b), 1.565 eV], the (001) MQW [(d), 1.59 eV], and the (110) QW [(c), 1.59 eV]. (Courtesy of Harris, Ref. 312.)

as the MQWs.³⁰⁵ The images confirmed the identification of the emission peaks. By using spatially resolved, polarization-dependent PL, Akiyama *et al.*³¹¹ have studied T-shaped QWRs as a function of the thicknesses of the constituent QWs. The difference in confinement energy of the 1D exciton (T-QWR) and the 2D excitons (QWs) was found to be largest when the constituent QWs had the same thickness.

A different sample, grown by the same technique, was investigated by NSOM in side view.³¹² In this case, the substrate consisted of a stack of QWs of varying widths. This was subsequently overgrown with a single QW, where the T-shaped QWRs are formed at the intersections between the QWs of the stack and the single QW, Fig. 42. By utilizing far-field excitation and near-field collection, luminescence from all QWs/QWRs structures should be possible to detect. However, due to the finite tip aperture size, in this case 250 nm diam, the individual QWs of the QW stack were not resolved, Fig. 43. The luminescence intensity from the single QW and the MQW was quenched close to the T-QWRs. This was interpreted as diffusion of carriers generated in the QWs, subsequently trapped in the T-QWRs, where they recombine. This interpretation was also validated by a study of the intensity using geometry arguments. The diffusion length of the carriers was thus determined to be 0.5 μm in the single QW and 0.7 μm in the MQW.³¹²

In recent studies of a strained T-QWR sample, the local nature of excitation of luminescence of NSOM was utilized to its full extent.^{313–315} MBE was used to grow the substrate, containing a stack of 30, 15, 7.5, 3.8, and 1.8 nm thick, strained $\text{In}_{0.10}\text{Ga}_{0.90}\text{As}$ QWs in between 1 μm thick GaAs barriers on a (001) GaAs substrate. In the second growth

step, two GaAs QWs of 8 and 3.5 nm thicknesses, embedded in 20 nm $\text{Al}_{0.3}\text{Ga}_{0.7}\text{As}$ barriers and a 5 nm $\text{Al}_{0.3}\text{Ga}_{0.7}\text{As}$ top layer were grown on the cleaved edge. In this way, ten single strained QWRs were formed on the cleaved edge above the strained QWs in the substrate. By using NSOM excitation and recording the full PL spectra in each pixel, the authors obtained a large four-dimensional data set (x, y, λ, I). In this way, clear spatially and spectrally resolved recombination from the different strained QWs were obtained. Similar spectra and images were obtained also for the GaAs/AlGaAs QWs. Data obtained from the different strained QWRs were employed for assessing the predicted enhancement of absorption for QWR structures.^{316,317} By integrating PLE intensities for the QW and strained QWR, correcting for the geometrical area of the structures, a factor of 3 in absorption enhancement of the QWR over the QW was found. However, as pointed out by the authors,³¹³ this enhancement result is only preliminary, and a more systematic study has to be performed in order to confirm this result.

The main drawback of the conventional T-shaped QWRs is the limited number of QWRs produced in a growth run. A way to overcome this limitation is to grow the structures on the sides of mesas, preferably with {111}-oriented sides.³¹⁸ Also, in this case a T-shaped QWR is formed, where monochromatic CL imaging was used to optimize the growth condition and spot-mode spectra identified the presence of a QWR peak.^{318,319}

4. Growth on ridges and mesas

The direct growth of QWRs and QDs can also be done on nonplanar substrates, where a pattern has been intentionally introduced either by etching of trenches and pits or ridges and mesas, or by mesa and ridge growth on masked substrates. The basic concept is to use the differences in growth rates of the thin QW/QWR/QD layer on the different crystallographic facets that develop during growth. The QWRs are formed on top of the ridges, in the bottom of the trenches, or at the edges of the trenches. The QDs are formed at the bottom of the pits or on the top of the mesas.

The growth of QWRs on ridges is usually done by MBE and the material systems include GaAs QWRs in between AlGaAs barriers,^{320–324} GaInAs QWRs in between GaAs barriers,³²⁵ and GaInAs QWRs in between InP barriers.^{326,327} All the structures show a low-energy peak, originating from the center of the ridge, where monochromatic CL imaging has been used to identify the spatial origin. The width of the QWR has been investigated as a function of growth temperature,³²⁸ and the peak position of the emission from the QWR has been correlated with the width of the ridge,³²⁰ where a redshift (larger vertical thickness) was observed as a result of a reduction of the width of the QWR.

An example of QWRs grown on mesas is shown in Fig. 44. The substrate used in this study contained a grating of $<2 \mu\text{m}$ wide mesas, separated by 2–3 μm deep, rectangular trenches. The structure was originally determined by (001) planes on top of the mesas and bottom of the trenches and {110} planes on the sides of the trenches. During the MBE growth, the tops of the mesas develop {111} planes, bordering the (001) top plateau. The formation of the QWRs relies

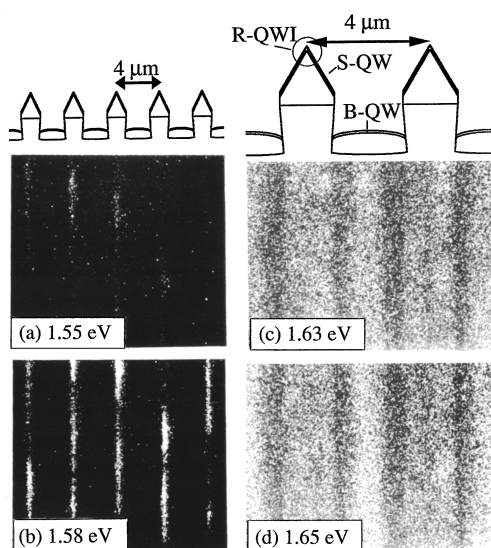


FIG. 44. Illustration of the technique of fabricating QWRs by growing a thin layer on a ridge. The $2\ \mu\text{m}$ wide GaAs ridges were overgrown by GaAs to form triangular ridges, defined by $\{111\}$ sides and narrow $\{001\}$ top facets. A thin AlGaAs layer was grown on the triangular ridges to form the lower barrier, followed by a thin GaAs layer, covered by a second AlGaAs layer as the upper barrier. The thin GaAs layer grew on both the $\{111\}$ sides of the ridges; the $\{001\}$ areas between the ridges; and on the $\{001\}$ top of the ridges. Monochromatic top-view CL imaging reveals three characteristic features: The B-QW in between the ridges at $1.65\ \text{eV}$, the S-QW on the $\{111\}$ sides of the ridges at $1.63\ \text{eV}$, and the top QWR (R-QW) at $1.58\ \text{eV}$. By varying the detection around the $1.58\ \text{eV}$ of the QWR inhomogeneities in the QWR were revealed, as illustrated by the image obtained at $1.55\ \text{eV}$, where traces of the QWR can be observed. The temperature was $10\ \text{K}$ during the imaging. (Courtesy of Wada, Ref. 329.)

on the differences in the relative growth rates of the $\{111\}$ and $\{001\}$ planes of mainly GaAs, where the $\{001\}$ is a faster growing plane. This leads to a narrowing of the plateau on top of the mesa during GaAs growth. After growth of a sufficiently thick layer, the structure is mainly determined by the $\{111\}$ sidewalls, with a self-limiting width of the $\{001\}$ top plateau, given only by the growth conditions. This width is, typically, $10\ \text{nm}$.^{329,330} The growth of a thin AlAs layer does not significantly widen this plateau and the subsequent deposition of a GaAs QW leads to areas of two different thicknesses on the mesa. The QW on the $\{001\}$ plateau is thicker than on the $\{111\}$ sides. This means that the $\{001\}$ -oriented QW segment is surrounded by AlAs on two sides and by thinner QWs on the remaining two sides, which means that this segment forms a QWR. Figure 44 shows a series of monochromatic CL images, in top view of this structure. Three areas of the QW can be identified: the QWR at $1.58\ \text{eV}$, the side QW at $1.63\ \text{eV}$, and finally, the QW in the trenches in between the mesas at $1.65\ \text{eV}$. An interesting feature can be observed at a lower energy ($1.55\ \text{eV}$) than the QWR, where a few areas of the $\{001\}$ segment emits luminescence at this energy. This reflects variations in the width of the $\{001\}$ plateau. A wider plateau leads to less quantization in the $\{001\}$ plane, which in turn leads to emission at a lower energy. This reveals a problem in this fabrication technique, where a variation in the width of the $\{001\}$ plateau leads to a variation in the peak energy position of the QWR.³²⁹

One of the more fundamental properties of QWR is the polarization of the emission and absorption, as discussed above and in appendix B. This means that the excitonic emission from a QWR is, to some degree, polarized along the axis of the QWR.³³¹ This has been the subject of a study of QWRs grown on ridges. In this case, the sample was a GaAs/AlGaAs MQW structure, grown on GaAs ridges, oriented along the $[100]$ direction, on a $\{001\}$ -oriented substrate. Initially, a wide-period SLS was deposited to reduce the width of the $\{001\}$ ridge top. As the MQWs were deposited, the width of the top of the $\{001\}$ ridge was further reduced until the structure was defined by $\{110\}$ planes on the sides. Above this point, denoted pinch-off, the structure consisted of two $\{110\}$ MQW and below this point, it was a stack of $\{001\}$ QWs with varying thickness, forming QWRs, connected to $\{110\}$ QWs. The emission at $87\ \text{K}$ from the structure was divided into three categories: (i) the bulklike emission from the wide period SLS ($>1.55\ \text{eV}$), (ii) the side MQW ($1.50\ \text{eV}$), and (iii) the QWR-like structures below pinch-off on the ridges ($1.52\ \text{eV}$). When the emission from the QWR was divided into the parallel and perpendicular polarization, there was a distinct anisotropy. The emission polarized parallel to the QWR was $\sim 7\%$ more intense than the emission polarized perpendicular to the QWR.³³¹ Polarization dependence of ridge-type QWRs has also been observed by $\mu\text{-PL}$,³³² where the polarization was found to be oriented along the QWRs. The strained GaInAs QWRs in this case were grown on the cleaved $\{110\}$ edge of a wide-period SLS of AlGaAs/GaAs. First, a layer of GaAs was grown selectively on the GaAs of the SLS, where the oxidized AlGaAs acted as a mask. The GaAs formed triangular mesas and the GaInAs QWRs were formed, similar to the GaAs QWRs discussed above.

A different approach to produce the mesas is to write trenches in a GaAs substrate and overgrow the substrate with InP. The InP grows preferentially in the trenches forming the triangular mesas. A QWR can then be formed on the InP mesa tops by depositing a GaInAs QW, followed by an InP capping layer. Under the right growth conditions, the GaInAs forms a QWR on the top of the mesa. The formation of QWRs was confirmed by polarization anisotropy in PL, similar to the case discussed above, and the identification of the spatial origin of the QWR line was done by monochromatic CL imaging.³³³

If the pattern consists of mesas rather than extended ridges, the resulting structure is a pyramid. On a $\{001\}$ -oriented substrate, square mesas with the base oriented along the $\langle 110 \rangle$ direction, the pyramid has mainly two $\{111\}$ A and two $\{111\}$ B side facets.³³⁴ If the base is oriented along the $\langle 100 \rangle$ direction instead, the pyramids are more complex, with $\{310\}$ and $\{110\}$ facets.^{331,335} On a $\{111\}$ A-oriented substrate, the triangular mesas have $\langle 110 \rangle$ -oriented bases, and the sides are determined by $\{113\}$ A facets.³³⁶ In all three cases the top facet has the same orientation as the substrate [i.e., $\{001\}$ and $\{111\}$, respectively], where the thicker QD is formed. This was confirmed by monochromatic CL imaging in top view.^{321,331,335–338} In time-resolved CL studies, the feeding of the QD from sidewall QWs was established in this type of structure.^{331,337,339} In a study of the effect of the width of the

top (001) facet of a GaAs/AlGaAs structure, μ -PL was employed.³⁴⁰ As the width was reduced from 1.4 to 0.2 μm , a redshift of the emission was found. This was attributed to an increase in the GaAs QW thickness. The μ -PLE was also studied, where both the light-hole and heavy-hole excitons were clearly observed. The splitting between the two peaks was found to reduce with mesa width, consistent with the thickness increase of the QW.

The mesas can also be fabricated by growth on masked substrates. The pattern can be a simple mask of, e.g., Si_3N_4 or SiO_2 in which growth windows have been opened up and during growth a mesa is formed in the window.^{341,342} With this technique the mesas form directly without any etching. This leads to QWRs similar to the ones grown on etched mesas.³⁴³ Growth on patterned substrates can be used to fabricate QDs. A substrate, covered by a mask with a number of growth windows is used for this purpose. This is, typically, a $\{111\}$ A-oriented GaAs substrate,^{344,345} with a mask containing triangular windows. The mesas that grow in these windows are defined by several planes, where the set of planes depends on the growth conditions. This leads to pyramid-shaped mesas, and at a certain stage of the growth, the AlGaAs mesa has a flat top, with a suitable area for the formation of a QD when a thin GaAs layer is grown on top of it. Like in the case of QWRs grown on top of mesas, the CL spectra contain several peaks. Using monochromatic CL imaging the spatial origin was identified, with the QD at the lower-energy position.^{334,346}

A recently developed technique to fabricate individual pixels for displays is related to growth on mesas. The aim was to grow the emitting material (e.g., GaInP) directly on a micromachined silicon tip, forming a light-emitting diode. The 1–2 μm diam emitters were arranged in matrixes with $\sim 5 \mu\text{m}$ spacing, with individual contacts. Monochromatic CL imaging was used to confirm that the emission really originated in the GaInP deposited on the Si tips.³⁴⁷

5. Growth in grooves and recesses

Growth on V-grooved substrates is, so far, the most promising way to fabricate QWRs with high quality, with a high degree of reproducibility, and clearly resolved 1D subbands.^{348,349} This type of structures has, therefore, attracted the most attention, either as single grooves or a regular grating.³⁵⁰ Growth of a thin QW on a grooved substrate results in different areas of the QW having different thicknesses. Each of these thicknesses will have a different peak energy position of the emission. In the analysis of the spatial origin of the emission, monochromatic CL imaging is a very useful tool. A typical V-groove structure is in the lattice-matched GaAs/AlGaAs material system grown on a patterned GaAs substrate using MOCVD,³ although GaInAs/AlGaAs,³⁵¹ GaPAs/AlGaAs,³⁵² and GaInAs/InP³⁵³ have been demonstrated. The growth conditions of the AlGaAs are chosen so that two sets of planes dominate the growth: typically, quasi- $\{111\}$ planes as sidewalls of the grooves and a (001) plane between them. At the bottom of the grooves, the meeting point of the two $\{111\}$ planes is rounded, rather than pointed. The radius of curvature of this rounding is given by the growth conditions and increases

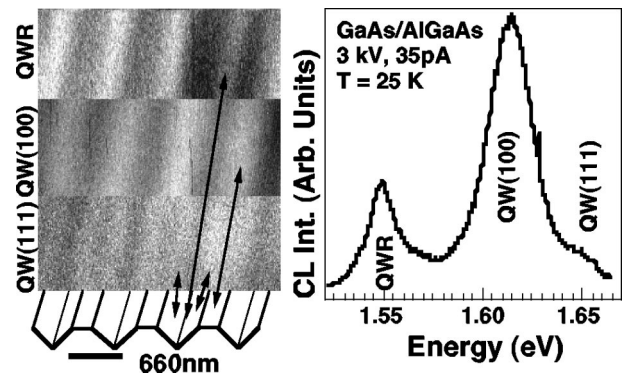


FIG. 45. Illustration of the technique of fabrication of QWRs by growth on a V-grooved substrate. A thin GaAs layer was embedded in a AlGaAs barrier. The structure was mainly limited by $\{111\}$ planes, defining the Vs and (001) planes on the ridges between the Vs. The QWR was formed at the bottom of the V. The average spectrum at 25 K consisted of two peaks, where one peak had a high-energy shoulder. Monochromatic top-view CL imaging revealed that the low-energy peak originated in the QWR, the high-energy peak originated in the (001) QW, and the high-energy shoulder on the high-energy peak originated in the $\{111\}$ QWs. An important feature was that the double periodicity of the $\{111\}$ QW can be resolved, displaying a resolution better than half the period of 660 nm (Ref. 400).

with Ga content in the AlGaAs layer.³⁵⁴ As a result of this, a crescent-shaped thickening is formed, when growing a thin GaAs layer on top of an AlGaAs barrier. When embedded in between AlGaAs barriers, the crescent-shaped region forms a QWR with a relatively large separation of the subbands, caused by the additional confinement in the QWR. A quasi-1D subband spacing of up to 45 meV has been reported for this type of QWR.^{354,355} In other samples, macro-PL linewidths down to 5.4 meV at 5 K have been reported.³⁴⁹ Using μ -PL,³⁵⁶ it has been demonstrated that this type of PL linewidth can be caused by emission from spatially localized states, similar to that found in thin QWs.²³⁶ In the QWR case,³⁵⁶ a 1 ML thickness difference in the vertical direction was calculated to introduce a ~ 9 meV difference in emission energy. Over a 1 μm segment, the QWR exhibited a series of $\geq 200 \mu\text{eV}$ wide peaks, distributed over 10 meV. The interpretation was that these lines originated in shallow QD-like states, caused by the slight corrugation of the QWR along the V groove.³⁵⁶

An additional effect of the growth on a V groove is that the Ga has a tendency to segregate to the bottom of the V groove while growing an AlGaAs layer. This leads to the formation of a Ga-rich sheet spanned by the direction of the groove and the growth direction. Typically, this has a width in the range of 10–50 nm and it forms a VQW.³⁵⁷ The effect of all the different areas is a complicated system with many emission peaks; at least three from the QW/QWR and one from each of the VQW and the barrier. In most cases, all these peaks are not fully resolved, and the individual peaks can behave differently because they originate in different areas. The identification of the spatial origin of the different peaks is an area where the local probe techniques can be very useful.^{267,351,353,358–362} Figure 45 shows a typical spectrum of a QWR sample, averaged over several periods of the structure. The identification of the peaks was confirmed by monochromatic imaging in both top and side view. Figure 45 also

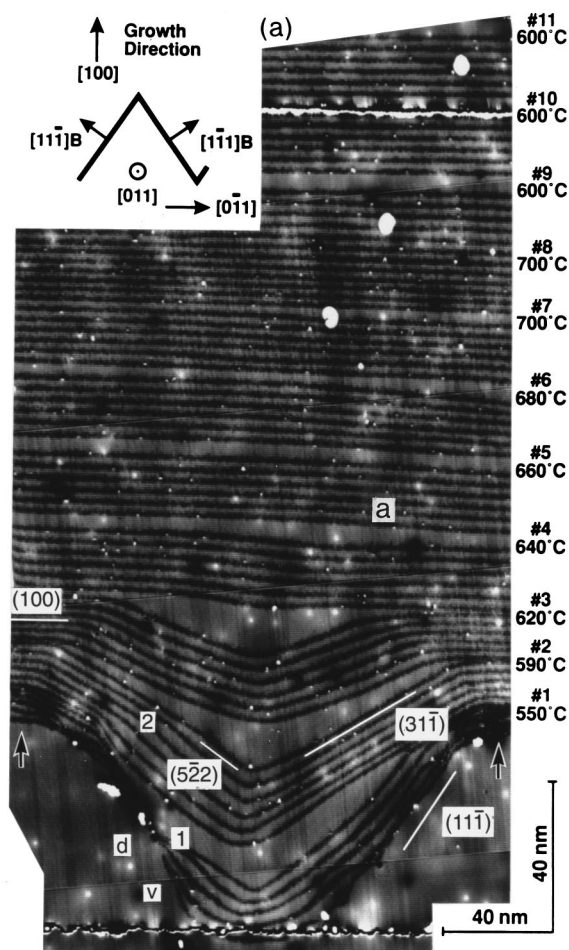


FIG. 46. STM image of GaAs QWRs embedded in $(\text{AlAs})_4(\text{GaAs})_8\text{SLSs}$. The range of the grey scale is $0.8 \text{ \AA}$. The 4 MLs of AlAs appear darker than the surrounding 8 MLs of GaAs. White hillocks about 2.5 nm in diameter labeled “d” represent electrically active dopants in the top layers. Atomically localized black and white dots are cleavage-induced vacancies “v” or adatoms “a,” respectively. Note that the scale along the $[100]$ direction is expanded by a factor of 1.4 compared to the $[011]$ direction. (Courtesy of Pfister, Ref. 363.)

shows top-view images of the three emission energies.

The very high spatial resolution of STL makes the technique suitable for studying variations in individual QWRs in a stack of several QWRs.^{67,363} The sample in this case was a sample grown by MBE on a holographically defined grating of V grooves. The sample consists of a stack of several GaAs QWRs, grown in between AlGaAs barriers and the whole structure is surrounded by a SLS on each side. Side-view STM studies were performed in UHV (better than 10^{-11} mbar) giving atomic resolution of a piece of the MQW sample, Fig. 46. The STL data were collected using another piece of the sample in a different system having a slightly higher base pressure (better than 5×10^{-10} mbar). This higher base pressure meant that the surface became contaminated after the cleaving, making it impossible to obtain atomic resolution in this equipment. The STL data were collected in the panchromatic mode. The resulting STL image is shown in Fig. 47, where it can be observed that the light was generated at the lower-band-gap GaAs QWRs having a varying intensity, although the nominal thicknesses are equal.

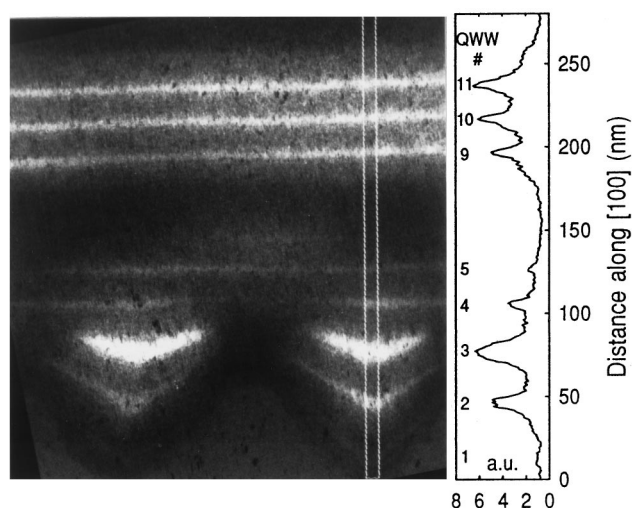


FIG. 47. Room-temperature panchromatic STL image of a $300 \times 300 \text{ nm}^2$ cross section of a piece of the same sample as in Fig. 46. The image shows two periods of the structures. The luminescence profile on the right shows the intensity in the boxed area, averaged horizontally. The luminescence from QWR 1 is heavily suppressed due to increased nonradiative recombination, presumably due to a higher relative density of electrically active defects at, or close to the interface between the substrate and the epitaxial layers, as can be seen in Fig. 46. These defects are close enough to QWR 1 to reduce the STL intensity. Also, for the QWs different luminescence intensities can be observed. This was attributed to different doping concentrations, as well as QW thicknesses as a result of different growth temperatures (cf. Fig. 28). The lateral luminescence variations were attributed to slight inhomogeneities within the layers. (Courtesy of Pfister, Ref. 363.)

The authors argue this to be caused by a difference in the density of electrically active sites at the grating-to-epilayer interface, close to the QWR 1. These sites were revealed by the side-view STM study, and were argued to lead to an increased rate of nonradiative recombination. The difference in doping concentration and thicknesses of QWR 4 through QWR 11 were also manifested in the STL intensities from these QWRs. The thickness was directly reflected in the observed shape of the bright areas in the images and the intensity was found to increase with doping level. The latter is an effect of the number of available majority carriers for the injected minority carriers to recombine with. The reason for the large contrast between the QWR (light) and SLS (dark) regions can be explained by considering the reduced electron injection efficiency in the SLS region due to the lower excess energy of the carriers in this region. It is known from other studies that the luminescence intensity increases when the excess energy above the conduction-band energy increases.^{62–64,67,72,210} The observed lateral variations were attributed to slight variations in the thickness within the different QWs.

Top-view STL has been used to study MOCVD-grown QWRs grown on V-grooved substrates.³⁶⁴ In this case, the sample was a single-layer QWR and the period was 765 nm . When injecting carriers directly into the QWR a narrow spectrum was obtained, with very little signal from the surrounding features. However, when injecting in between two QWRs, the QWR emission was rather weak. This study was done using a *p*-type semiconducting tip (GaP), which means that holes were injected into (or electrons were extracted

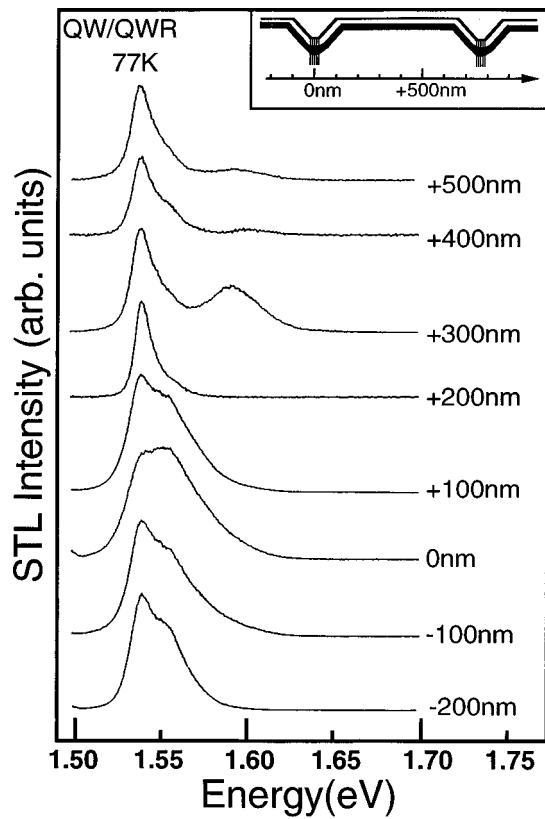


FIG. 48. A series of STL top-view spectra recorded at 77 K. The sample was a single GaAs QWR structure, grown in between AlGaAs barriers on a V-grooved substrate, with a 765 nm period. In the series of spectra, the tip was moved perpendicular to the QWRs in steps of 100 nm. The STL exhibits two emission peaks, 1.54 eV of the QWR and 1.59 eV of the QW. At all positions the QWR peak is dominating the spectra, whereas the QW peak is only visible in the spatial position between the QWRs. This demonstrates the very efficient carrier transfer to the QWR. The spectra also demonstrate an important feature of QWRs. As the tip approaches the QWR and the excitation density increases, emission from the second subband of the QWR appears on the high-energy side of the QWR peak. With the tip positioned directly above the QWR (0 nm), the emission from the first and second subbands are equivalent in intensity (Ref. 364).

from) the valence band of the sample. In another study a metal tip was employed under the same bias condition as for the semiconducting tip above. In this case the STM tip was moved laterally in steps of 100 nm perpendicular to the QWRs, and a spectrum was acquired at each point, Fig. 48. When approaching the QWR, the excitation density in the QWR increased and the emission from the first excited state could be observed on the high-energy side of the QWR peak. The same band-filling effects were found in a CL experiment, where the probe current was increased over three orders of magnitude.³⁶⁴

Since these structures are very complex with different areas connected and possibly exchanging carriers, they are very well suited for analysis of the time evolution of the different peaks involved.^{60,365} The combined spatial and temporal resolution, together with the possibility of high local excitation density, makes CL very suitable for this purpose. By simply varying the probe current it is possible to obtain information on saturation effects and to obtain information about the subband spacing.^{302,366} This is illustrated in Fig. 49(a). At a low probe current (510 pA), only one peak could

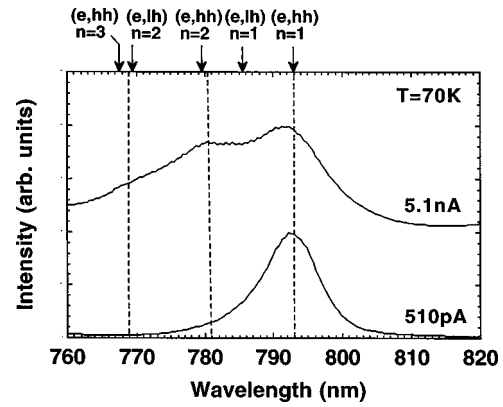


FIG. 49. When a V-groove QWR is excited with different probe currents in CL, band filling effects can be observed. At 0.51 nA only one peak can be observed, whereas at 5.1 nA a second peak and a trace of a third peak can be observed. These peaks correspond to the first, second, and third lateral subbands in the QWR. The energy separation of the subbands is ~ 21 meV. The equienergy spacing of the subbands is typical for the crescent-shaped QWR, where the confining potential is almost parabolic. (Courtesy Grundmann, Ref. 366.)

be observed. At a higher probe current (5.1 nA), a shoulder corresponding to the second subband ($n=2$) of the QWR, as well as a trace of the third subband ($n=3$), could be seen. The subband splitting ($n=1$ to $n=2$) was found to be 20 meV, in agreement with theoretical calculations of the subband energy³⁶⁶ using the shape of the QWR as determined by TEM. In a study of the temporal behavior, the relaxation from the excited to the first subbands, as well as carrier temperature can be studied. The pulse was long enough to establish a quasiequilibrium, where the first subband was saturated. Immediately after the excitation pulse the second

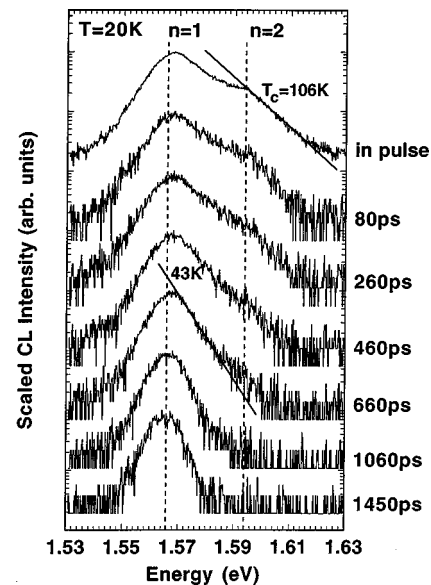


FIG. 50. Time-resolved CL spectra from the QWR in Fig. 49. The pulse is long enough for an equilibrium to be reached, and two subbands can be observed. From the slope of the high-energy side, the carrier temperature can be estimated to be 106 K, which is 86 K above the lattice temperature of 20 K. After the pulse, the second subband decays, whereas the first subband remains constant for 400 ps before it decays. This clearly indicates that the carriers in the second subband relax down to the first. During the relaxation, the carrier temperature is reduced. (Courtesy Grundmann, Ref. 366.)

subband peak started to decrease, with a time constant of 200 ps. In contrast, the first subband remained saturated for 400 ps before it started to decrease with a time constant of 500 ps. It was concluded that the second subband had two channels for losing carriers: the “intrinsic” radiative process and the transfer to the first subband, whereas the first subband had only the radiative channel. In a series of time-delayed spectra, Fig. 49(b), the faster decrease of the peak of the second subband was clearer. With increasing delay, the high-energy side of the peak decreased much faster (half the time constant) than the main peak. From the slope of the high-energy side of the peak, the carrier temperature could be obtained (see Fig. 50). In the pulse, the carrier temperature was found to be 106 K, which was relaxed to the 20 K of the lattice in ~ 900 ps, mainly caused by scattering of the holes by acoustic phonons.³⁶⁶

The V-grooved QWRs have also been imaged by spatially resolved PL.³⁶⁷ The materials system is GaAs/AlGaAs and the QWRs were obtained by selective growth on V-grooved substrates. The fabrication of the QWRs was slightly different, where the starting point was a substrate, masked by stripes. Growth initially formed triangular mesas between the stripes and after continued growth, the mesas merged, resulting in a triangular grating of V grooves.³⁶⁸ In this case a streak camera was used, where the sample emission was focused on the entrance slit of the camera. As a result, a time-resolved 1D image was obtained. The QWR was oriented either parallel or perpendicular to the slit. The diffusion length was found to be longer along a QWR than between QWRs, as expected. It was also found that the diffusion length in the QWRs was longer than in normal QWs. The same group has also imaged the spatial distribution in 2D using a microscope and a CCD camera for detection, in continuous mode.³⁶⁹ The diffusion length along the QWR was found to be $\sim 4 \mu\text{m}$ for a QWR with a lateral width of 15 nm. The diffusion length increased for wires with a smaller lateral width within the size limits of 30 nm and 15 nm. A further decrease in lateral width, however, resulted in a shorter diffusion length, which was attributed to an increased contribution of interface scattering in smaller QWRs. A high mobility along QWRs was predicted already in 1980.³⁷⁰ The argument is basically that the probability of elastic scattering in QWRs is reduced since, when the original state has a wave-vector $\mathbf{k}_i$ it can only scatter to $-\mathbf{k}_i$ in one dimension. In higher dimensions there is a continuum of allowed final states for elastic scattering. The technique where imaging streak cameras are used is very quick, direct, and will revolutionize the measurements of carrier diffusion. Watabe *et al.*³⁶⁷ have also begun initial studies of transport between QDs in this way.

A third approach of growth on a nonplanar surface is to form the QWRs in the corners between the ridges and the grooves, sometimes referred to as L-shaped QWRs. An example of this type of structure was reported in Ref. 371. The substrate used was (111)-oriented GaAs, which had $5 \mu\text{m}$ wide mesas along a $\langle 110 \rangle$ direction, separated by 20 nm deep and $5 \mu\text{m}$ wide trenches. In this case, the QWRs were formed on the sidewalls of the mesas. This was confirmed by monochromatic CL imaging in side view.³⁷¹

The approach of V-grooved substrates can be used to fabricate QDs. In this case the pattern is not an infinitely long groove, but $2\text{--}3 \mu\text{m}$ long grooves. The growth on these types of substrates results in even more complicated structures than the corresponding QWR structures, but the pattern of the emission peaks is similar. In between the grooves a normal QW is formed; the QW on the sides of the grooves is slightly thicker, lower in energy. Finally, at the bottom of the groove the thickest region of the QW forms a QD. The identification of the various peaks was done using CL imaging.³⁷² A potentially less complicated way of making QDs in pits is to use a (111)*B*-oriented substrate and to etch triangular pits, determined by (111)*A* sides.^{264,373,374} Growth of a QW layer can lead to the formation of a thicker region at the bottom of the pit. This is a result of a different profile of the barrier and the QW at the bottom of the pit. This can either be AlGaAs barriers and a GaAs QD,²⁶⁴ or GaAs barriers and a GaInAs QD.^{373,374}

It is not only lattice-matched and strained III–V materials that have been grown on V-grooved substrates. Also, thin layers of SiGe in between Si barriers lead to the formation of QWRs when grown on V-grooved substrates.^{375,376} The CL spectra from these structures showed three peaks and their respective phonon replicas. Monochromatic imaging revealed that one peak originated in the areas in between the V grooves and that the other two peaks were related to the grooves.^{375,376}

6. Stranski–Krastanow growth of quantum dots

Recently, the technique of self-assembling of QDs directly in the growth step has received much attention.^{377–384} The growth techniques range from MBE³⁸⁵ via CBE³⁸⁶ to MOVPE.³⁸⁷ The formation of the QDs relies on the relaxation of a highly mismatched layer by formation of 3D islands rather than misfit dislocations, as in the case of the more moderately mismatched layers. The latter case has been discussed in the previous section III B 1 on misfit dislocations and strain in III–V and II–VI heterostructures. The growth of a highly mismatched layer initially forms a thin, homogeneous, and flat film. Further growth builds up strain in the layer, where this strain can be minimized by the formation of 3D islands in a growth mode known as Stranski–Krastanow (SK).^{382,388–390} The formation of these 3D islands can be further enhanced by employing a growth interruption after depositing the strained layer. These 3D islands are coherent and generally dislocation free and they grow on top of a very thin 2D layer, denoted as the wetting layer. They are also of similar shape and size, given by the growth conditions. Typically, the area density of these QDs is $10^{10}\text{--}10^{11} \text{ cm}^{-2}$.³⁷⁹ In this growth regime, the typical QD contains $10^3\text{--}10^4$ molecules and the energy position of the ground state of single QDs differing in size by one molecule is on the verge of the detection limit in these types of luminescence experiments.³⁷⁹ The typical 0D-level spacing is 20–30 meV, and due to the true 0D nature of these QDs, they exhibit little or no thermal broadening.³⁷⁹ The homogeneous linewidth of QDs has recently been observed to have a weak dependence on temperature.³⁹¹ This was attributed to temperature-dependent interactions of the carriers in the QD

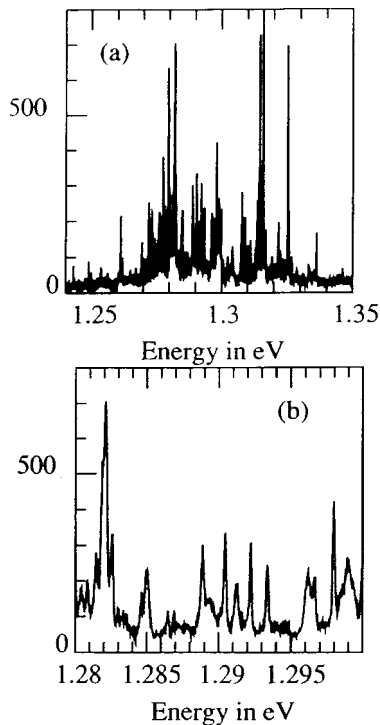


FIG. 51. The emission spectra from a mesa containing a number of InAs QDs in between GaAs. The emission consists of sharp lines, shown in a large energy range (a), and in a smaller energy range (b). QDs are expected to have sharp emission lines and each of the individual lines shown in the figure was attributed to a cluster (or a QD) of InAs. In this case the spatial selection was done by etching out the relevant structure and selecting the emission from one of the mesas by an optical microscope. The number of lines detected was found to scale with the area of the mesas, supporting the correlation between the individual lines and individual QDs (Ref. 394).

with phonons and photons rather than thermal excitation into nearby excited states, as observed in QWs. The very weak temperature dependence means that the luminescence spectrum of an ensemble of QDs gives the statistical fluctuation in the size and spectrum of a few QDs revealing their individual energy positions.

The material systems for these SK-QDs are generally SiGe in between Si barriers,³⁹⁰ (Ga)InAs in between GaAs,³⁷⁹ In(P)As in between InP, InAlAs in between AlGaAs,³⁷⁸ InP in between GaInP,³⁷⁷ or even complicated QD structures like GaInP/InP/GaInAs/InP/GaInP.³⁹² Seemingly, only layers grown in compression form these types of QDs. In the InP/GaInP material system the growth is bimodal, with two populations of QDs. One population consists of fully developed QDs with similar shapes and sizes. The other population consists of smaller, partially formed QDs, which are less homogeneous in size. Typically, these fully developed QDs have a very low area density of $\sim 10^8 \text{ cm}^{-2}$.³⁷⁷ These fully developed QDs exhibit less quantization than the smaller type, and the 0D-level spacing is in the order of a few to ten meV.

It is highly desirable to be able to obtain local spectroscopy of individual QDs at low excitation density. CL spectroscopy (see above) generates ~ 700 e-h pairs per incident electron, caused by the high acceleration voltage, which also gives the electrons a high energy. In CL there is a high carrier density with significant excess energy, which can give

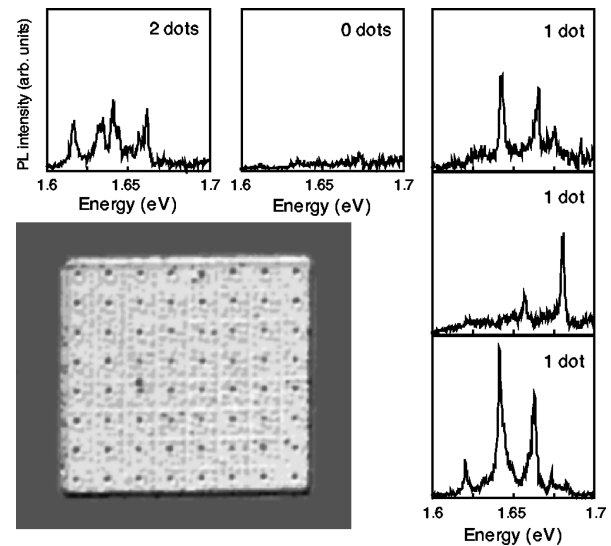


FIG. 52. PL spectra from InP QDs in GaInP barriers. The spectra were recorded from holes in a gold mask, illustrated by the image. From most of the holes there is emission from one QD, although from some holes there is emission from two QDs and in some holes there is no QD emission. The emission consists of several lines (cf. Fig. 40). The emission was collected by an optical microscope (Ref. 384).

rise to carriers with a high temperature, compared to the lattice temperature. The high carrier density will easily saturate the QDs through state filling. From time-dependence studies of high and low excitation power density PL spectra of InP QDs in between barriers of GaInP, it was found that the carrier cooling was fast (the lattice temperature was reached in ~ 10 – 100 ps) at low excitation power density, as is observed in QWs,³⁹³ but much slower (more than 1 ns) at high excitation power density³⁸⁰ where state-filling phenomena become dominant.

Marzin *et al.*³⁹⁴ have measured spatially resolved PL from QDs, with a lateral size of ~ 20 nm, in the GaInAs/GaAs system, where the spatial resolution has been obtained by the etching of mesas where the lateral dimension was down to 200 nm. A mesa with a lateral size of 500 nm was estimated to contain 100 QDs and the emission spectra obtained showed narrow-line emission from the structure. This narrow-line emission shown in Fig. 51 was interpreted as originating in individual QDs.³⁹⁴ The disadvantage of dry etching mesas is that the luminescence intensity of the etched structures tends to be quite low, which is possibly due to a damaged surface and sidewalls.

A different way to select a small area of the sample is to cover the sample with an opaque mask, with μm -sized holes. This was demonstrated for a gold mask, where the holes were defined by simply bonding a gold wire and subsequently removing the wire along with some of the metal mask.¹⁴⁷ The mask with the holes can also be defined by lithography, followed by metallization and lift-off.³⁸⁴ The materials system in this case was GaInP with InP QDs, where the expected number of fully developed QDs in one μm^2 is about 1. The QDs were fabricated by the Stranski-Krastanow growth technique using MOVPE.³⁷⁷ In addition to the fully developed QDs there are many, small, partially formed QDs of different sizes. The fully developed QDs

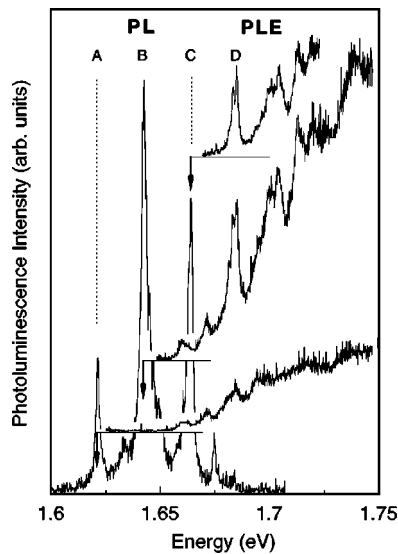


FIG. 53. PL spectrum from one single InP/GaInP QD. The emission consists of several lines. By recording PLE spectra using these lines, it can be seen that the lines share common excited states. This proves that they originate from the same QD. The spectra were recorded through a microscope and in addition selected by holes in a metal mask (c.f. Fig. 52) (Ref. 384).

have a height of ~ 15 nm and a lateral extension of 40–50 nm, as observed by TEM.^{16,395} The QDs are shaped in the form of truncated pyramids with a hexagonal base. There is only weak lateral confinement (of the order of 1 meV in the valence band and 10 meV in the conduction band) in the fully developed QDs.³⁹⁶ These QDs are not electronically coupled so this is the level spacing in single QDs.

Figure 52 shows representative emission spectra from holes in the gold mask. It can be seen that the emission peak from one QD has a linewidth of ~ 1 meV, and consists of several emission lines, similar to the case for interdiffused QDs (see Sec. III C 2). The emission linewidth from a large

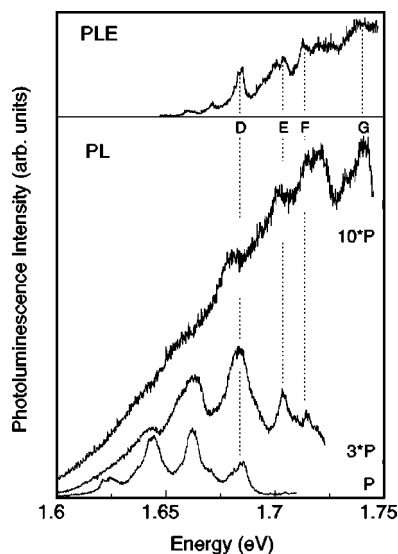


FIG. 54. The effect of increasing excitation power density on the spectra shown in Fig. 53. At higher excitation densities the emission lines are broader and discrete emission lines are seen at higher energies. Both are an effect of state filling in the QD. For comparison, a PLE spectrum is also shown, which identifies the same excited states of the QD (Ref. 384).

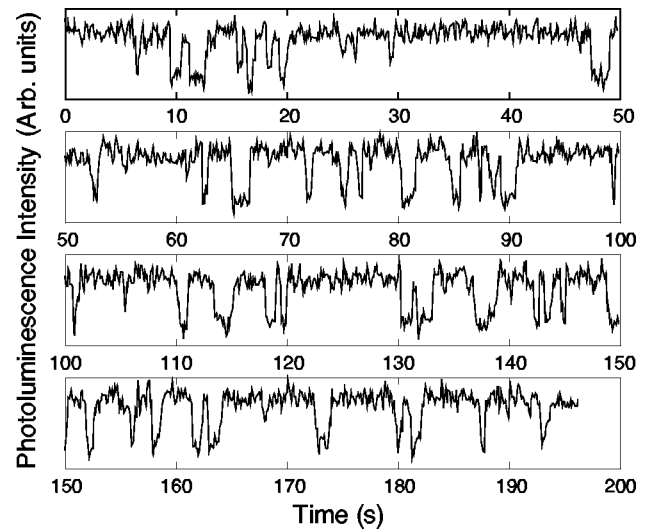


FIG. 55. A trace of the emission intensity of an InP/GaInP QD as a function of time. This particular QD exhibits switching between two states, on a time scale of seconds. Most QDs in this sample did not exhibit this type of switching. The cause of the switching is unknown, but the statistic is consistent with random telegraphic noise (Ref. 397).

number of QDs is around 15 meV. The narrower linewidth in spatially resolved PL was attributed to a lower number of carriers present during photoexcitation than during CL, where many carriers may cause state filling and consequent line broadening. Figure 53 shows the emission spectrum and the PLE spectrum of one QD in detail.³⁸⁴ It is remarkable that the emission from one QD consists of several lines (denoted as the ground manifold). By measuring PLE and detecting on each of these emission lines it was found that there was no observable transfer between the states in the ground manifold. At higher energies, however, there are absorption features which are common to the emission lines in the ground manifold. It was, therefore, concluded that the emission lines originate from one single QD. The effect of increasing excitation power density is shown in Fig. 54. There is an initial broadening of the peaks and higher peaks quickly become populated due to the Pauli exclusion principle. Time-resolved measurements give a time decay of ~ 0.7 ns in these QDs. No clear evidence of biexcitons was observed in Ref. 384. It was also concluded from the time-dependent measurements, as well as from the PLE measurements, that the efficient relaxation of the carriers to the ground manifold was not due to carrier–carrier scattering, as suggested in Ref. 285.

It has been found that the emission from some InP/GaInP QDs exhibits switching between two states on a time scale of seconds.³⁹⁷ This could be observed as an apparent switching off and on of the emission. The switching was only observed for a few QDs, about one in one thousand, and the switching is consistent with random telegraph noise. Figure 55 shows a trace of the intensity as a function of time. The switching was attributed to a modulation of the recombination, rather than to a modulation of the capture.³⁹⁷

The emission from the partially formed InP/GaInP QDs shows a very narrow linewidth in PL at high spectral resolution, compared with the fully formed QDs. The partially

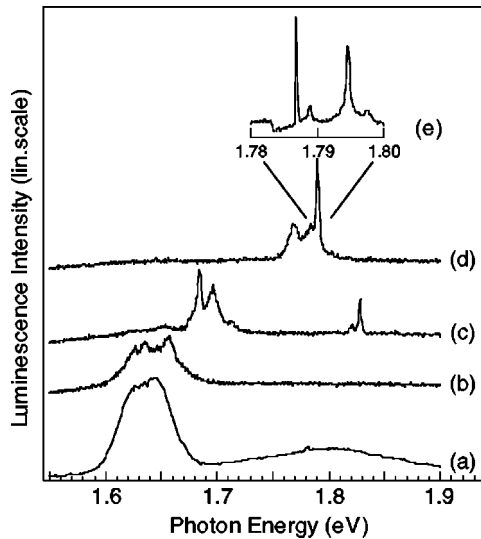


FIG. 56. (a) PL spectrum of an InP/GaInP QD sample. The emission at 1.65 eV corresponds to the fully developed QDs, whereas the emission from the partially formed QDs is observed in the wavelength region 1.7–1.9 eV. (b)–(d) Tunnel-induced spectra taken at different spatial positions (separated by several at least tens of μm) at a bias of 4 V. At some positions (b) only the fully developed QDs were detected and at other positions (c) and (d), only emission from the partially formed QDs was detected. In this case, the emission was split into several narrow lines. The inset shows a high spectral resolution STLS spectrum, with an emission linewidth down to 200 μeV . This is much less than the $k_B T$ of 7 meV (at 77 K), but close to the instrument resolution of 150 μeV . The absence of thermal broadening reflects the 0D nature of the emission (Ref. 75).

formed QDs have a small height (~ 1 nm) and a lateral extension of ~ 30 – 40 nm. There is, however, a large variation in the lateral extensions within one sample. These QDs are isolated, and they are formed on top of the QW (the wetting layer), which is otherwise very flat. The sharpest emission line from these QDs has a linewidth of ~ 40 μeV .¹⁴⁷ This is due to the small size of these QDs, which will give significant lateral quantization. There are no nearby excited states to which the exciton may thermalize. This argument is, however, also true for the fully formed QDs. The reason for the difference in linewidth for fully formed QDs and partially formed QDs is unresolved. It is also significant that QDs formed in the InAs/GaAs usually have a very narrow linewidth.^{378,379,394} It may be noted that both types of InP/GaInP QDs are seen in the same sample, making it rather unlikely that inhomogeneous effects (such as charging and decharging of nearby donors) are responsible for the differences in the linewidths.

The potential of studying novel effects of low-dimensional phenomena using PL at high spatial resolution was demonstrated by measurements of the photon emission at different excitation power densities.¹⁴⁷ At low excitation power density only a few lines were observed in the emission from partially formed InP/GaInP QDs. At higher excitation power densities a complicated behavior was observed. Some emission lines exhibited a super-linear dependence, which reminisces the intensity behavior of biexcitons. In one group of lines a transfer of emission intensity from the low-energy component to the high-energy component was observed with increasing excitation power density. It is clear

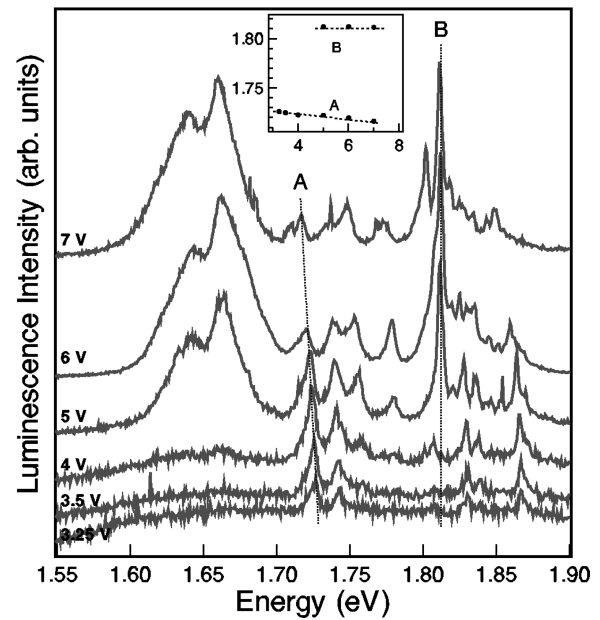


FIG. 57. Series of top-view 77 K STLS spectra at different tip-sample biases. The sample is a single-layer, InP QD in between GaInP barriers. The sample contains fully developed QDs (~ 1.65 eV) and partially formed QDs (1.70–1.90 eV). Passing a threshold bias of 4–5 V, the 1.65 eV emission, as well as the 1.70–1.90 eV emission, became strong. Especially the peaks around 1.80 eV became very strong. The threshold was attributed to the onset of impact ionization of electron-hole pairs (or excitons). As a consequence of the larger effective diffusion length for excitons, luminescence from more distant QDs was also detected. Since these QDs were more distant from the tip position than the partially developed QDs also present at low bias, the local tip-induced band bending affected these QDs less, and a local Stark shift was, therefore, anticipated to affect only the QDs close to the tip position. The Stark shifts of two of the lines, A and B originating from the partially formed QDs, are shown in the top panels. A is close to the tip and B is far away from the tip (Ref. 75).

that the behavior of partially formed QDs is very different than the behavior of fully formed QDs. More studies are clearly needed, in particular, it is necessary to know the structure of the partially formed QDs in detail in order to draw firm conclusions. We also expect that theoretical simulations are necessary in order to identify the states responsible for the emission lines.

Samples in the InP/GaInP QD series with dual-sized QDs have also been studied by STLS.^{74,75,147} The 77 K spot-mode spectra (see Fig. 56) were recorded as a function of lateral position, as well as a function of applied bias (see Fig. 57). For low biases, luminescence was obtained from small QDs in every position. At some lateral positions luminescence from the fully developed QDs was also detected. In this sample the density of fully developed QDs was in the order of one per μm^2 , as determined by AFM and TEM/SEM. The absence of signal from the fully developed QDs in most of the spot-mode spectra indicates that the lateral diffusion of the injected carriers (in this case, electrons) was less than 1 μm . However, when the bias was increased further e-h pairs were created, with the subsequent formation of excitons. In this case it was possible to detect the luminescence from the fully developed QDs for tip positions where this could not be observed at low bias. Furthermore, with increasing bias some of the luminescence lines exhib-

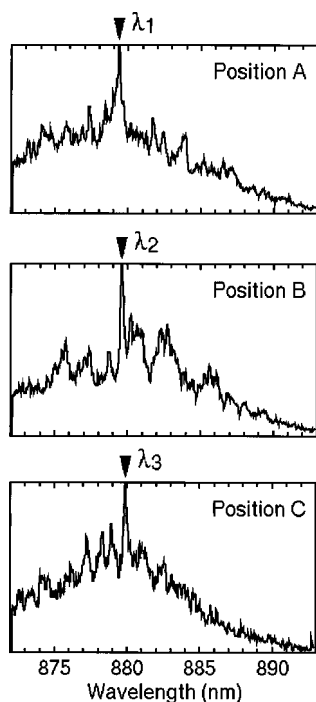


FIG. 58. Three top-view, spot-mode spectra of a sample with InAs QDs in GaAs barriers at 5 K. The nominal thickness of the InAs layer was 0.6 nm. The spots were separated by less than 5 μm and selected from bright spots in three CLWI images. The energy separation of the main peaks in the spectra corresponds to a variation in the dot size of six InAs molecules. Note the very high spectral resolution ($<150 \mu\text{eV}$, limited by the detection system). (Courtesy Grundmann, Ref. 379.)

ited a shift of spectral position while some did not. This was explained using a Stark shift model, where the spectral shift was argued to increase with the band bending.^{74,75} The QDs situated close to the tip were, therefore, more affected by the local band bending, resulting in a larger shift closer to the tip. The QDs that emerged as a function of the increased lateral excitonic diffusion did not experience the local band bending, and consequently, did not exhibit any shift. Using a simple model the actual band bending was calculated and the 3D extension was found to be $\sim 1 \mu\text{m}$.⁷⁴ With the knowledge of the extension of the band bending and the appearance of unaffected emission peaks, the authors concluded that the excitonic diffusion length was of the same order.⁷⁴ This was supported by a comparison with the spatial resolution in CL images of similar samples.^{74,377}

As discussed above, one of the complications of many CL experiments on QDs is the relatively high excitation density, which normally causes broadening of the emission peaks. A higher excitation density gives a higher carrier temperature, resulting in a larger broadening. In the case of QWRs this can be observed as a broadening of the peaks, but more notably as emission from higher subbands.^{302,366} This can be observed in spectra from the fully developed QDs, where the emission peaks are broader in the CL spectra than the corresponding PL spectra, even though the CL spectra are from single QDs.³⁷⁷ It has recently been shown that these QDs can exhibit very narrow linewidths (0.6–0.15 meV)^{378,379} in CL experiments, and not just in PL experiments. This is illustrated in Fig. 58. Furthermore, it was

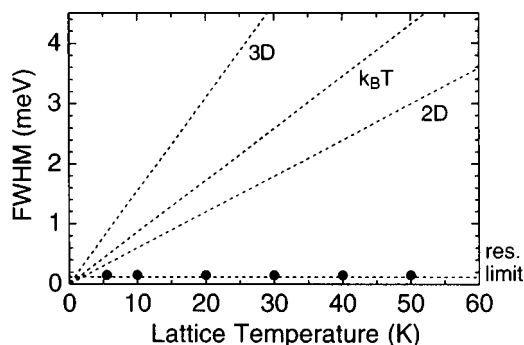


FIG. 59. Temperature dependence of the linewidth (full width at half maximum) of the single emission lines in the sample used in Fig. 58. Over the whole temperature range (5–50 K) the linewidth stays below the resolution limit of the detection system. For comparison, the expected broadening for 3D and 2D systems, as well as $k_B T$ are indicated in the plot. The absence of thermal broadening is consistent with the 0D nature of the QDs. (Courtesy Grundmann, Ref. 379.)

demonstrated, Fig. 59, that the linewidths were independent of sample temperature,³⁷⁹ a further result of the 0D nature. The problem of isolating single QDs still remains even with CL as the size of the generation volume and the diffusion of the carriers together with the density of QDs do not allow this. CL experiments in a STEM using thin TEM samples have an advantage in the ability to image the single QDs in the normal STEM mode. It was, therefore, possible to attribute emission peaks with the various single QDs and it was concluded that the spatial resolution allowed the identification of single QDs emitting at the same energy, provided they are separated by 0.5 μm or more.³⁷⁸

D. Impurities in low-dimensional structures

A subject that is related to the reduced size of the QWs/QWRs/QDs is the effect of individual impurity atoms in these structures.³⁹⁸ The interest in impurities is of a fundamental nature as well as motivated by applications. As an example, we can consider an array of QDs with the dimensions $5 \times 100 \times 100 \text{ nm}^3$ and doping level of $\sim 10^{16} \text{ cm}^{-3}$. Statistically, every second QD will contain an impurity, resulting in a different electronic structure, even if the QDs are identical in size. In a more well-controlled system of a single 5 nm GaAs QW, with a doping level of $\sim 10^{15} \text{ cm}^{-3}$, the average interimpurity distance is $\sim 500 \text{ nm}$ in the QW plane, which is just on the verge of the spatial resolution of CL in a QW. The QWR structures grown on V-grooved substrates are ideal for studying the effect of single impurities.³⁹⁹ The interesting part is the planar (001) QW rather than the QWR itself. A single layer grown on a submicron grating can result in stripes of (001) QWs with widths of 2–500 nm and a spacing of the impurities of 400–1000 nm along the QW stripes. This is a scale, where it is possible to resolve the emission from single impurities, provided that they are spectrally separated from the QW emission. This has been demonstrated in the case of single carbon acceptors (C_{As}) in a 450 nm wide and 5 nm thick QW stripe.³⁹⁹ Figure 60 shows the average spectra at two different probe currents. At very low probe currents there is a definite shoulder on the low-energy side of the main peak. At a higher probe current, this

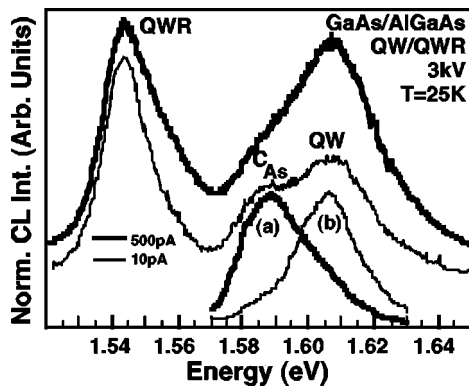


FIG. 60. At a normal CL probe current of 500 pA and a sample temperature of 25 K, the spectrum of a V-groove QWR structure exhibited two peaks: one from the QWR and one from the QW. By reducing the probe current to 10 pA, a feature appeared on the low-energy side of the QW peak, which was identified as the emission via carbon acceptors in the QW. In the spot-mode spectra, the two features were isolated: (a) a spectrum with almost only C_{As} emission and (b) only intrinsic QW emission (Ref. 399).

shoulder is clearly saturated. Images of the C_{As} peak show bright spots, corresponding to single C_{As} . These bright spots show a reversed contrast compared with the image using the main QW peak. However, the dark stripes corresponding to the positions of the QWRs can be observed as dark lines in both images. The inset in Fig. 60 shows two spectra obtained in spot mode. A bright spot (a) and a dark spot (b) in the image of the C_{As} . In both cases, the spectra are dominated by a single peak, as most (>90%) of the recombination occurs via a single C_{As} (a) and via free-exciton recombination (b) in the QW.

The effect of impurities can be observed in the QWRs of the same sample. Due to the smaller volume of the QWR, the separation between the impurities is much larger. Figure 61(c) shows a monochromatic top-view image of an area, containing several QWRs appearing as bright lines. Figure 61(b) shows the same area, but obtained at a 25 meV lower

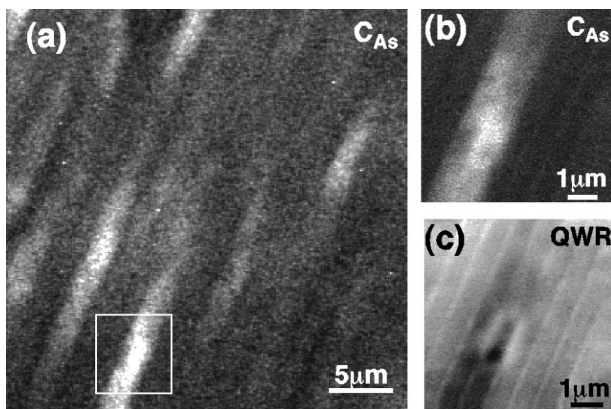


FIG. 61. Single impurities can also be found in the QWR of the sample of Fig. 60 by detecting the emission at an energy 25 meV below the QWR peak. (a) The bright spots corresponding to the single C_{As} in the QWR are separated by several tens of μm along the QWR. (b) and (c) show a higher magnification of the area outlined in white in (a). (b) In the image of the C_{As} emission, a single C_{As} can be observed. (c) shows the same area, but detecting the intrinsic QWR emission, where the individual QWRs can be observed. The images were recorded at 25 K (Ref. 400).

energy. Here, one bright spot can be observed. The spot is centered at one of the QWRs and is caused by one single C_{As} in the QWR. The shape of the bright spot illustrates the collection area for this part of the QWR, limited by the two adjacent QWRs (physically limited by the VQWs of the adjacent grooves). Along the groove, the size of the bright spot is determined by carrier diffusion along the QWR, several μm in this case. Figure 61(a) shows a larger area of the sample, illustrating the positions of several impurities. In the QWR the spacing is several tens of μm .⁴⁰⁰

IV. FUTURE DEVELOPMENTS

Local probe techniques share the same type of detection systems and they will all benefit from developments in this area. Since there are two general problems with these techniques in spatial stability and excitation density variations it is important to reduce the acquisition time. The main contribution to this is the use of parallel detection, mainly by using diode arrays or CCD cameras, in conjunction with standard monochromators. A more dedicated method is to use a CCD camera and a monochromator, with the detection in a very high order ($n \approx 100$) and a prism to disperse the orders. The 2D image on the CCD camera contains the spectral information over a very large range. The spectrum can be reconstructed using this 2D image. This setup is referred to as an echelle monochromator, which can reduce the time to record a complete spectrum. Another way to reduce the acquisition time is to optimize the light collection, both by increasing the solid angle and removing possible sources of losses between the sample and detector. A typical example of the latter is to replace a fiber bundle with mirrors, which can improve the sensitivity by an order of magnitude or more.

An important area for future studies will be more sophisticated experiments on a very localized scale such as time-dependent measurements in combination with streak cameras having a time resolution below 5 ps at the present time (and probably much better in the near future). We also expect a strong development of femtosecond spectroscopy on a very local scale.

More structures will be grown, specially designed for local probe studies, with the layers near the surface or even uncapped and with the interesting objects spatially separated enough to distinguish their individual signals. This typically means that in a sample the QDs should be separated by more than 1 μm . The idea of uncapped structures (or side-view studies) is that the surface, with all its surface states, will act as a sink for diffusing carriers, effectively killing all diffusion.⁶¹

In the case of $\mu\text{-PL}$ the technique is already at its resolution limit, so the development on the instrument side will be minor. However, the simplicity of the technique will increase its usefulness. With a more efficient and sensitive detection system, more and more of the types of measurements currently done in the conventional PL mode will soon be implemented in its $\mu\text{-PL}$ equivalent.

With the development in the imaging $\mu\text{-PL}$ area alternative methods for luminescence generation will be used. Today, luminescence imaging is generally used for studying

near-field patterns from laser diodes. This type of imaging of luminescence is also expected to become increasingly important in testing of integrated circuits where, e.g., hot luminescence from the switching in complementary metal-oxide semiconductor (CMOS) circuits can be recorded. This has recently been demonstrated.⁴⁰¹

Most of the work in improving the NSOM has concentrated on the fabrication of probe tips. The most favored approach is the tapered single-mode fiber tip, coated by various metals. The goals are both to reduce the losses and to improve the throughput of the tip, as well as to reduce the size of the aperture and thereby increase the spatial resolution. Developments in the area of etched fiber tips have significantly improved the signal intensities. With an improvement of three orders of magnitude over conventional pulled fiber tips, the acquisition time for an image is significantly reduced.^{115,121} The evolution of metallized tips can combine the NSOM with the STM, where the height regulation can be done with the constant current mode of the STM.¹²⁶ This in turn can give detailed topographical maps of the surface and spectral information simultaneously. The height regulation is another topic that has received attention lately. It is crucial for the resolution to keep the aperture at a constant height without risking to crash the tip into the surface.

One field that inevitably will attract more attention is the development of new low-temperature versions of the NSOM. The natural driving forces are the improved spectral resolution and the significantly higher signal intensity that result from the lowered temperature. The main obstacle to overcome is the complexity of these systems, where the already intricate designs of the room-temperature NSOM has to be incorporated in a temperature-stable and low-drift version.

With more and more NSOMs in operation there will be development in related methods. Here, we will only mention some of these techniques. By detecting the emission in the far-infrared range, it is possible to make images of the temperature distribution of devices while in operation, with a resolution of $\sim 5^\circ\text{C}$. If the device is a laser, the true near-field pattern can be recorded with a very high resolution. NOBIC, the near-field equivalent of the OBIC and EBIC techniques has also been demonstrated.³⁵

The trend for CL is clear with work being done at lower and lower acceleration voltages. Ten years ago most work was done at the highest possible voltage of the SEM, usually 30 keV, whereas today it is very seldom that the work is done above 10 keV. Most of the really high-resolution work is done at 1–3 keV and at low probe currents, typically, 100 pA or less.

Another future prospect is that the variation of acceleration voltage, and thereby the depth of the excitation, will play an important role in the study of complicated and layered structures, especially with the possibility to simulate the excitation volume. What is needed to complete this side of the CL technique is an investigation of how the acceleration voltage and distance from the surface influence the optimum resolution. It is not clear that a lower voltage gives the best resolution, as a too low voltage will give a larger portion of excitation via diffusion and not direct excitation. The diffu-

sion part will inevitably result in a broadened excitation, i.e., lower resolution.

Two different approaches to the STL have emerged during the last few years. Having the STM in an UHV environment makes it possible to perform studies on virtually contamination-free surfaces, such as cleaved surfaces. With this approach it is possible to obtain atomically resolved images and intensity maps simultaneously.³⁶³ A simpler approach, both from an instrumentation and operating point of view, is to have the STM in a protective atmosphere of, e.g., He gas. In this case the spatial resolution is generally slightly lower. Like all the other local probe techniques, the incorporation of a cooling stage is very desirable, as discussed above. The best solution is, therefore, a low-temperature STM which is operated in a UHV environment.

One of the major difficulties in the work with STL is to position the tip in the area of interest. This generally requires optical access to the sample, in order to observe the tip-sample region with an optical microscope for coarse positioning. Optical access is also essential for an efficient detection system. In a slightly different setup the STM can be included in a SEM, where the coarse positioning is done in the SEM mode. This also means that both STL and CL can be recorded, where ordinary CL imaging can be used to locate interesting features.

The possibility to make time-resolved measurements with STM has been demonstrated.^{402,403} However, a more suitable design would be an AFM with a conducting tip, where the positioning of the tip is done without current injection in contrast to the STM.

On the physics side there are plenty of problems, e.g., in the field of QDs. In which systems are biexcitons observable? The relaxation of carriers into QDs needs further investigation. Is there a bottleneck for the relaxation of the carriers as observed by the occurrence of multiple emission lines also at a low temperature?^{285,384} What is the contribution of carrier-carrier scattering to carrier relaxation? What is the effect of the strong coupling of carriers in QDs? What is the nature of the phonons? For QWRs similar questions arise. In addition, detailed investigations of the factors limiting the mobility of carriers along the QWR need to be investigated.

In summary, the future prospects lie mainly in developing more efficient and faster detection systems. The more mature techniques of CL and μ -PL will see limited developments of the instruments themselves, whereas the scanning probe-based techniques will probably see more developments on the instrument side as well.

V. SUMMARY

We have in this review article summarized the experimental situation concerning applications of local luminescence spectroscopic techniques for studies of low-dimensional semiconductor structures. Different experimental spectroscopic and imaging techniques have been described and compared from the points of view of mechanisms and instrumentation. A survey has been presented of published work where these techniques have been applied to

different problems of low-dimensional semiconductor structures. Finally, an outlook for future developments of this field of research has been presented.

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APPENDIX A: MAGNITUDE OF ABSORPTION

Absorption in bulk semiconductors is a well-known phenomenon and it is comparatively straight forward to verify experimentally.¹⁰⁴ However, in low-dimensional structures the bulklike absorption is no longer valid. To calculate the absorption, a quantum-mechanical approach is needed.¹ In the very common model of quantum structures (the $\mathbf{k} \cdot \mathbf{p}$ model⁴⁰⁴ in conjunction with the envelope function approximation⁴⁰⁵) the wave function of a confined electron Ψ can be described as a linear combination of zone-center (i.e., $\mathbf{k}=0$) Bloch functions u_j , multiplied by envelope functions F_j such that

$$\Psi = \sum_j F_j(r) u_j(r). \quad (\text{A1})$$

The number of functions j ($j=1,2,\dots,n$) in the expansion is arbitrary, but in practice j is usually at most 8, which involves six valence-band states and two conduction-band states (i.e., including spin). A practical method to solve the eigenvalue problem is given in Ref. 7 along with a detailed discussion. The Schrödinger equation, which has Eq. (A1) as a solution depends on $\mathbf{k}$ of the problem (unless the quantum structure is a QD, where $\mathbf{k}$ has lost its meaning).

If we restrict ourselves to less than eight bands ($j < 8$), it is necessary to solve two problems, one for the electrons, and one for the holes. It is often sufficient to ignore spin, which leaves us with only one band for the electrons and one or two bands for the holes. In the simplest approximation with only one band for the electron and one band for the holes we have

$$\Psi_e(r) = F_e(r) u_e(r), \quad (\text{A2})$$

for the electrons and,

$$\Psi_h(r) = F_h(r) u_h(r), \quad (\text{A3})$$

for the holes.

The absorption strength is proportional to the square of the matrix element:

$$W \sim \langle \Psi_f(r) | \partial/\partial x | \Psi_i(r) \rangle, \quad (\text{A4})$$

which becomes:

$$\begin{aligned} W &\sim \langle \Psi_f(r) | \partial/\partial x | \Psi_i(r) \rangle \\ &= \langle F_f(r) u_f(r) | \partial/\partial x | F_i(r) u_i(r) \rangle \\ &= \langle F_f(r) u_f(r) | F'_i(r) u_i(r) \rangle \\ &\quad + \langle F_f(r) u_f(r) | F_i(r) u'_i(r) \rangle, \end{aligned} \quad (\text{A5})$$

where in the last equality we have used a familiar rule for the differentiation of a product. If a function $\Gamma_n(r)$ varies so slowly that it may be taken to be constant over a unit cell and if $g_n(r)$ have the periodicity of the lattice, we have

$$\begin{aligned} \langle \Gamma_m(r) g_m(r) | \Gamma_n(r) g_n(r) \rangle \\ = \langle \Gamma_m(r) | \Gamma_n(r) \rangle \langle g_m(r) | g_n(r) \rangle. \end{aligned} \quad (\text{A6})$$

The envelope functions and their derivatives are the slowly varying functions, and the Bloch function and their derivatives have the lattice periodicity. If we use the relation in Eqs. (A6) and (A5) we get

$$\begin{aligned} W &\sim \langle F_f(r) | F'_i(r) \rangle \langle u_f(r) | u_i(r) \rangle + \langle F_f(r) | F_i(r) \rangle \\ &\quad \times \langle u_f(r) | u'_i(r) \rangle. \end{aligned} \quad (\text{A7})$$

For transitions between the valence band and the conduction band, we have different (orthogonal) Bloch functions in the initial and the final state (i.e., $u_f \neq u_i$), and the first term (which is responsible for intersubband transitions in the valence band or in the conduction band), therefore, becomes zero. The factor $\langle u_f(r) | u'_i(r) \rangle$ in the second term is constant for each semiconductor and is very similar for all direct band-gap III–V semiconductors. For transitions between the valence band and the conduction band we, therefore, get

$$W \sim \langle F_f(r) | F_i(r) \rangle. \quad (\text{A8})$$

Assuming equal hole and electron masses and equal depths of the potential wells, the overlap $\langle F_f(r) | F_i(r) \rangle$ is either 0 or 1, and does not depend on the width of the QW. The magnitude of the absorption due to transitions between the ground state of the valence-band well and the ground state of the conduction-band well can be calculated using Eq. (A8). With the appropriate constants, this is $\sim 1\%$ in direct band-gap III–V QWs. At high photon energy there will be more than one possible transition contributing to the absorption.

APPENDIX B: POLARIZATION

In order to include polarization effects it is necessary to include more than two terms in the expansion Eq. (A1). If we include six valence bands and two conduction bands, the eigenfunctions at $\mathbf{k}=0$ for bulk direct band-gap III–V semiconductors are given in the notation $|J, m_j\rangle$, where J is the angular momentum and m_j is the projection along the z axis by

$$\begin{aligned} \left| \frac{3}{2}, \frac{3}{2} \right\rangle &= \frac{1}{\sqrt{2}} (x + iy) \uparrow, \\ \left| \frac{3}{2}, -\frac{3}{2} \right\rangle &= \frac{1}{\sqrt{2}} (x - iy) \downarrow, \end{aligned}$$

$$\left| \frac{3}{2}, \frac{1}{2} \right\rangle = \frac{1}{\sqrt{6}} (x + iy) \downarrow - \sqrt{\frac{2}{3}} z \uparrow, \quad (\text{B1})$$

$$\left| \frac{3}{2}, -\frac{1}{2} \right\rangle = -\frac{1}{\sqrt{6}} (x - iy) \uparrow - \sqrt{\frac{2}{3}} z \downarrow,$$

$$\left| \frac{1}{2}, \frac{1}{2} \right\rangle = \frac{1}{\sqrt{3}} (x + iy) \downarrow + \sqrt{\frac{1}{3}} z \uparrow,$$

$$\left| \frac{1}{2}, -\frac{1}{2} \right\rangle = -\frac{1}{\sqrt{3}} (x - iy) \uparrow + \sqrt{\frac{1}{3}} z \downarrow.$$

These are the eigenfunctions for the valence band only. The x, y, z functions form a different basis, which is often used and is defined using these relations. The arrows indicate spin up ($\uparrow$) and spin down ($\downarrow$).

The states

$$\left| \frac{3}{2}, \frac{3}{2} \right\rangle, \quad \text{and} \quad \left| \frac{3}{2}, -\frac{3}{2} \right\rangle, \quad (\text{B2})$$

are the heavy-hole states

$$\left| \frac{3}{2}, \frac{1}{2} \right\rangle, \quad \text{and} \quad \left| \frac{3}{2}, -\frac{1}{2} \right\rangle, \quad (\text{B3})$$

the light-hole states, and

$$\left| \frac{1}{2}, \frac{1}{2} \right\rangle, \quad \text{and} \quad \left| \frac{1}{2}, -\frac{1}{2} \right\rangle, \quad (\text{B4})$$

belong to the spin-orbit split-off band.

If a crystal is interacting with an electromagnetic field, the momentum term $\mathbf{p}$ in the Hamiltonian is replaced by

$$\frac{\hbar^2}{2m} \left(\mathbf{p} - \frac{e}{c} \mathbf{A} \right)^2, \quad (\text{B5})$$

where $\mathbf{A}$ is the vector potential of the electromagnetic field, which is usually simplified to

$$\frac{\hbar^2}{2m} \mathbf{p}^2 - \frac{\hbar^2 e}{mc} \mathbf{p} \cdot \mathbf{A}. \quad (\text{B6})$$

It can be seen that the second term is the perturbation which is responsible for time-dependent phenomena such as absorption and emission. The vector potential $\mathbf{A}$ for a plane monochromatic wave is given by

$$\mathbf{A} = A_0 \boldsymbol{\epsilon} e^{i(\omega/c) \mathbf{n} \cdot \mathbf{x} - i\omega t} + \text{c.c.}, \quad (\text{B7})$$

where c.c. denotes complex conjugation, $\mathbf{n}$ is the propagation vector, and $\boldsymbol{\epsilon}$ is the polarization vector. In the electric dipole approximation we get

$$\mathbf{A} = A_0 \boldsymbol{\epsilon} e^{-i\omega t} + \text{c.c.}, \quad (\text{B8})$$

and consequently, the electric dipole operator becomes proportional to

$$A_0 \mathbf{p} \cdot \boldsymbol{\epsilon} e^{-i\omega t} + \text{c.c.} \quad (\text{B9})$$

The first term is responsible for absorption and the second term is responsible for emission. A particular component of $\mathbf{p}$, say p_x , has a nonvanishing matrix element only between $|x\uparrow\rangle$ and $|s\uparrow\rangle$ or between $|x\downarrow\rangle$ and $|s\downarrow\rangle$, due to symmetry, where $|s\uparrow\rangle$ and $|s\downarrow\rangle$ are the conduction-band states. A simple calculation of the overlap between $|s\uparrow\rangle$ and $|\frac{3}{2}, \frac{3}{2}\rangle$,

and between $|s\uparrow\rangle$ and $|\frac{3}{2}, \frac{1}{2}\rangle$ using this electric dipole operator shows that the transition strengths (given by the square of the overlap) should have a ratio of 3 with the highest transition strength for the light-hole state. In practice, the situation is more complicated since the states defined by Eq. (B1) are not the true eigenfunctions of a QW (see Appendix A). However, it is true that for lattice-matched QWs, the transition strength is highest for the light-hole state.

APPENDIX C: ACRONYMS

AFM	Atomic force microscope
ALE	Atomic layer epitaxy
AP-MOVPE	Atmospheric pressure metalorganic vapor phase epitaxy
BF	Backscattered electrons
CBE	Chemical beam epitaxy
CCD	Charge couple device
CL	Cathodoluminescence
CLWI	Cathodoluminescence wavelength imaging
DOS	Density of states
e-h pairs	electron-hole pairs
EBIC	Electron-beam-induced current
EDAX	Energy dispersive analysis of x rays
EELS	Electron energy-loss spectroscopy
EL	Electroluminescence
FIB	Focused ion beam
GS-MBE	Gas source molecular beam epitaxy
L_{diff}	Diffusion length
lHe	Liquid helium
LN ₂	Liquid nitrogen
LP-MOVPE	Low-pressure metalorganic vapor phase epitaxy
LPE	Liquid phase epitaxy
MBE	Molecular beam epitaxy
MC	Monte Carlo
ML	Monolayer
MOCVD	Metalorganic chemical vapor deposition
MOVPE	Metalorganic vapor phase epitaxy
NOBIC	Near-field optical beam-induced current
MQW	Multiquantum well
NSOM	Near-field optical microscope
OBIC	Optical beam-induced current
OMCVD	Organometallic chemical vapor deposition
PE	Primary electrons
PL	Photoluminescence
PLE	Photoluminescence excitation
PMT	Photomultiplier tube
PSTM	Photon scanning tunneling microscope
QD	Quantum dot
QW	Quantum well
QWR	Quantum wire
RIE	Reactive ion etching
SE	Secondary electrons

SEM	Scanning electron microscope
SLS	Superlattice
SQW	Single quantum well
STEM	Scanning transmission electron microscope
STL	Scanning tunneling luminescence
STM	Scanning tunneling microscope
TEM	Transmission electron microscope
UHV	Ultrahigh vacuum
VPE	Vapor phase epitaxy
VQW	Vertical quantum well
WDAX	Wavelength dispersive analysis of x rays
μ -PL	Microphotoluminescence
μ -PLE	Microphotoluminescence excitation
0D	Zero-dimensional
1D	One-dimensional
2D	Two-dimensional
3D	Three-dimensional

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